

STRICT HIERARCHY BETWEEN PARALLEL, SEQUENTIAL, AND INDEFINITE-CAUSAL-ORDER STRATEGIES

FOR CHANNEL DISCRIMINATION

JESSICA BAVARESCO, MIO MURAO, MARCO TÚLIO QUINTINO

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arXiv:2011.08300 [quant-ph]



THE TASK:

MINIMUM-ERROR CHANNEL DISCRIMINATION

STATE DISCRIMINATION

STATE DISCRIMINATION

CANDIDATES:

$$\rho_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \rho_2 = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

STATE DISCRIMINATION

CANDIDATES:

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INPUT:



STATE DISCRIMINATION

CANDIDATES:

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INPUT:



PROMISSE:



=



,



=

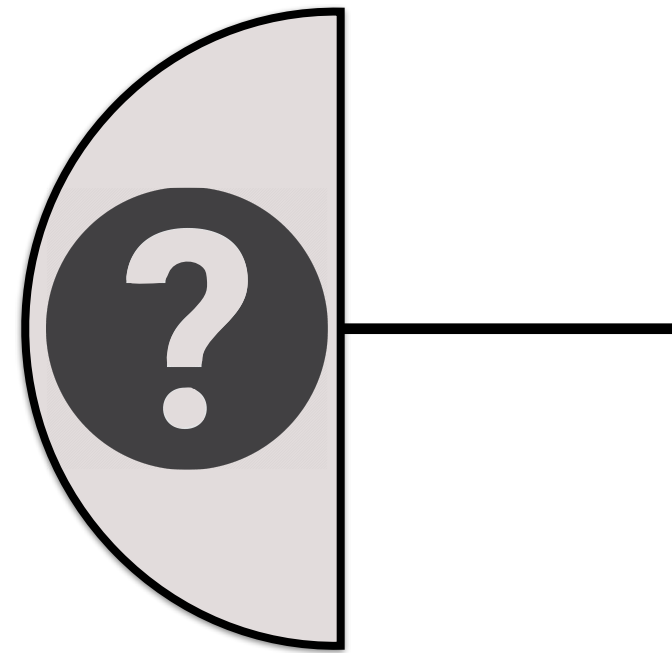


(with probability $p_1 = \frac{1}{3}$)

(with probability $p_2 = \frac{2}{3}$)

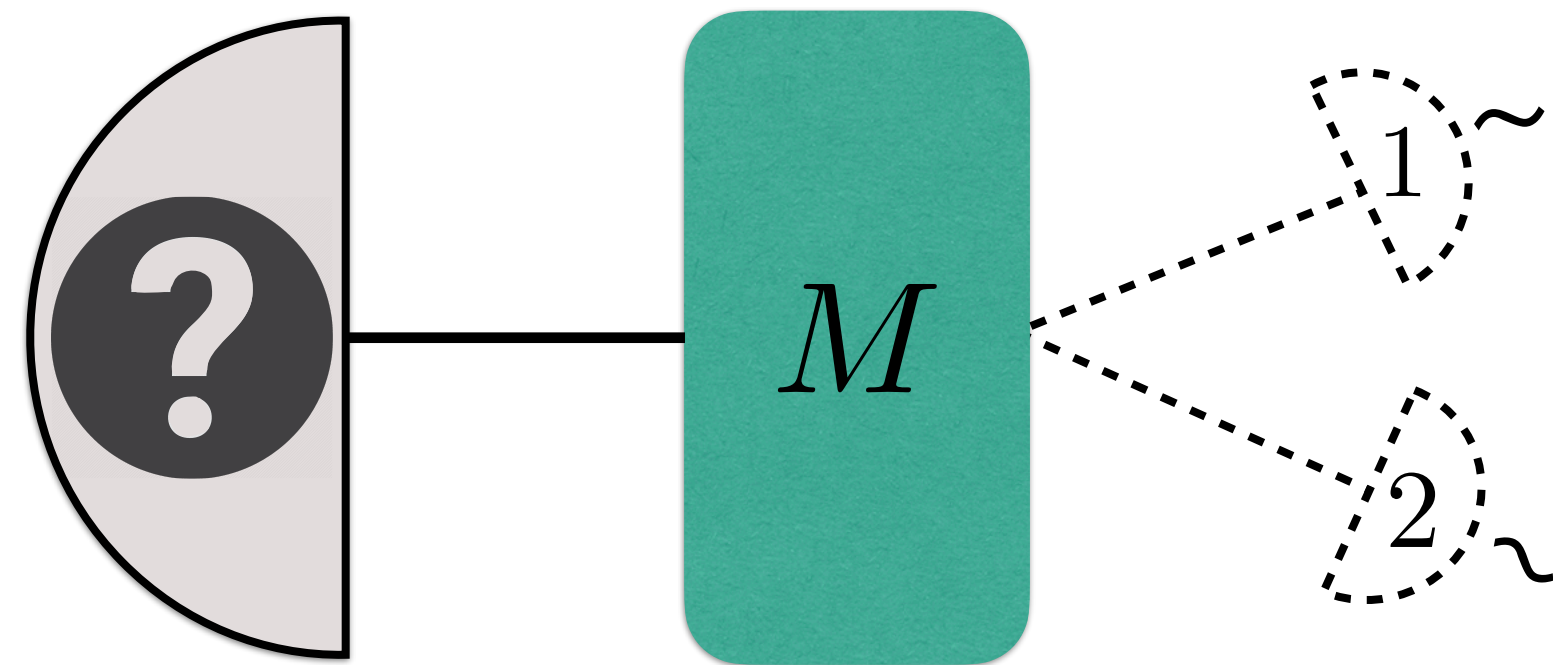
STRATEGY

ONE COPY!



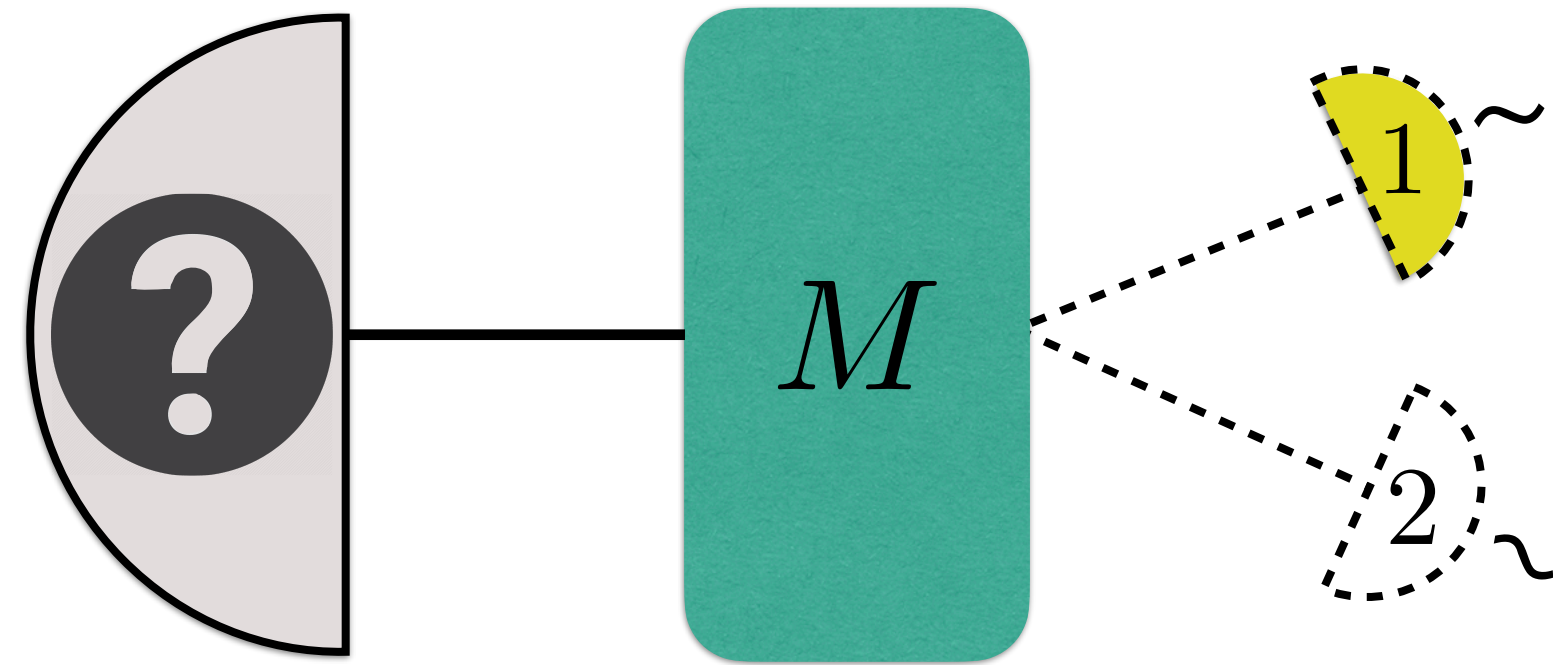
STRATEGY

ONE COPY!



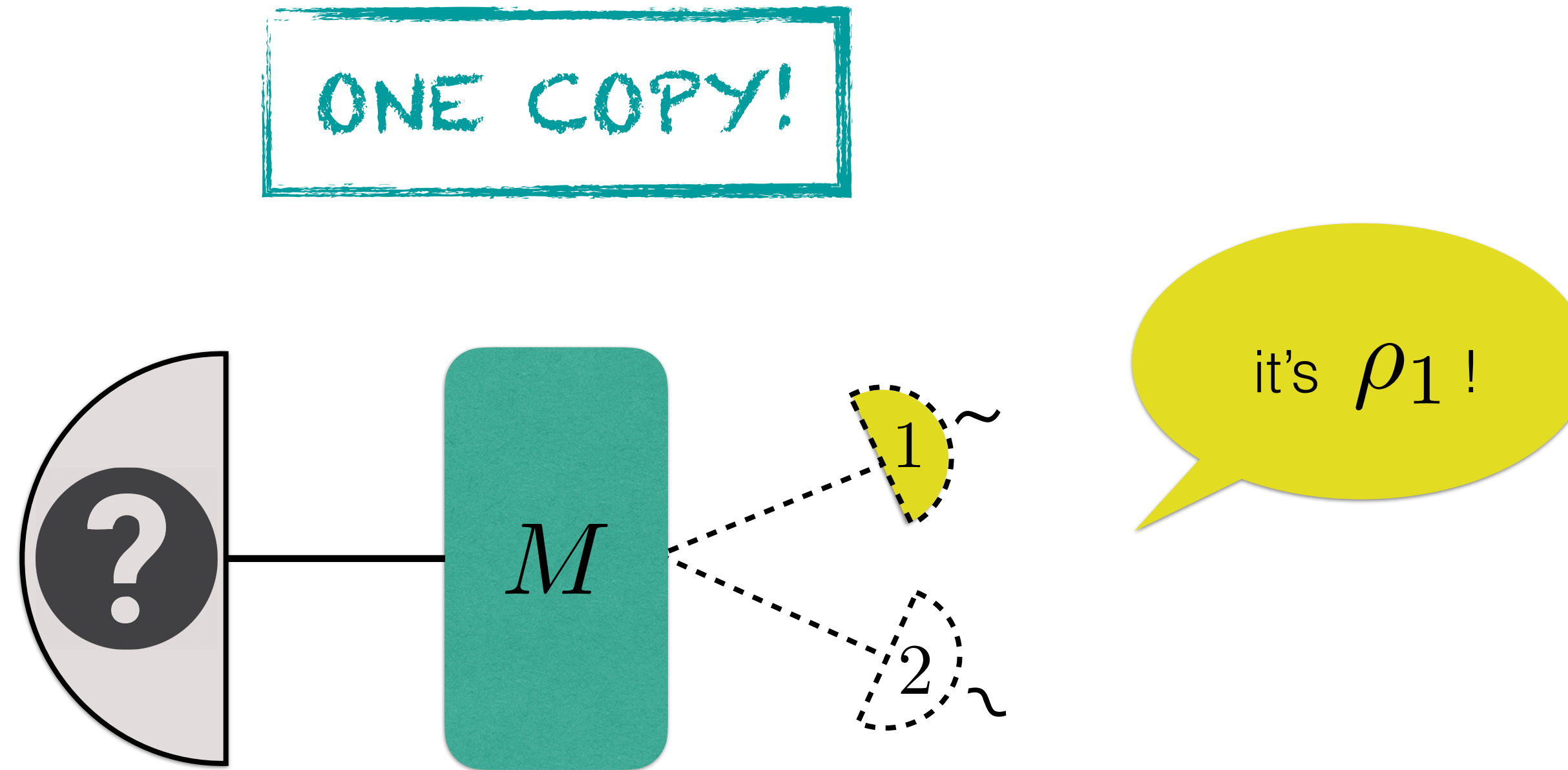
STRATEGY

ONE COPY!



it's ρ_1 !

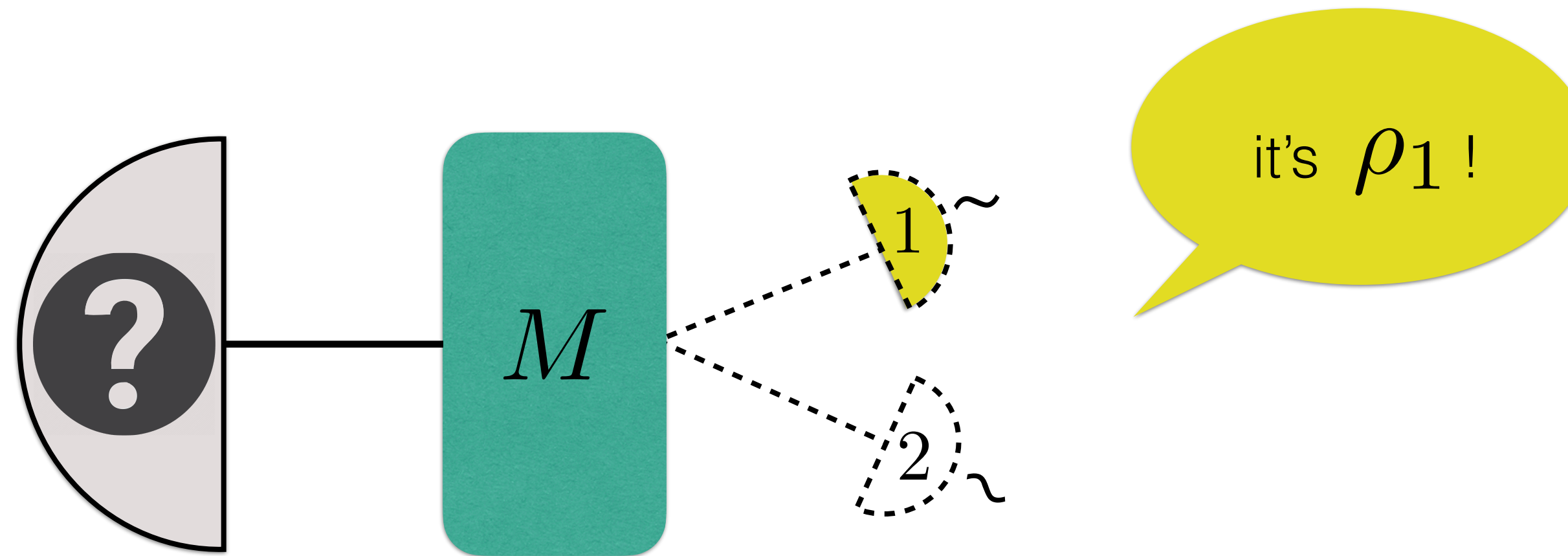
STRATEGY



$$\begin{aligned} p_{\text{succ}} &= p_1 p(1|\rho_1, M) + p_2 p(2|\rho_2, M) \\ &= p_1 \text{tr}(M_1 \rho_1) + p_2 \text{tr}(M_2 \rho_2) \end{aligned}$$

STRATEGY

ONE COPY!



$$\begin{aligned} p_{\text{succ}} &= p_1 p(1|\rho_1, M) + p_2 p(2|\rho_2, M) \\ &= p_1 \text{tr}(M_1 \rho_1) + p_2 \text{tr}(M_2 \rho_2) \end{aligned}$$

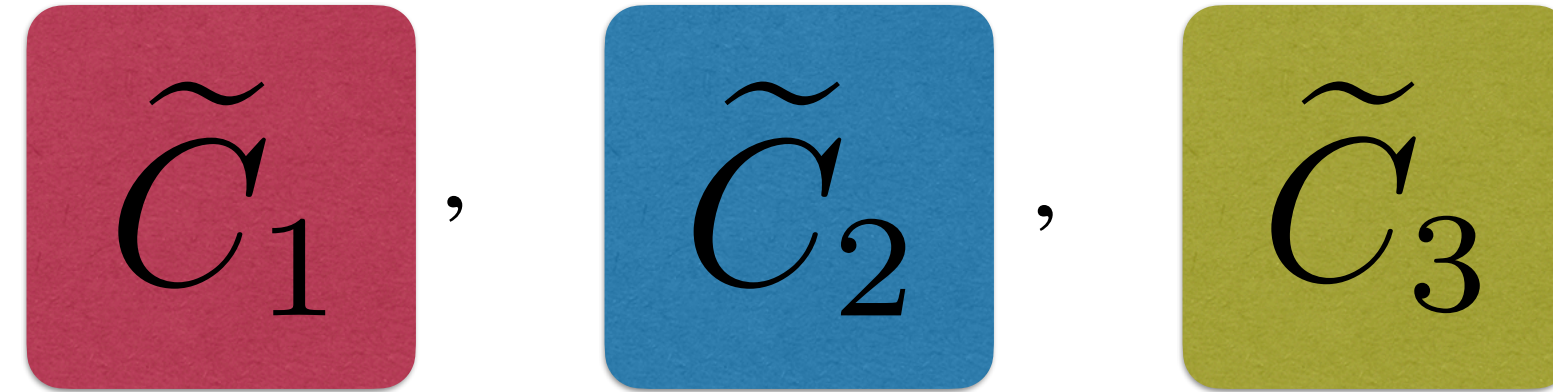
$$p_{\text{succ}}^* = \max_{\{M_1, M_2\}} p_1 \text{Tr}(M_1 \rho_1) + p_2 \text{Tr}(M_2 \rho_2)$$

✓ ACTUAL
THE TASK:

MINIMUM-ERROR CHANNEL DISCRIMINATION

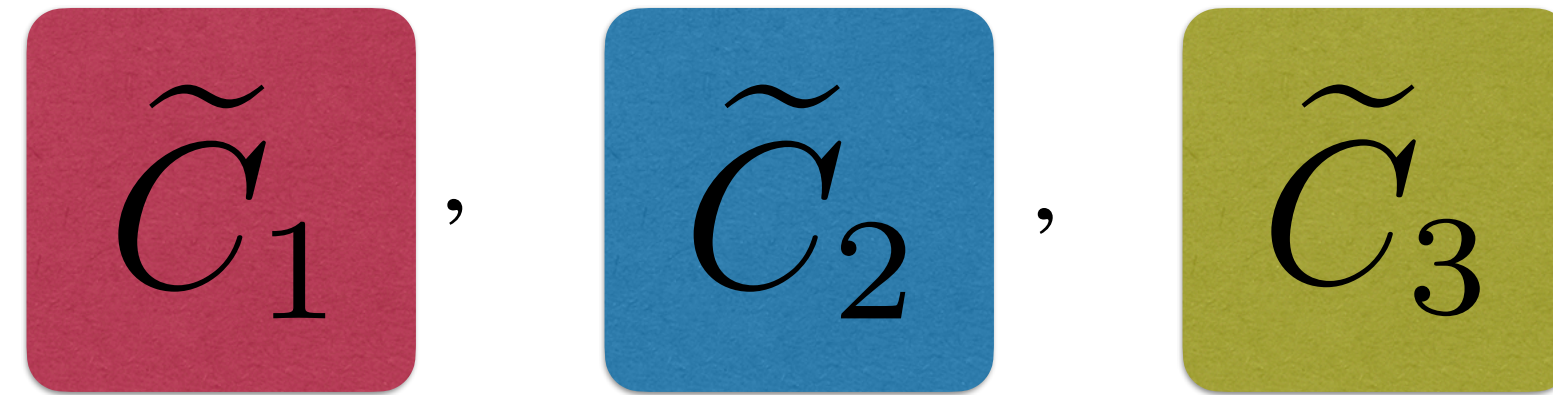
CHANNEL DISCRIMINATION

CANDIDATES:



CHANNEL DISCRIMINATION

CANDIDATES:

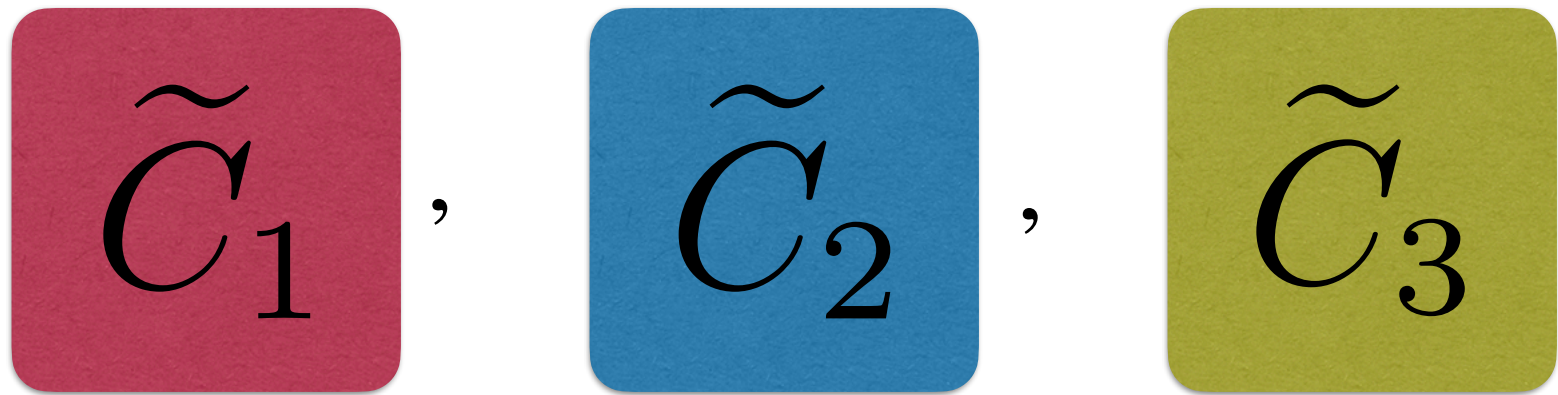


INPUT:



CHANNEL DISCRIMINATION

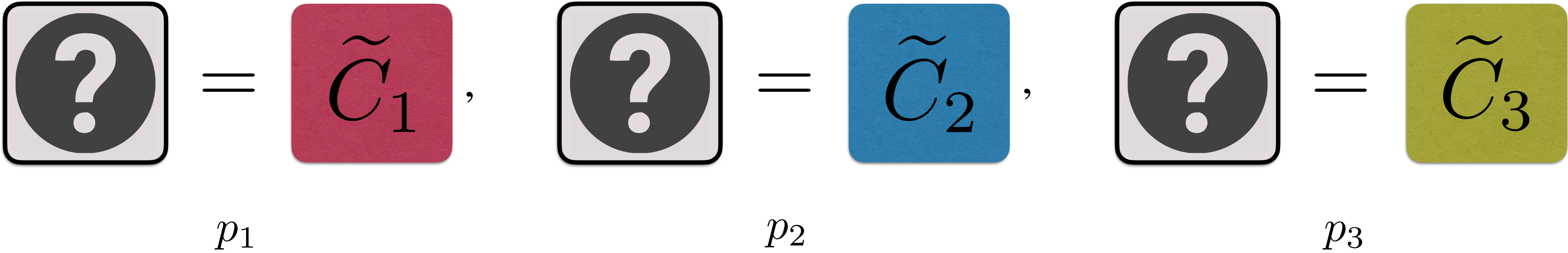
CANDIDATES:



INPUT:



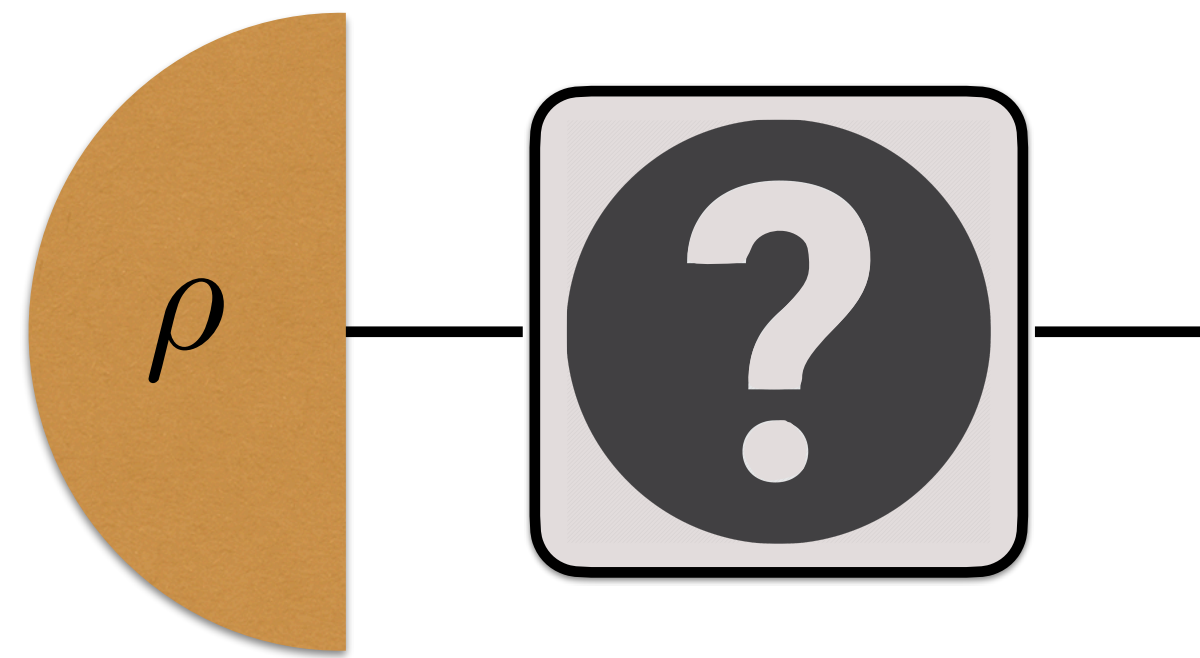
PROMISSE:



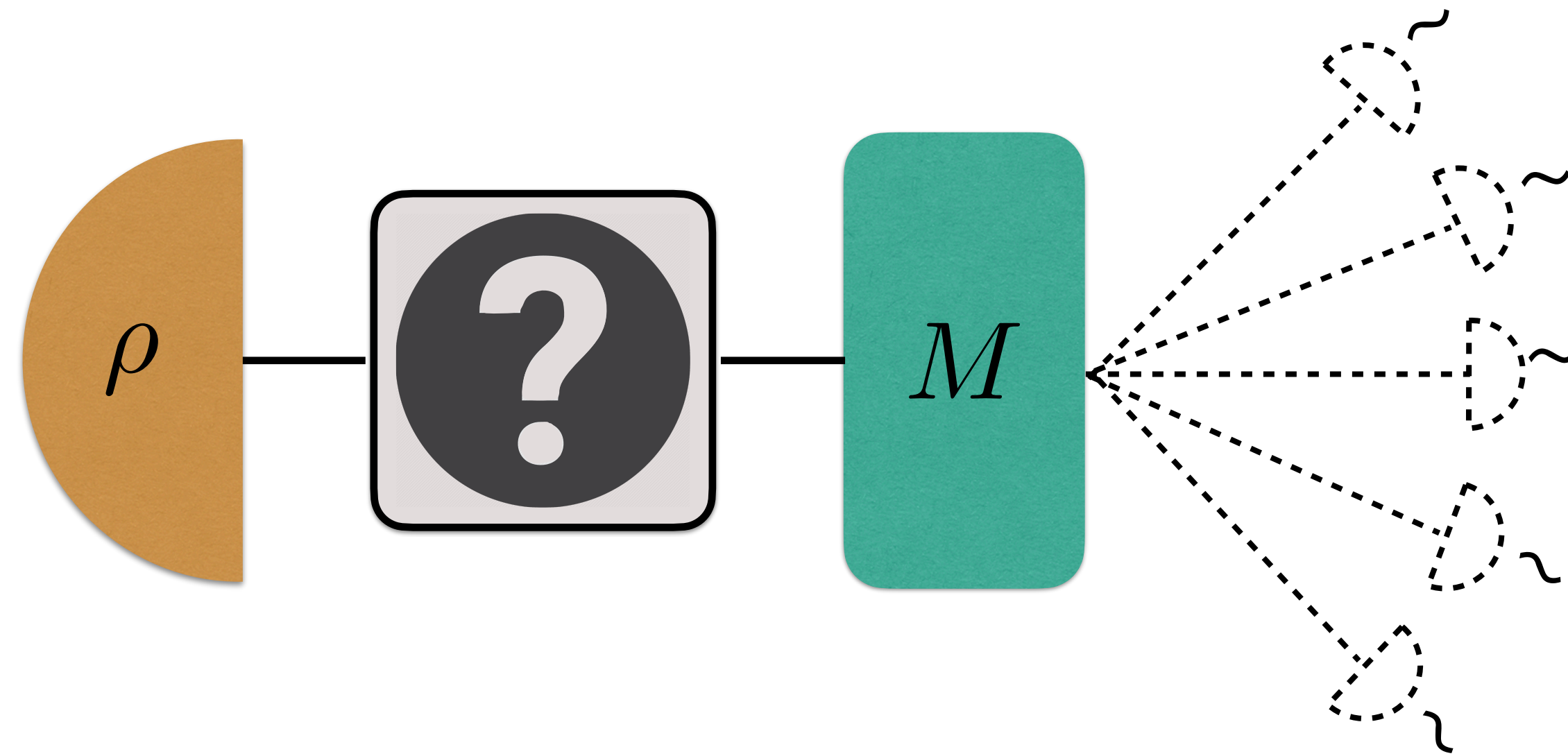
STRATEGY

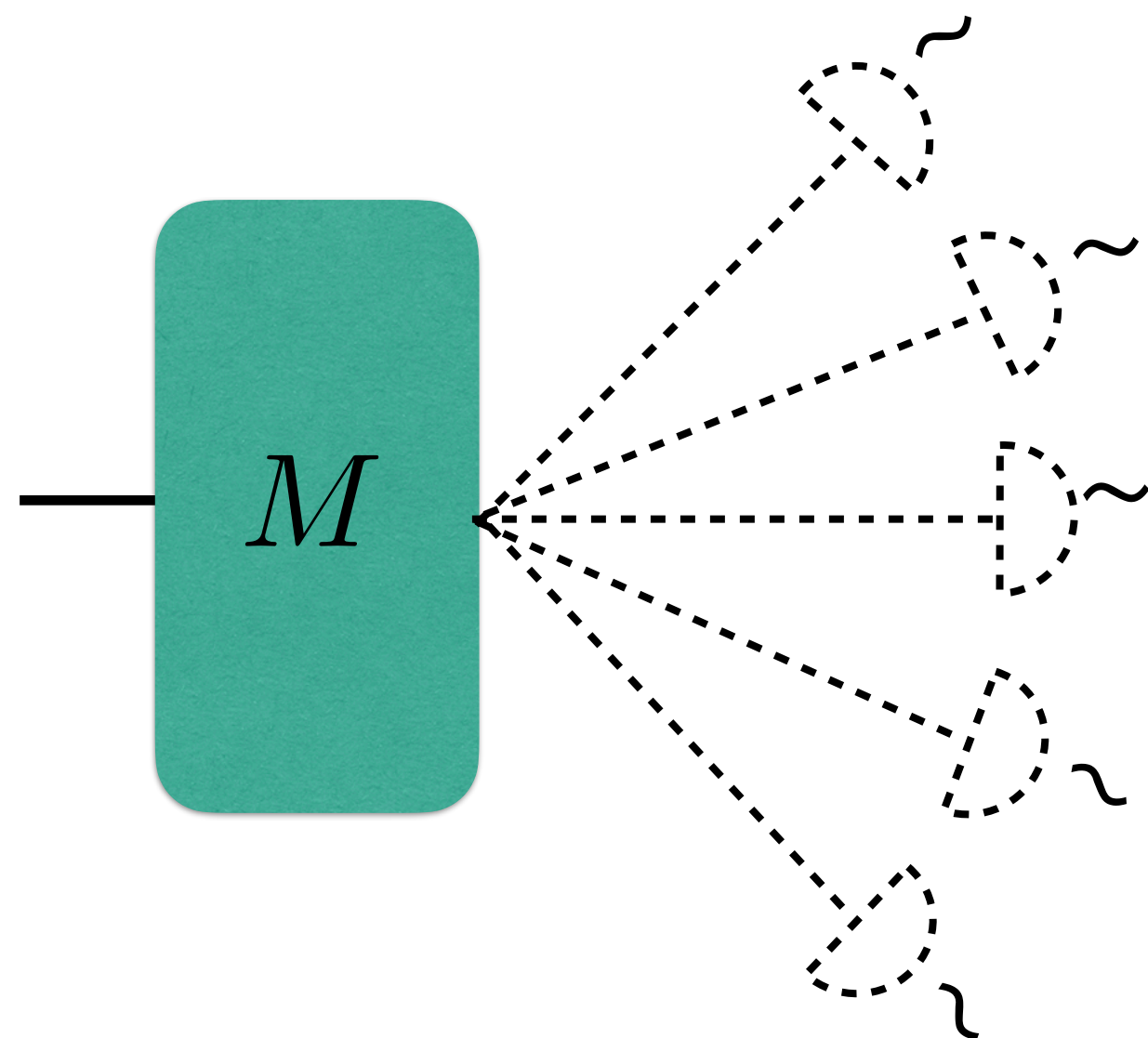


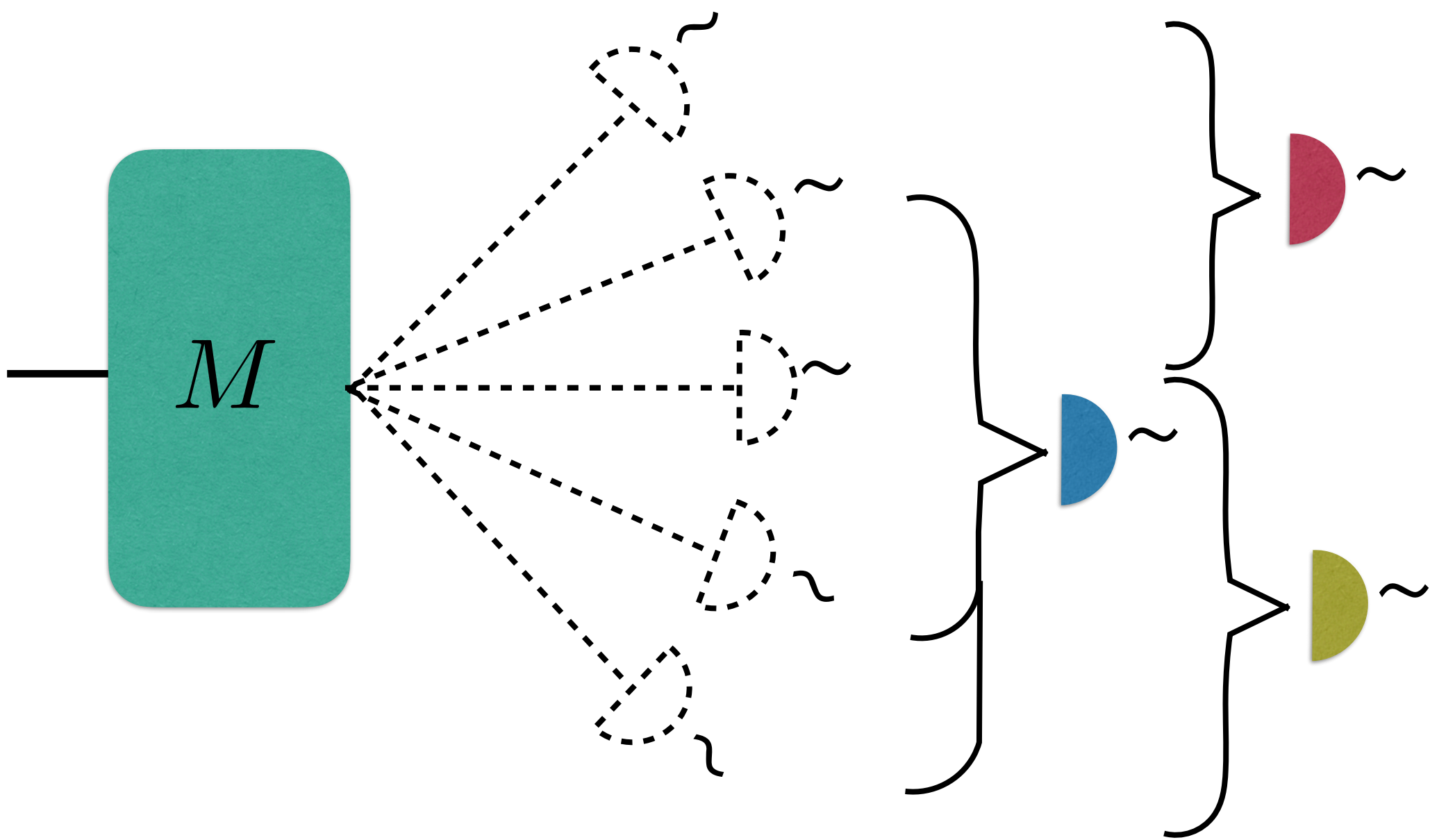
STRATEGY



STRATEGY



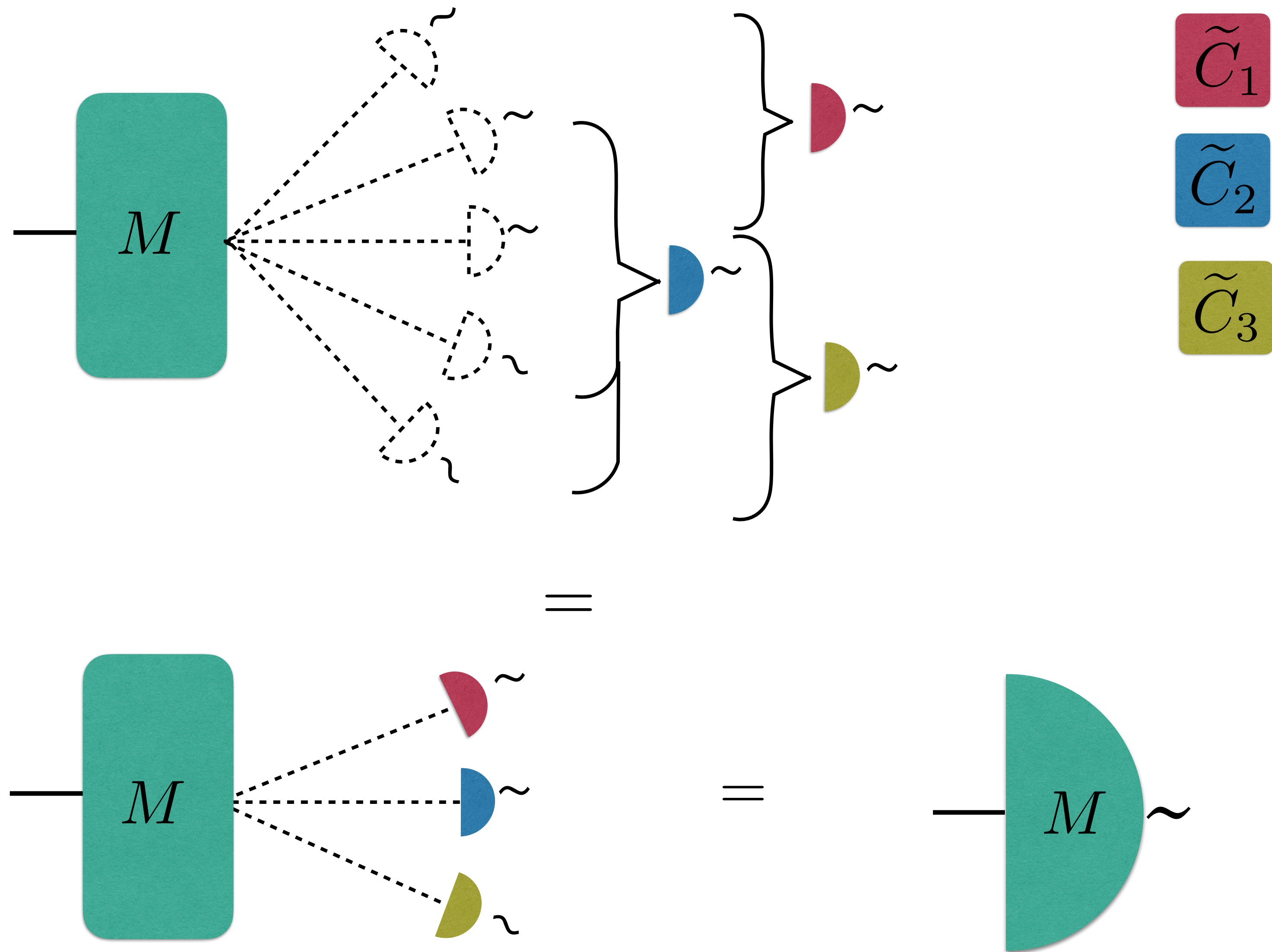




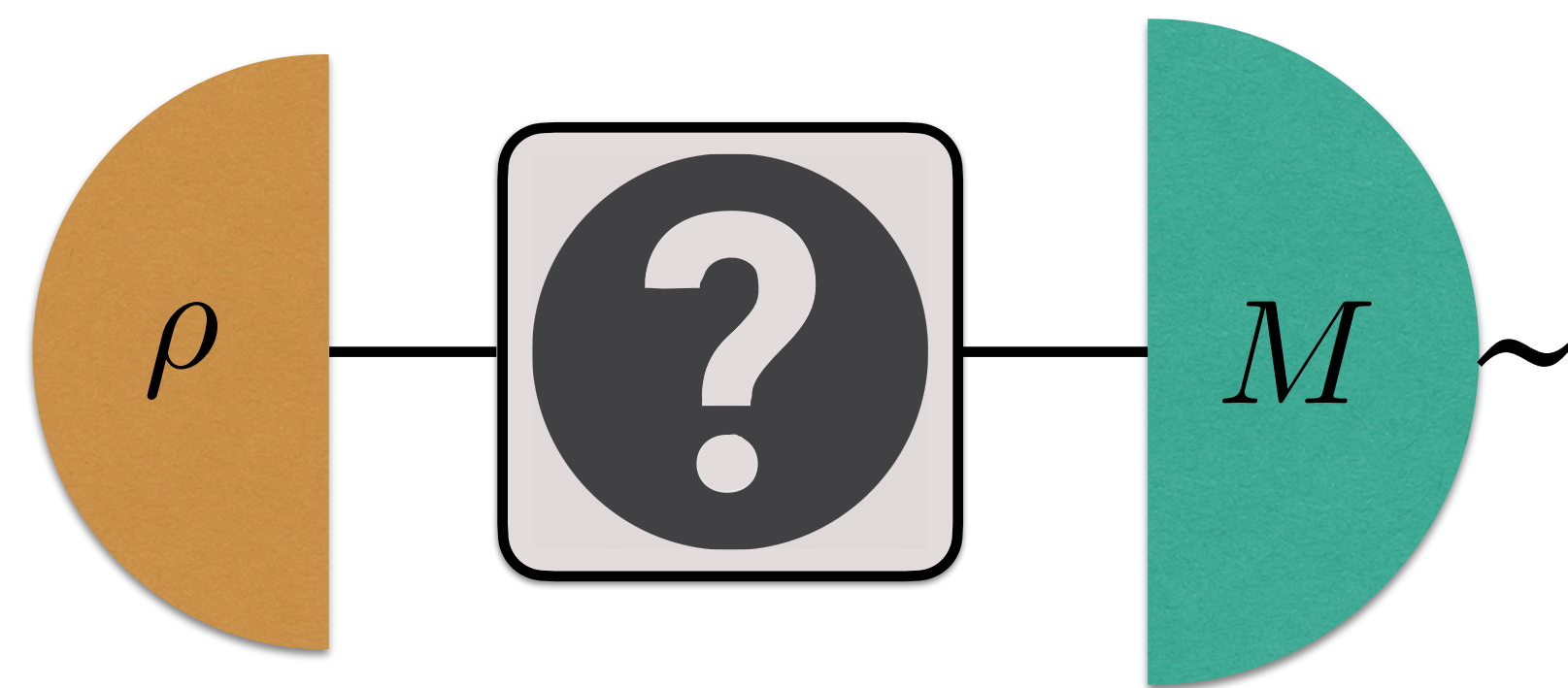
$\tilde{C}_1$

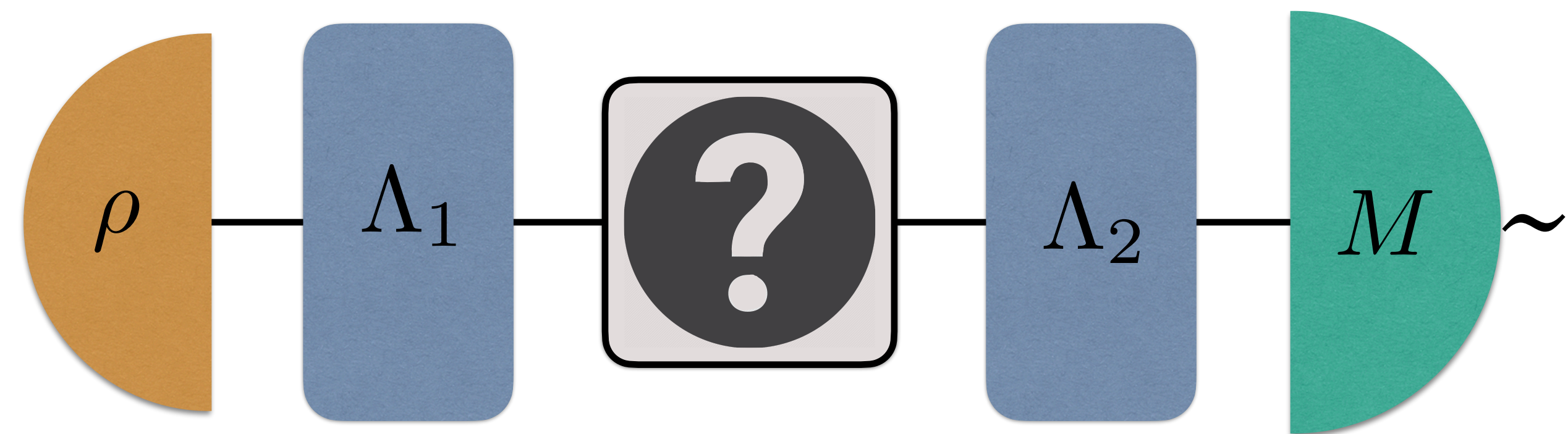
$\tilde{C}_2$

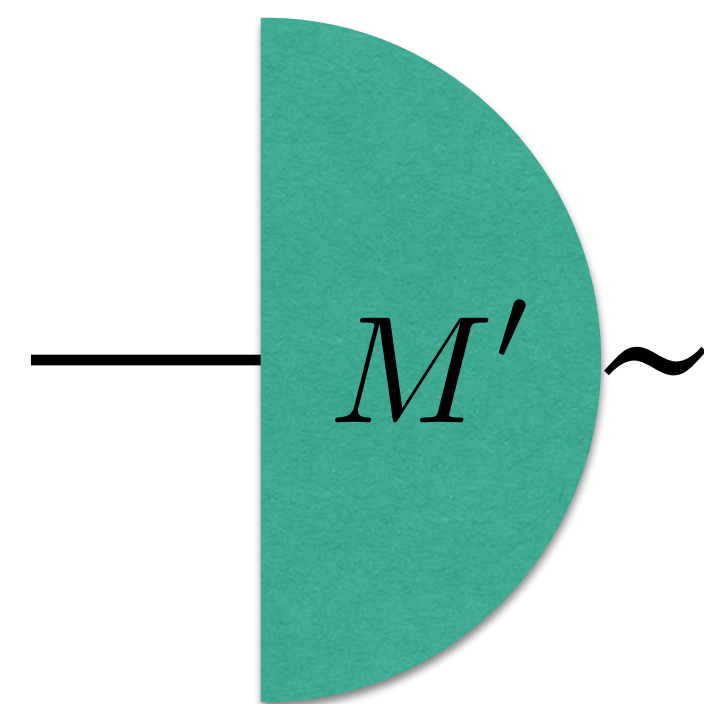
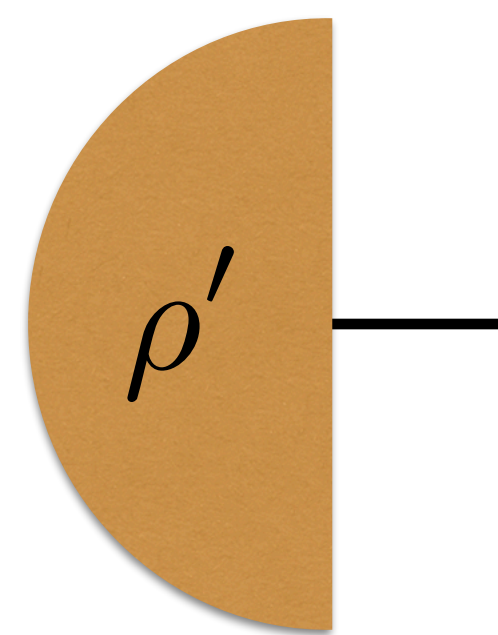
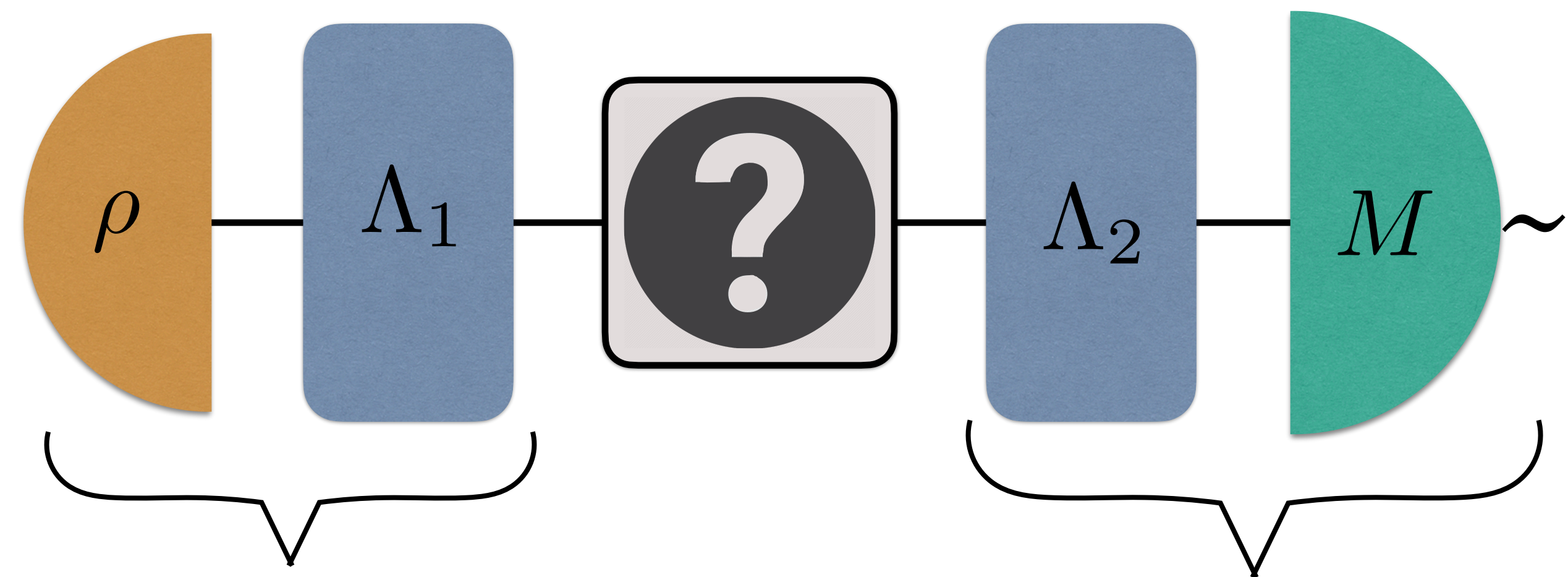
$\tilde{C}_3$

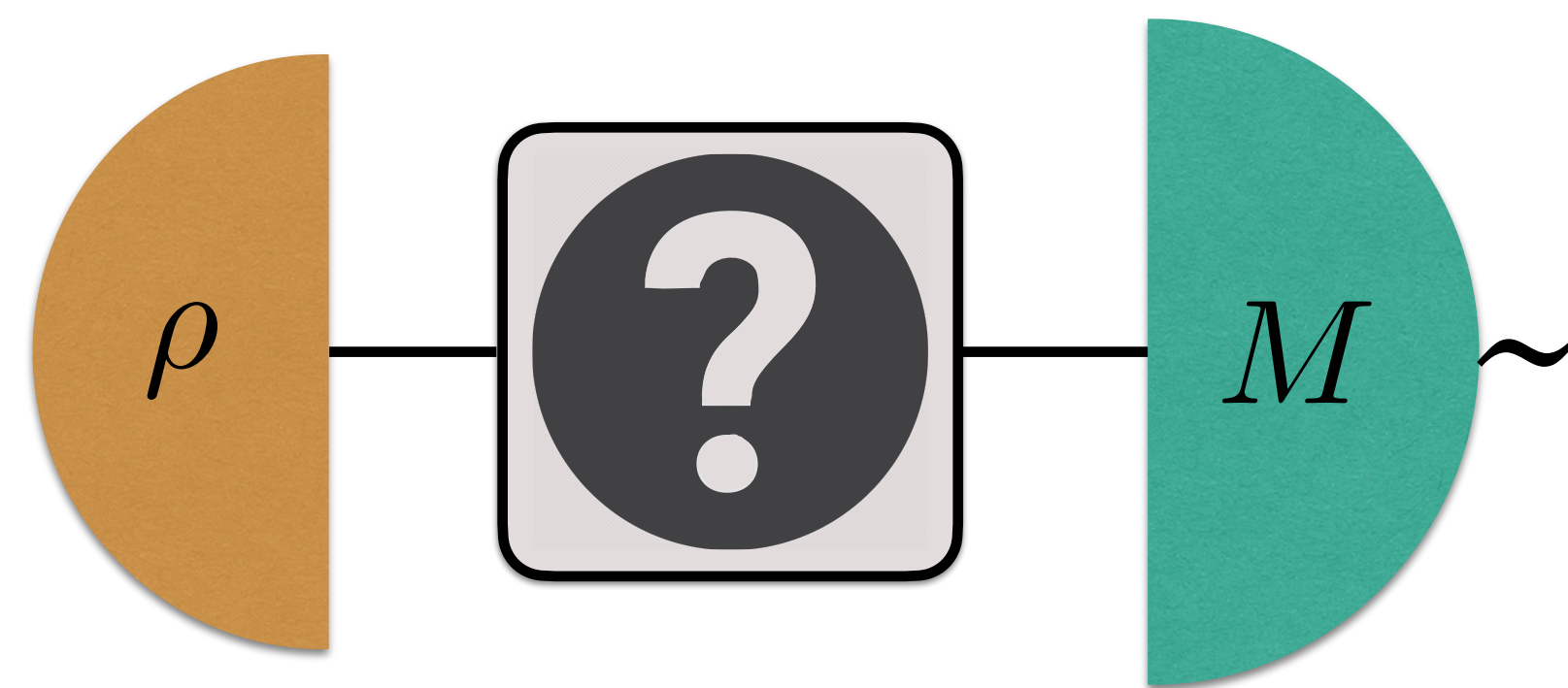


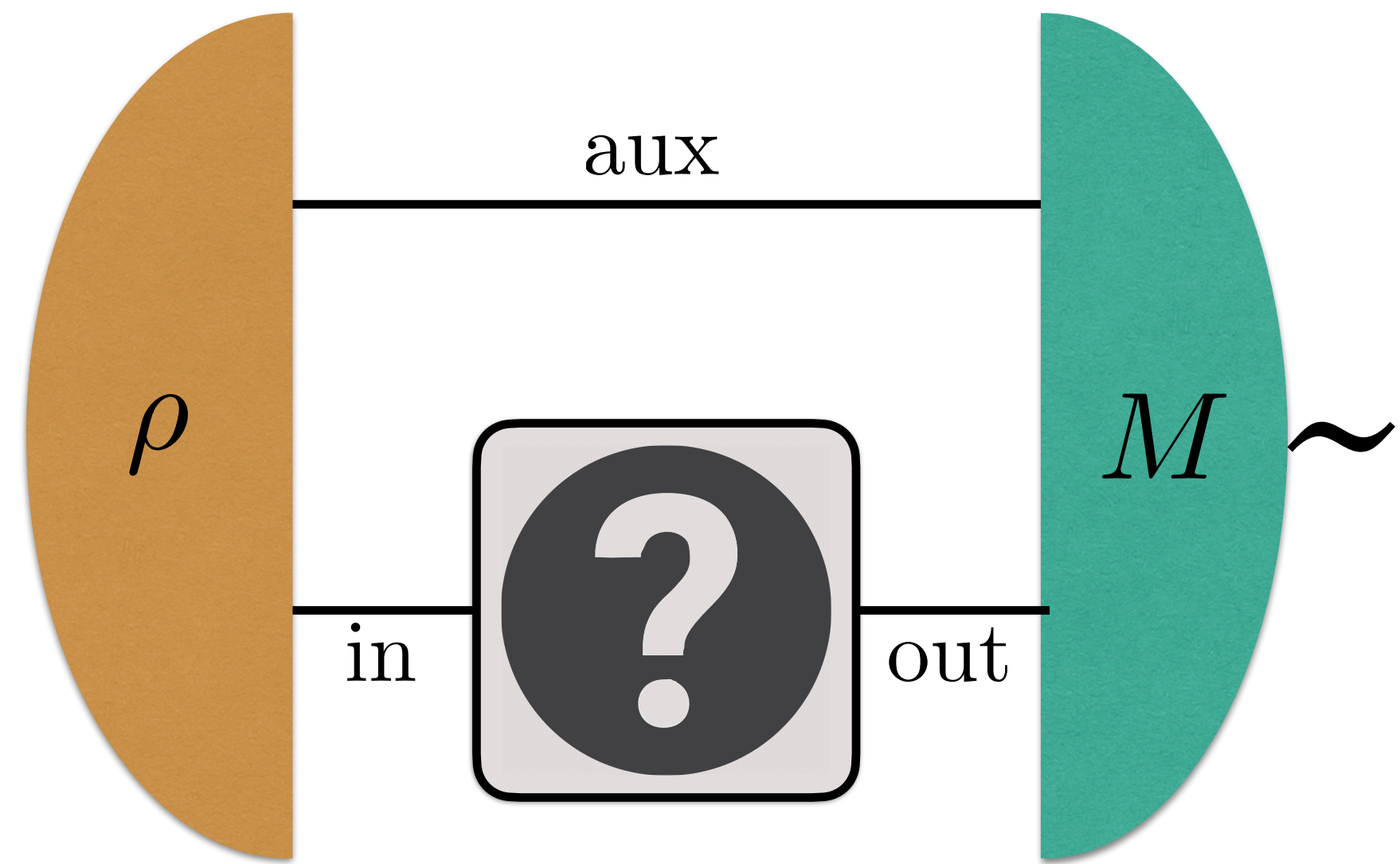
(as many outcomes as candidates)



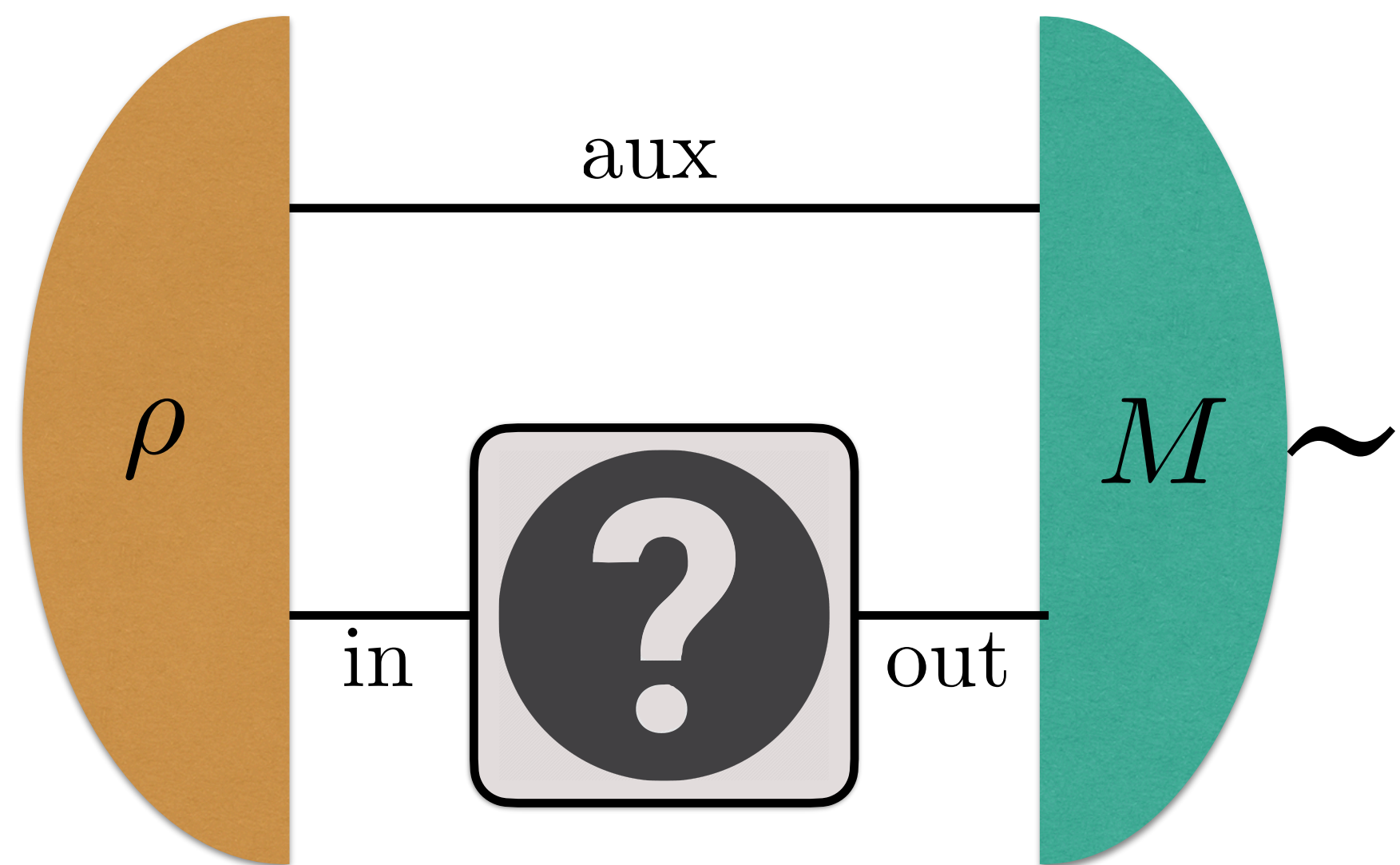




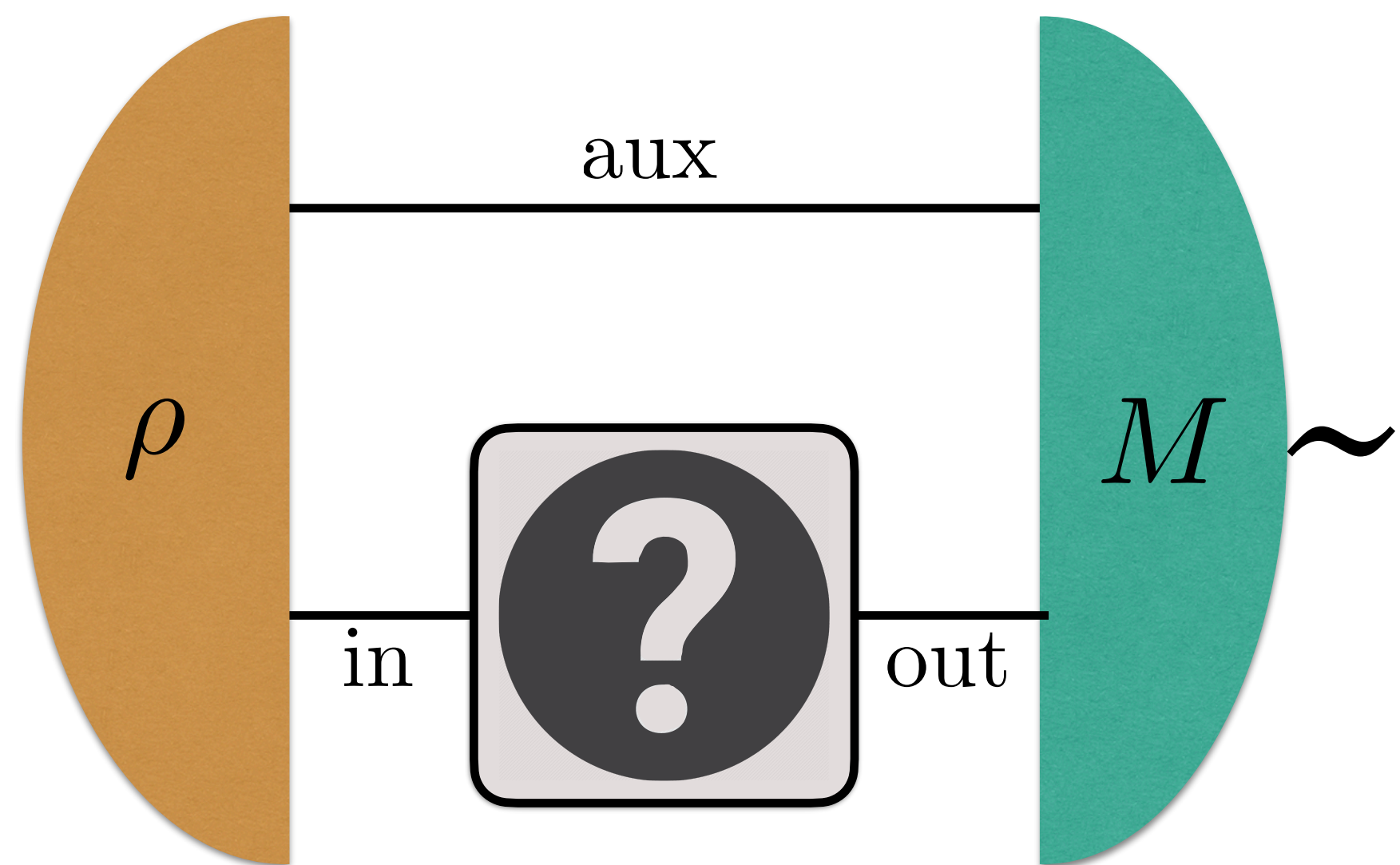




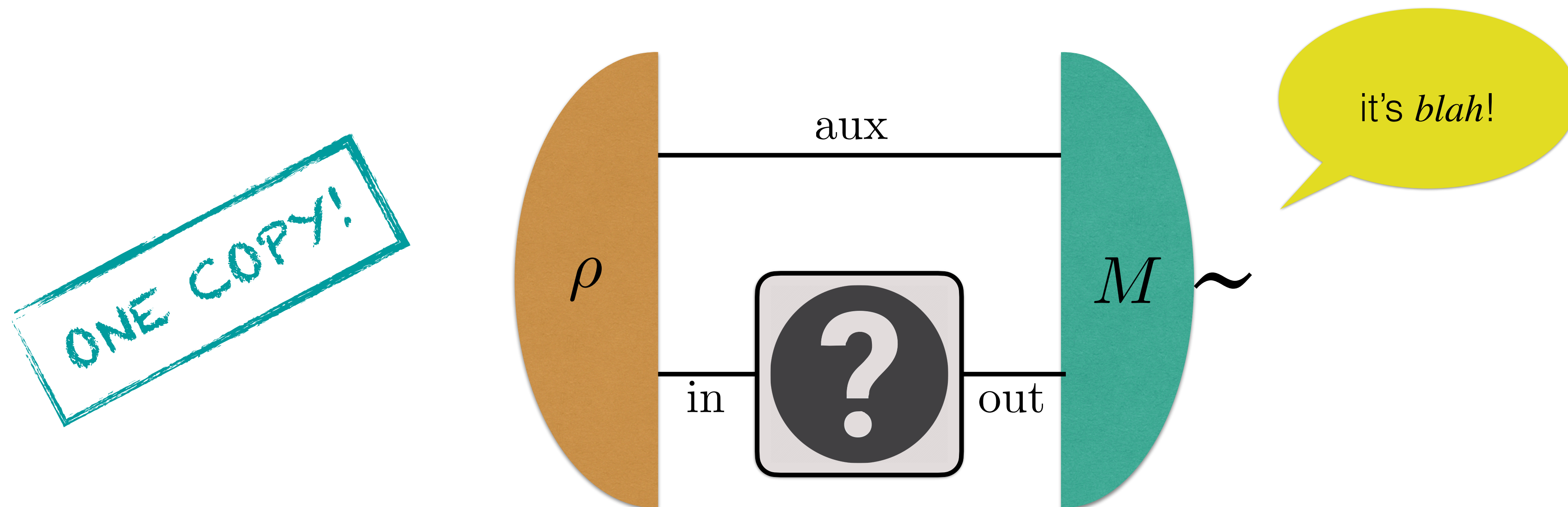
ONE COPY!



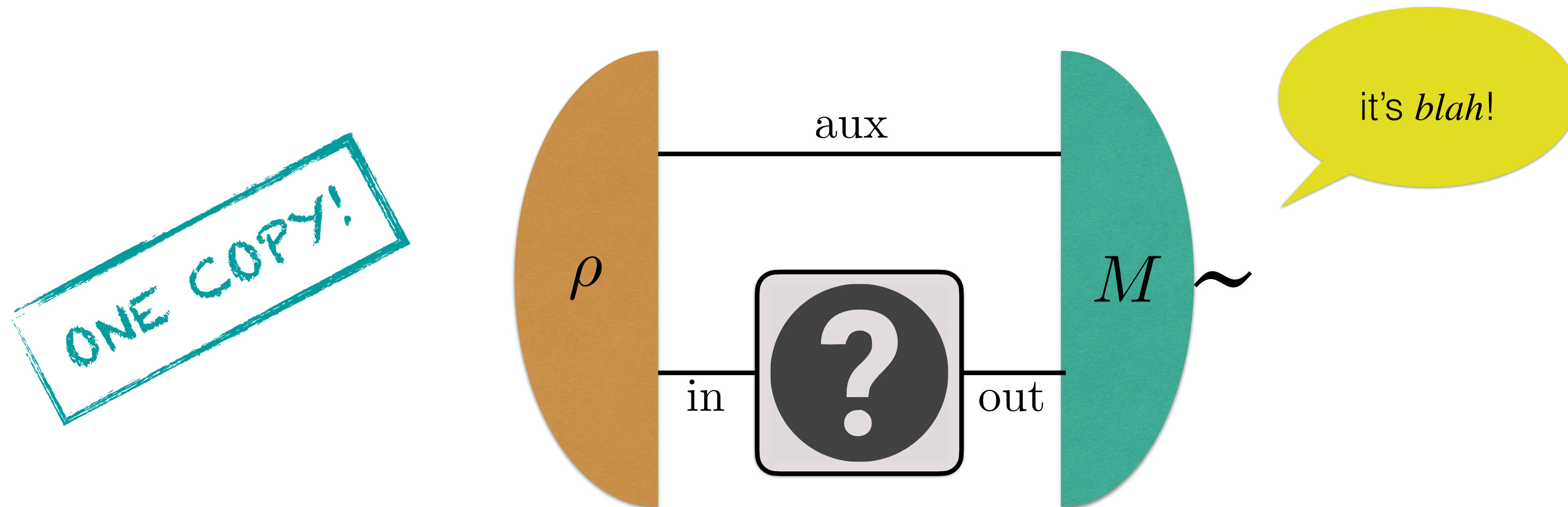
ONE COPY!



it's *blah*!



$$\begin{aligned}
 p_{\text{succ}} &= p_1 p(1|\tilde{C}_1, \rho, M) + p_2 p(2|\tilde{C}_2, \rho, M) + p_3 p(3|\tilde{C}_3, \rho, M) \\
 &= \sum_{i=1}^N p_i \text{Tr}[(\tilde{C}_i \otimes \tilde{\mathbb{I}})(\rho) M_i]
 \end{aligned}$$



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$$= \sum_{i=1}^N p_i \text{Tr}[(\tilde{C}_i \otimes \tilde{\mathbb{I}})(\rho) M_i]$$

$$P := \max_{\underline{\rho, \{M_i\}}} \sum_{i=1}^N p_i \text{Tr}[(\tilde{C}_i \otimes \tilde{\mathbb{I}})(\rho) M_i]$$

EXAMPLE

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ENSEMBLE:

$$\{p_i\} = \left\{ \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4} \right\}$$

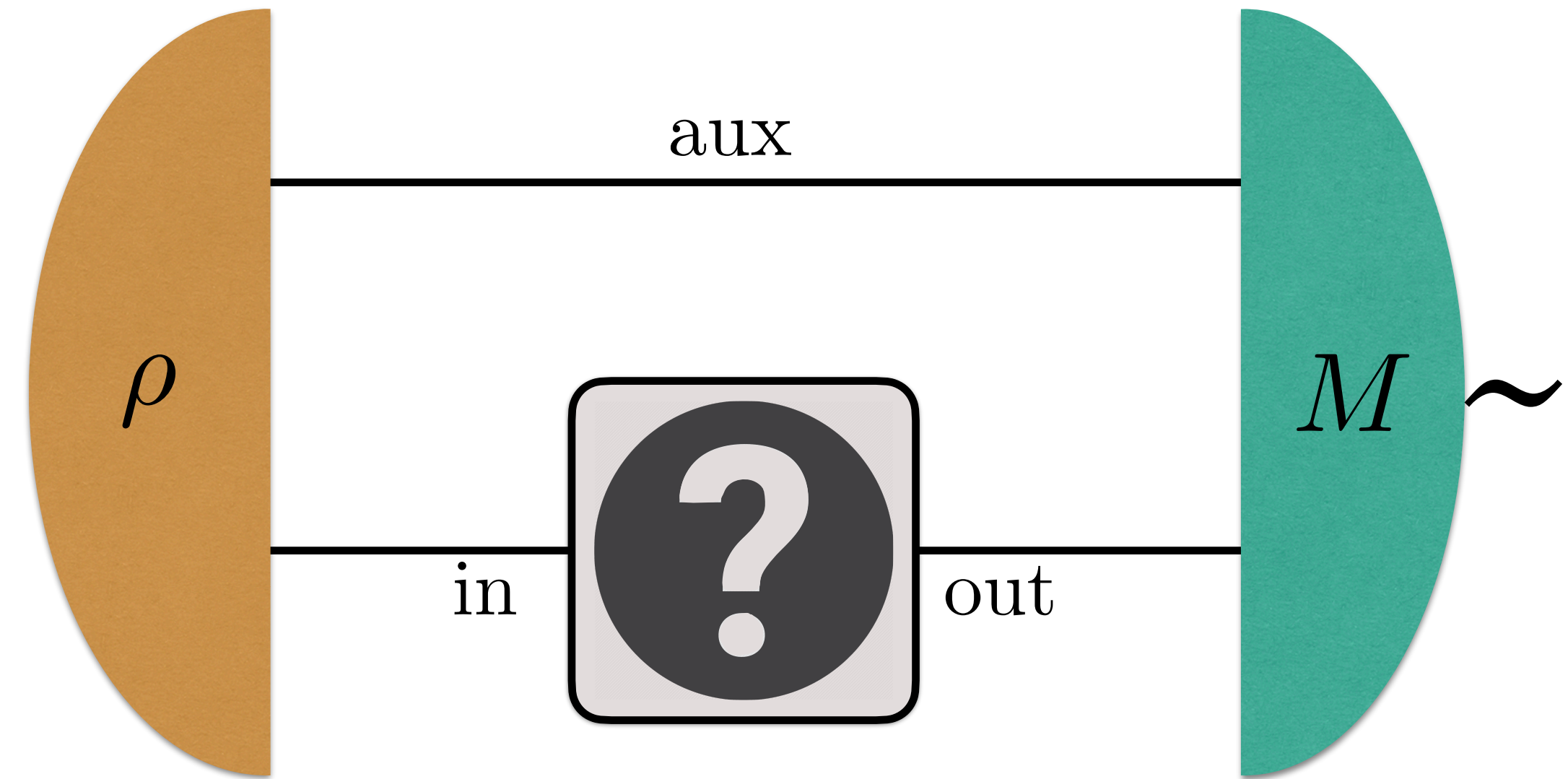
$$\{C_i\} = \{\mathbb{I}, \sigma_X, \sigma_Y, \sigma_Z\}$$

EXAMPLE

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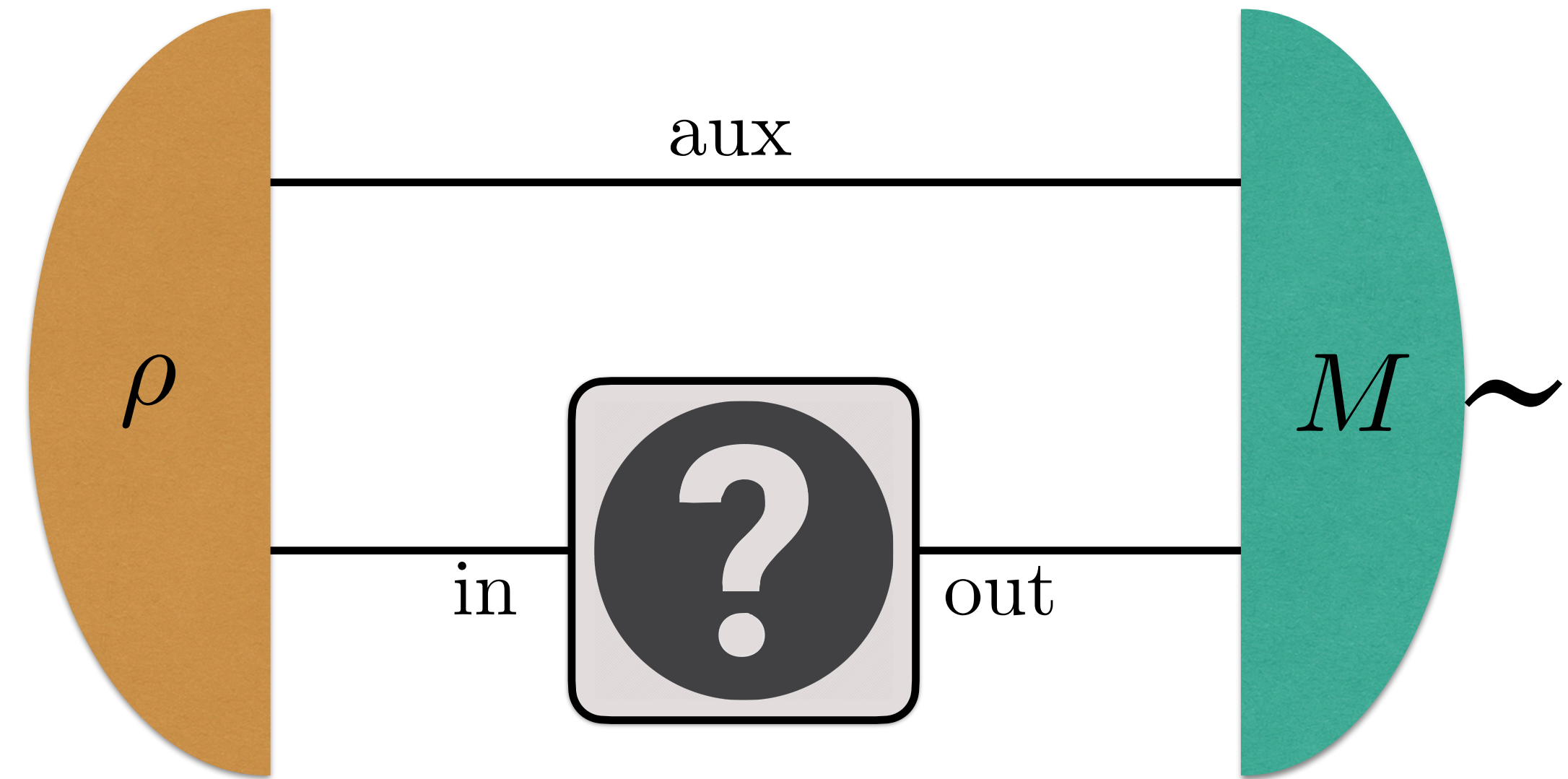
EXAMPLE

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$$\{C_i\} = \{\mathbb{I}, \sigma_X, \sigma_Y, \sigma_Z\}$$

STRATEGY:

$$\rho = |\Phi^+\rangle\langle\Phi^+|$$
$$\{M_i\} = \{|\Phi^+\rangle\langle\Phi^+|,$$
$$|\Phi^-\rangle\langle\Phi^-|,$$
$$|\Psi^+\rangle\langle\Psi^+|,$$
$$|\Psi^-\rangle\langle\Psi^-|\}$$



EXAMPLE

ENSEMBLE:

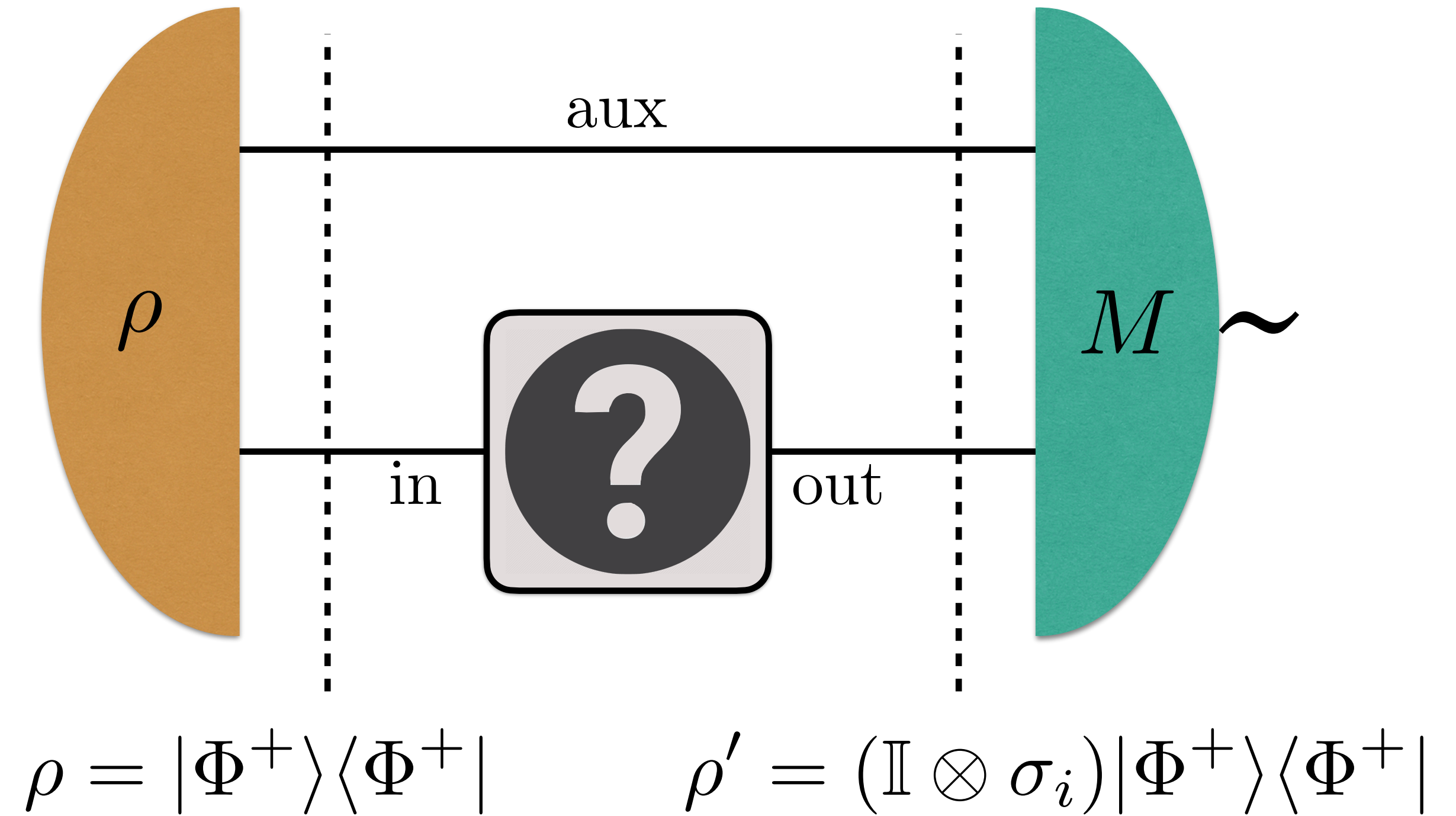
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EXAMPLE

ENSEMBLE:

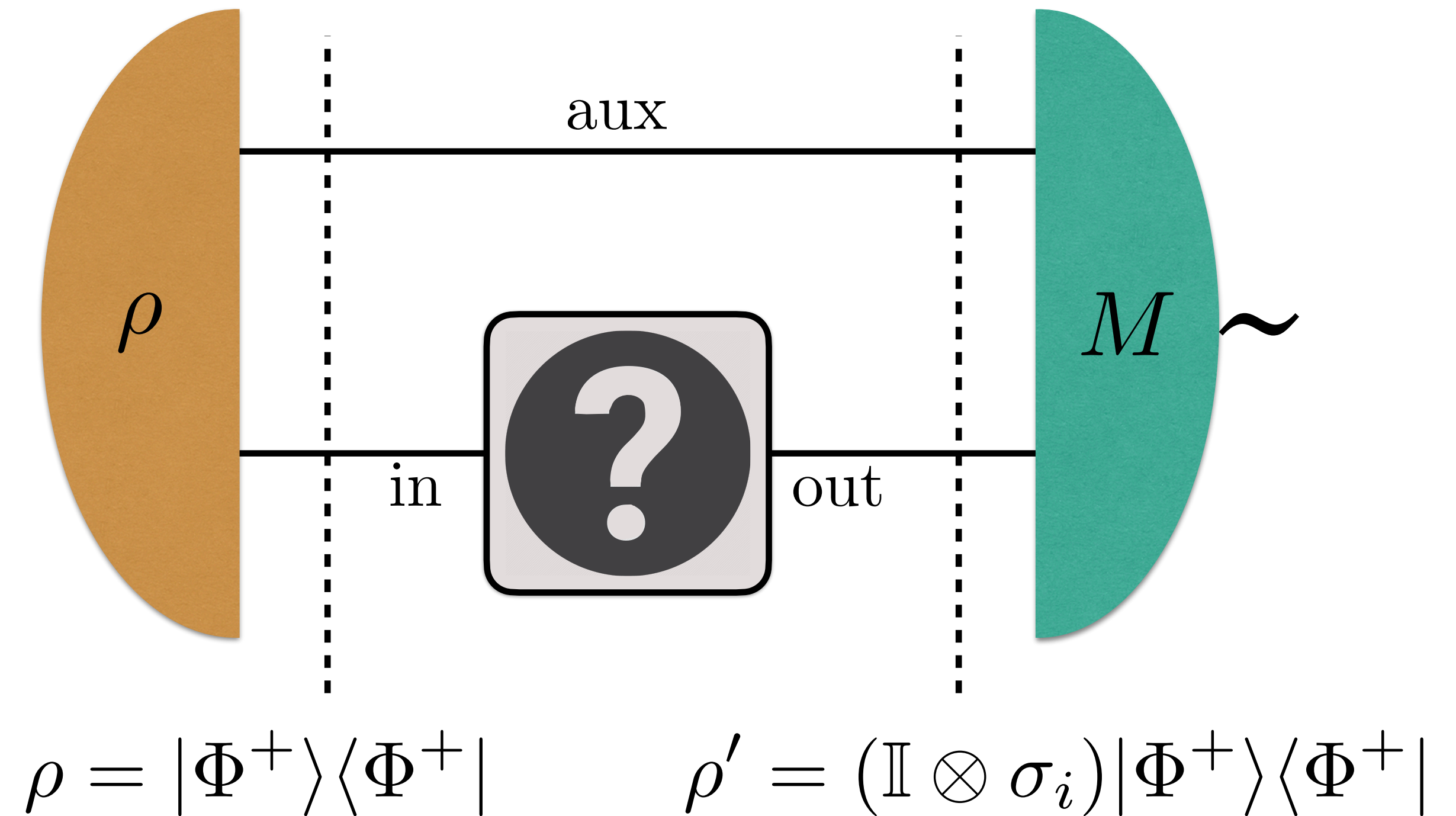
$$\{p_i\} = \left\{ \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4} \right\}$$

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$$(\mathbb{I} \otimes \mathbb{I}) |\Phi^+\rangle\langle\Phi^+| = |\Phi^+\rangle\langle\Phi^+|$$

$$(\mathbb{I} \otimes \sigma_X) |\Phi^+\rangle\langle\Phi^+| = |\Psi^+\rangle\langle\Psi^+|$$

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EXAMPLE

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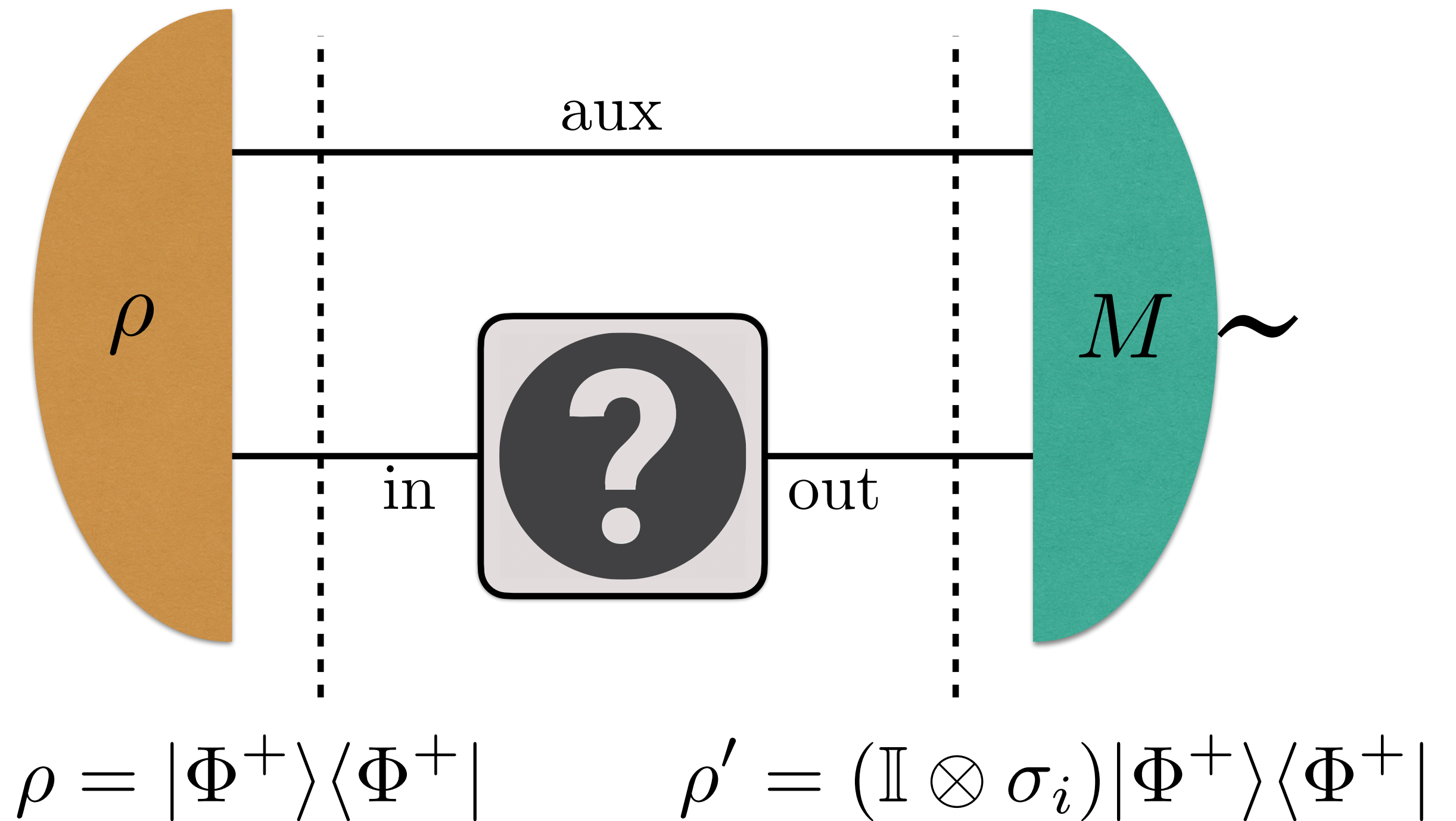
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$$p_{\text{succ}} = 1$$



$$\begin{aligned} (\mathbb{I} \otimes \mathbb{I}) |\Phi^+\rangle\langle\Phi^+| &= |\Phi^+\rangle\langle\Phi^+| \\ (\mathbb{I} \otimes \sigma_X) |\Phi^+\rangle\langle\Phi^+| &= |\Psi^+\rangle\langle\Psi^+| \\ (\mathbb{I} \otimes \sigma_Y) |\Phi^+\rangle\langle\Phi^+| &= |\Psi^-\rangle\langle\Psi^-| \\ (\mathbb{I} \otimes \sigma_Z) |\Phi^+\rangle\langle\Phi^+| &= |\Phi^-\rangle\langle\Phi^-| \end{aligned}$$

EXAMPLE

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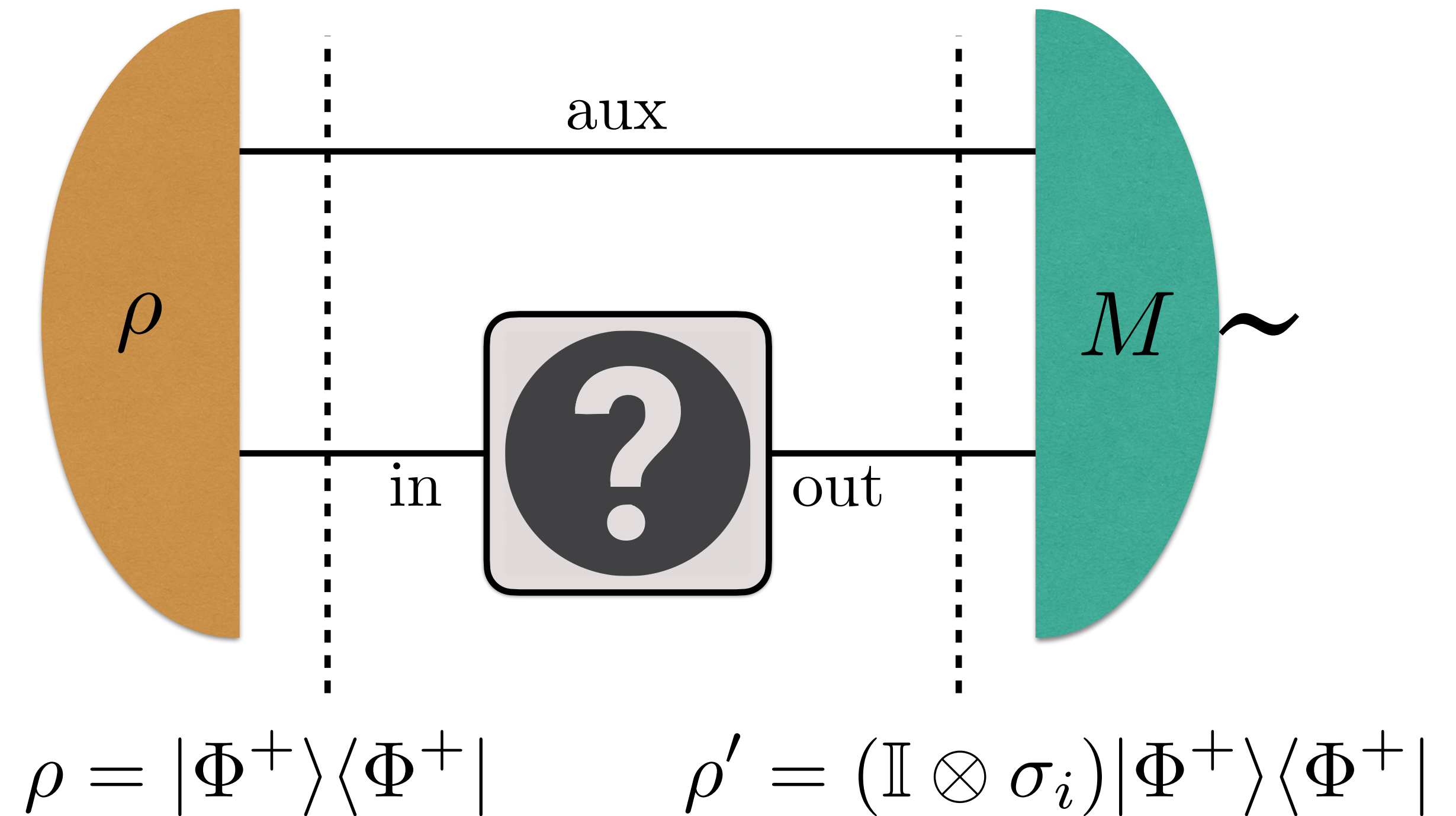
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$$\begin{aligned} (\mathbb{I} \otimes \mathbb{I}) |\Phi^+\rangle\langle\Phi^+| &= |\Phi^+\rangle\langle\Phi^+| \\ (\mathbb{I} \otimes \sigma_X) |\Phi^+\rangle\langle\Phi^+| &= |\Psi^+\rangle\langle\Psi^+| \\ (\mathbb{I} \otimes \sigma_Y) |\Phi^+\rangle\langle\Phi^+| &= |\Psi^-\rangle\langle\Psi^-| \\ (\mathbb{I} \otimes \sigma_Z) |\Phi^+\rangle\langle\Phi^+| &= |\Phi^-\rangle\langle\Phi^-| \end{aligned}$$

CHOI-JAMIOŁKOWSKI ISOMORPHISM

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$$\tilde{C} : L(H^I) \rightarrow L(H^O) \quad \mapsto \quad C \in L(H^I \otimes H^O)$$

CHOI-JAMIOŁKOWSKI ISOMORPHISM

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CHOI-JAMIOŁKOWSKI ISOMORPHISM

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$$C := (\tilde{\mathbb{I}} \otimes \tilde{C})(|\Phi^+\rangle\langle\Phi^+|)$$

$\tilde{C}$ is a CPTP map, then $C \geq 0$

$$\text{Tr}_O C = \mathbb{I}^I$$

$$P := \max_{\rho, \{M_i\}} \sum_{i=1}^N p_i \operatorname{Tr} \left[(\tilde{C}_i \otimes \tilde{\mathbb{I}})(\rho) M_i \right]$$

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MAP

$$\begin{aligned}
 P &:= \max_{\rho, \{M_i\}} \sum_{i=1}^N p_i \operatorname{Tr} \left[(\tilde{C}_i \otimes \tilde{\mathbb{I}})(\rho) M_i \right] \\
 &= \max_{\rho, \{M_i\}} \sum_{I=1}^N p_i \operatorname{Tr} \left[(\rho^{\text{in,aux}} \otimes \mathbb{I}^{\text{out}})(C_i^{\text{in,out}} \otimes \mathbb{I}^{\text{aux}})(M_i^{\text{aux,out}} \otimes \mathbb{I}^{\text{in}}) \right]
 \end{aligned}$$

MAP

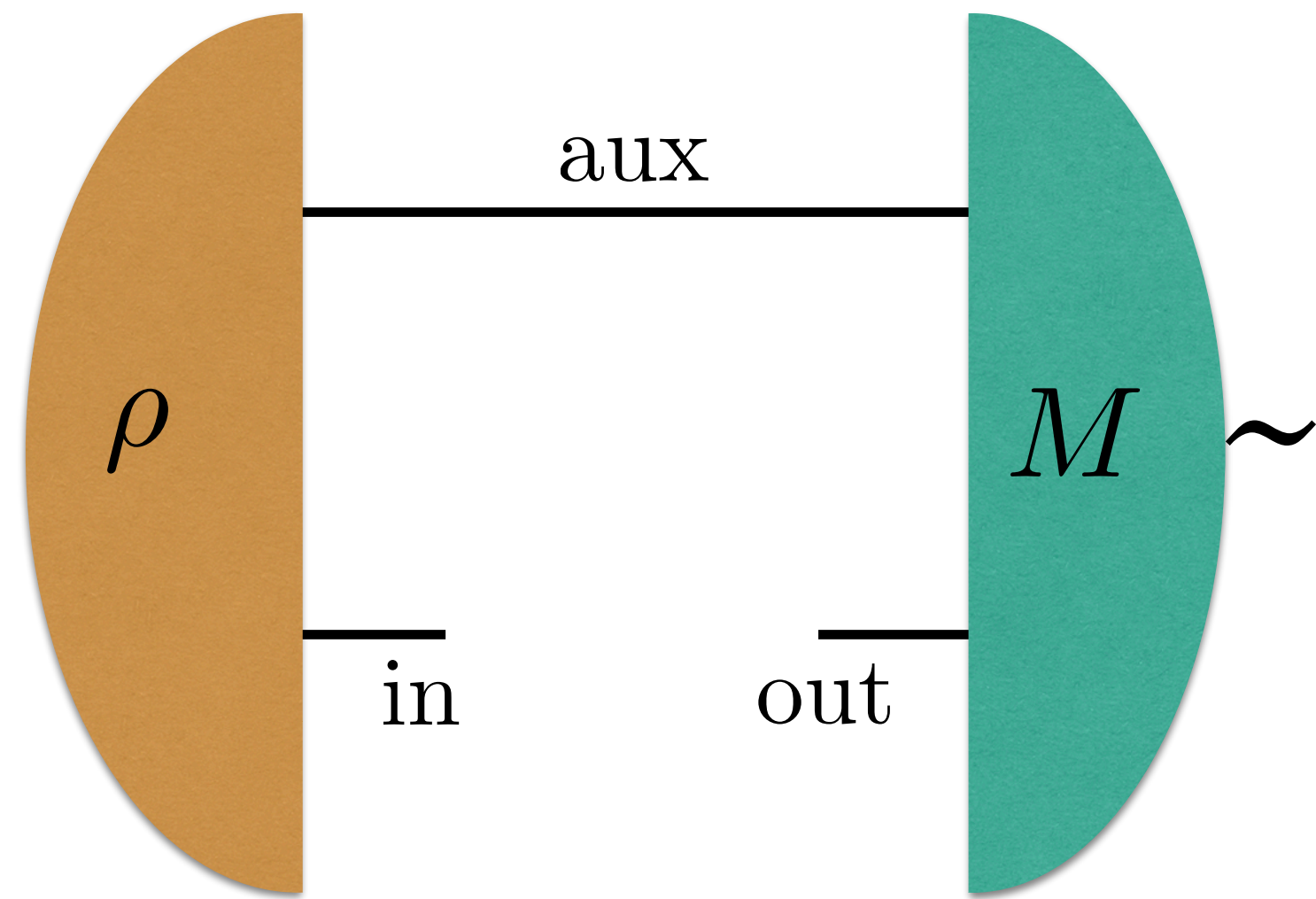


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 \end{aligned}$$

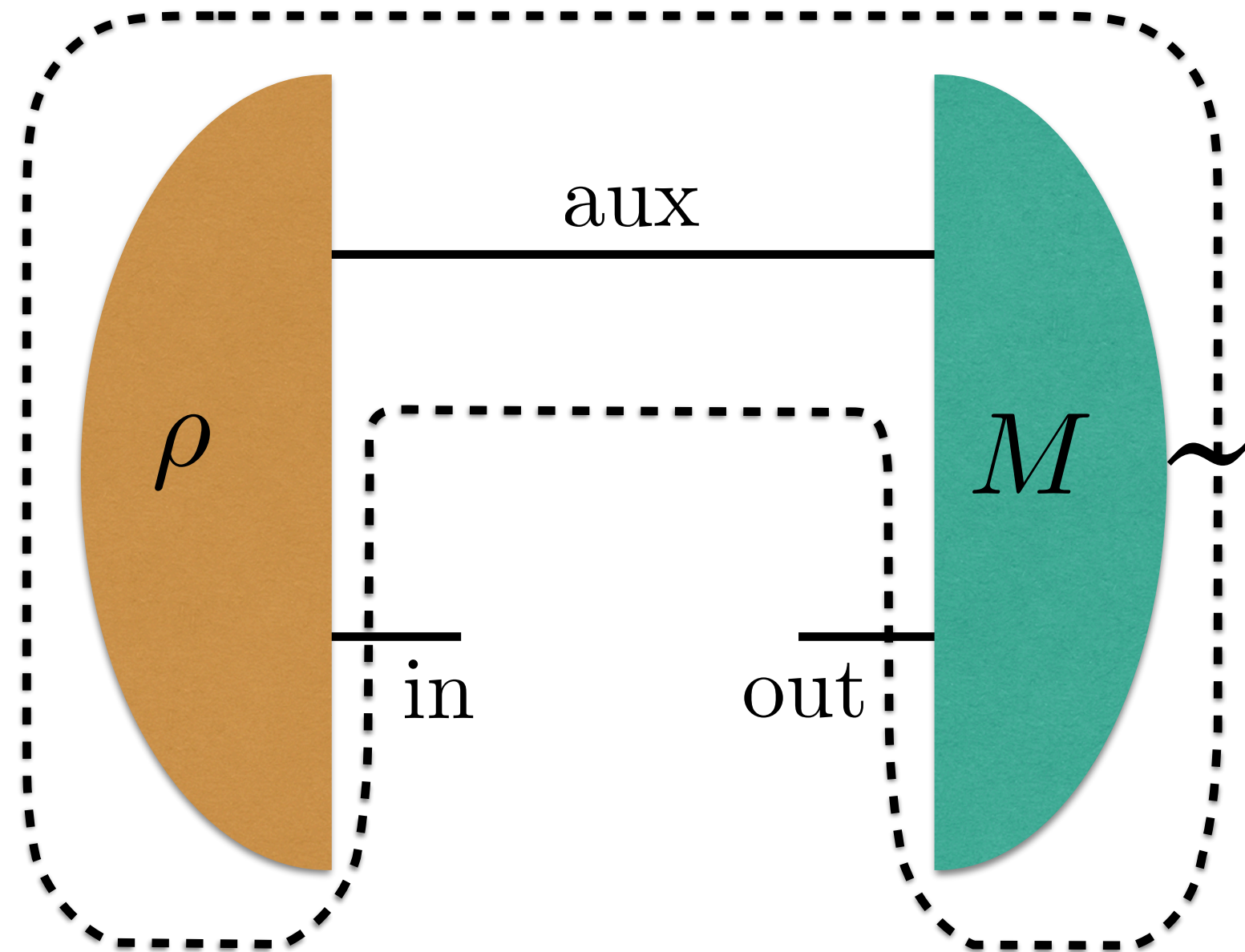
MAP

“CHOI STATE”

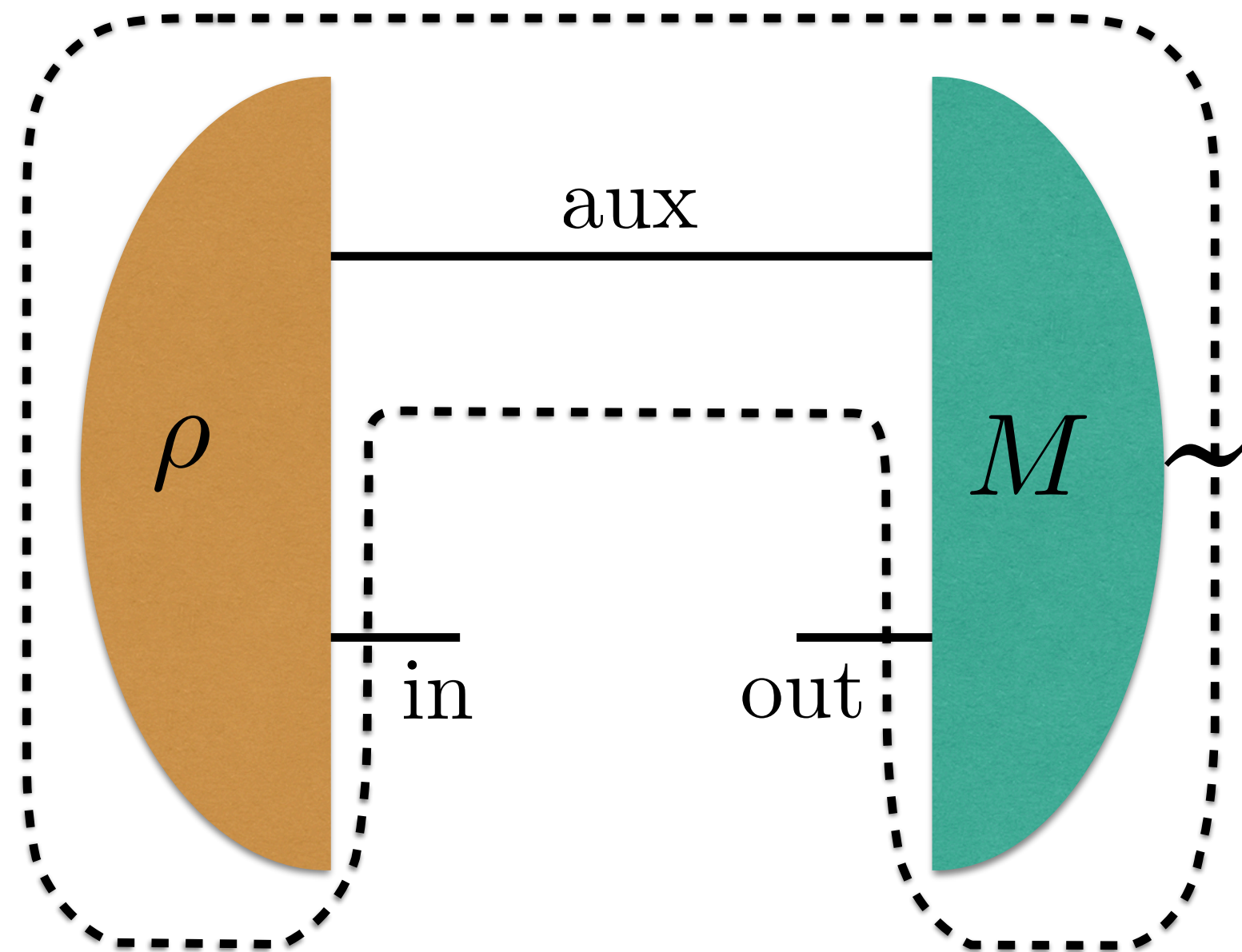
TESTERS



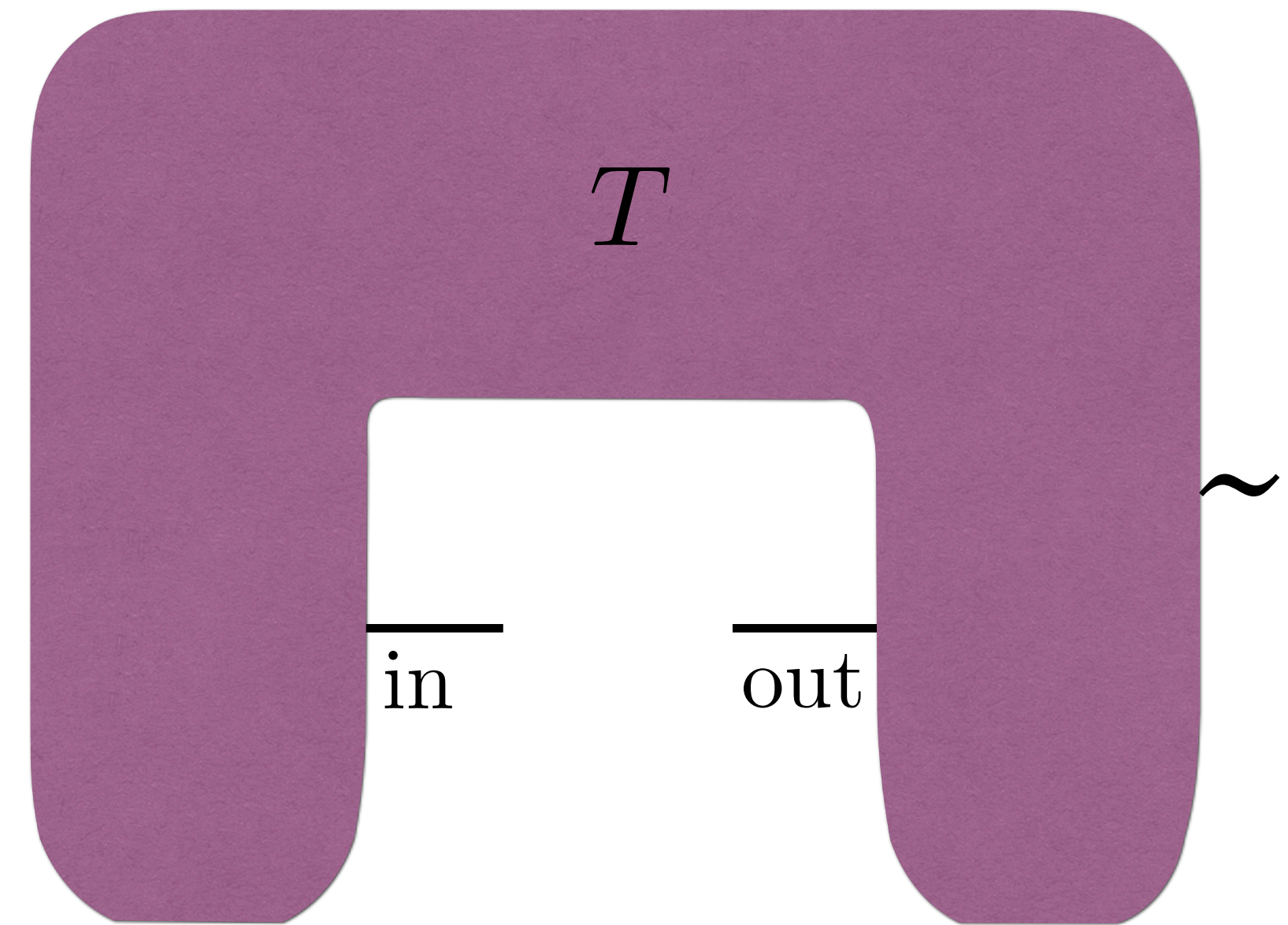
TESTERS



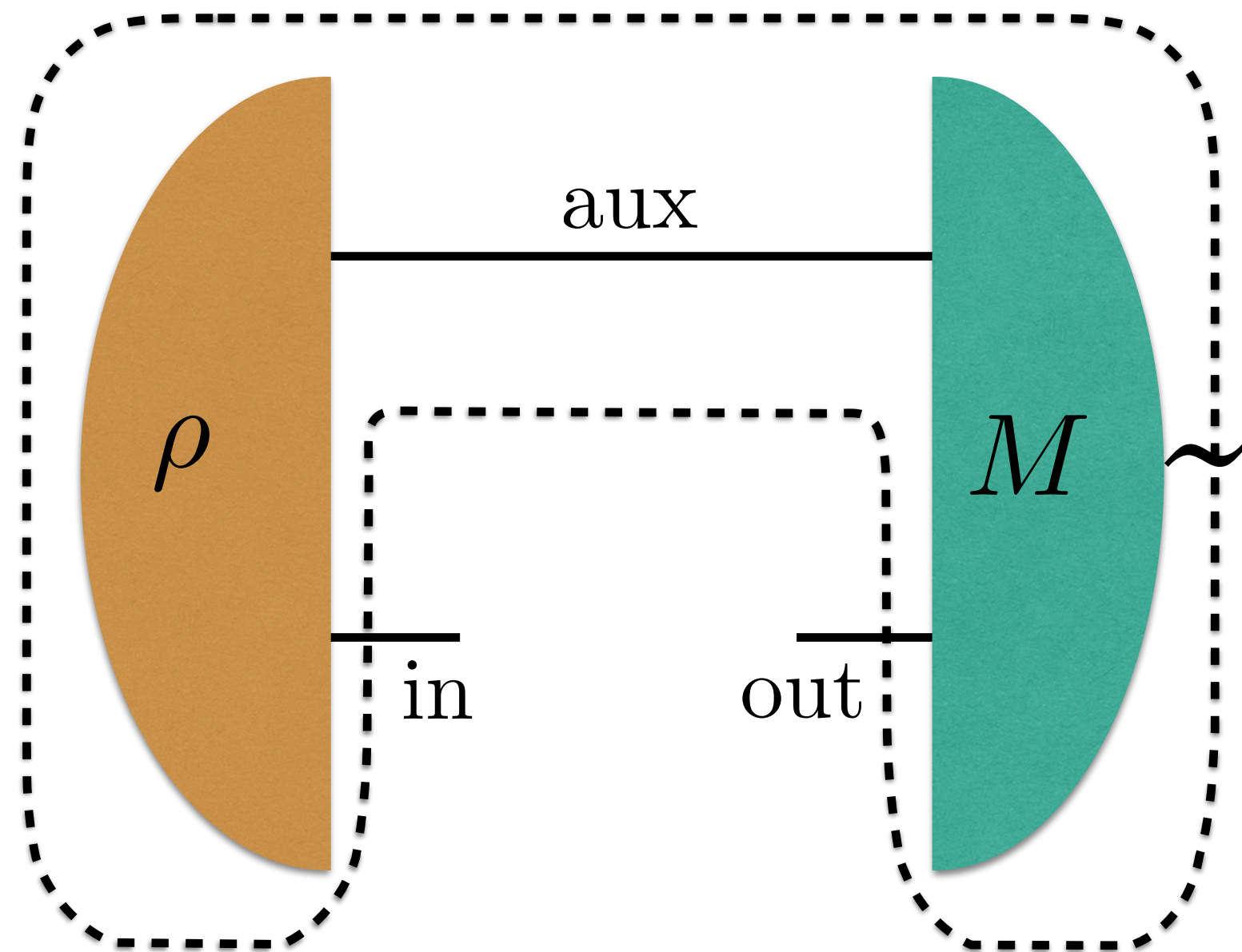
TESTERS



=



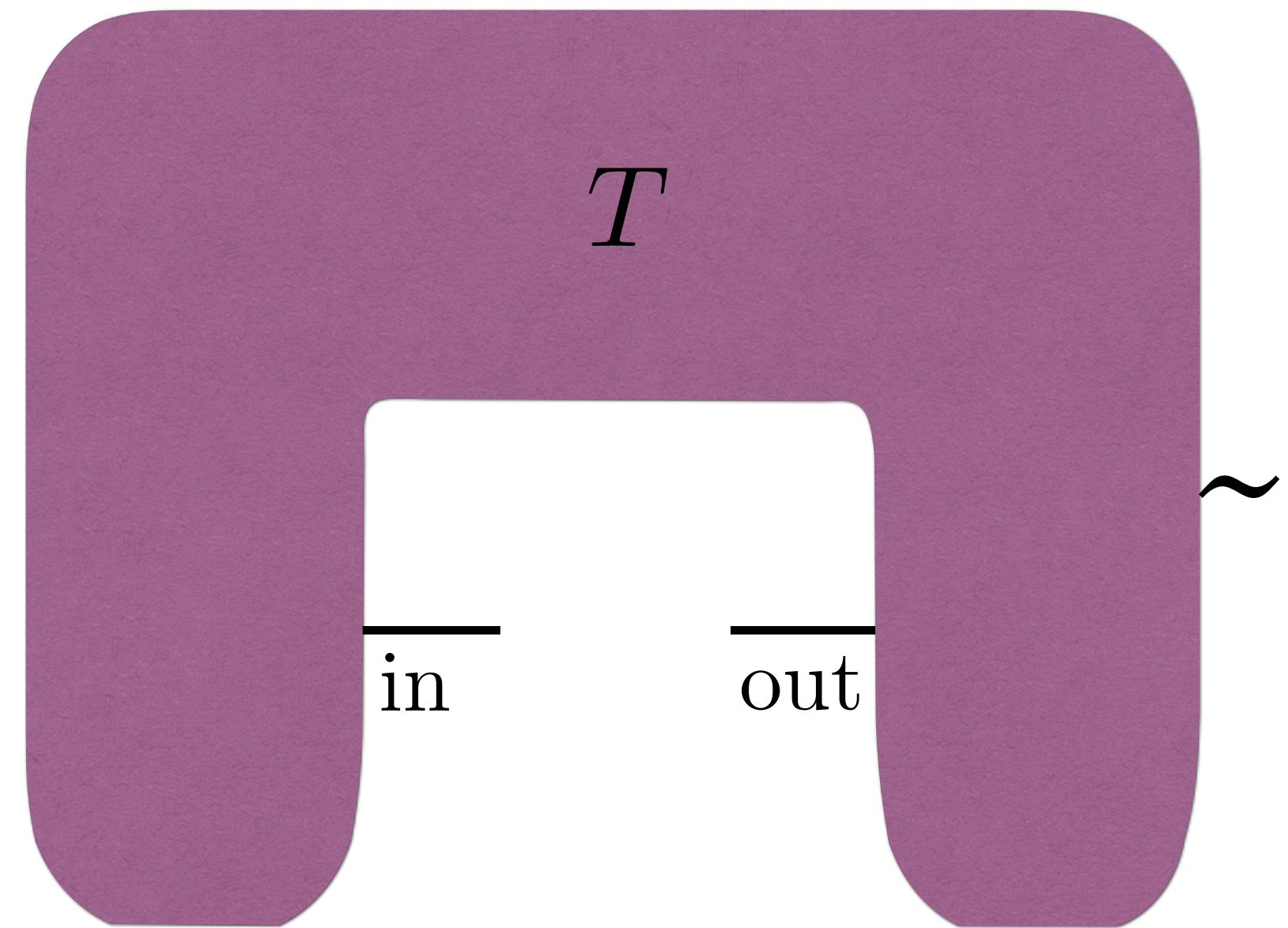
TESTERS



$$\rho \in L(H^{\text{in}}, \text{aux})$$

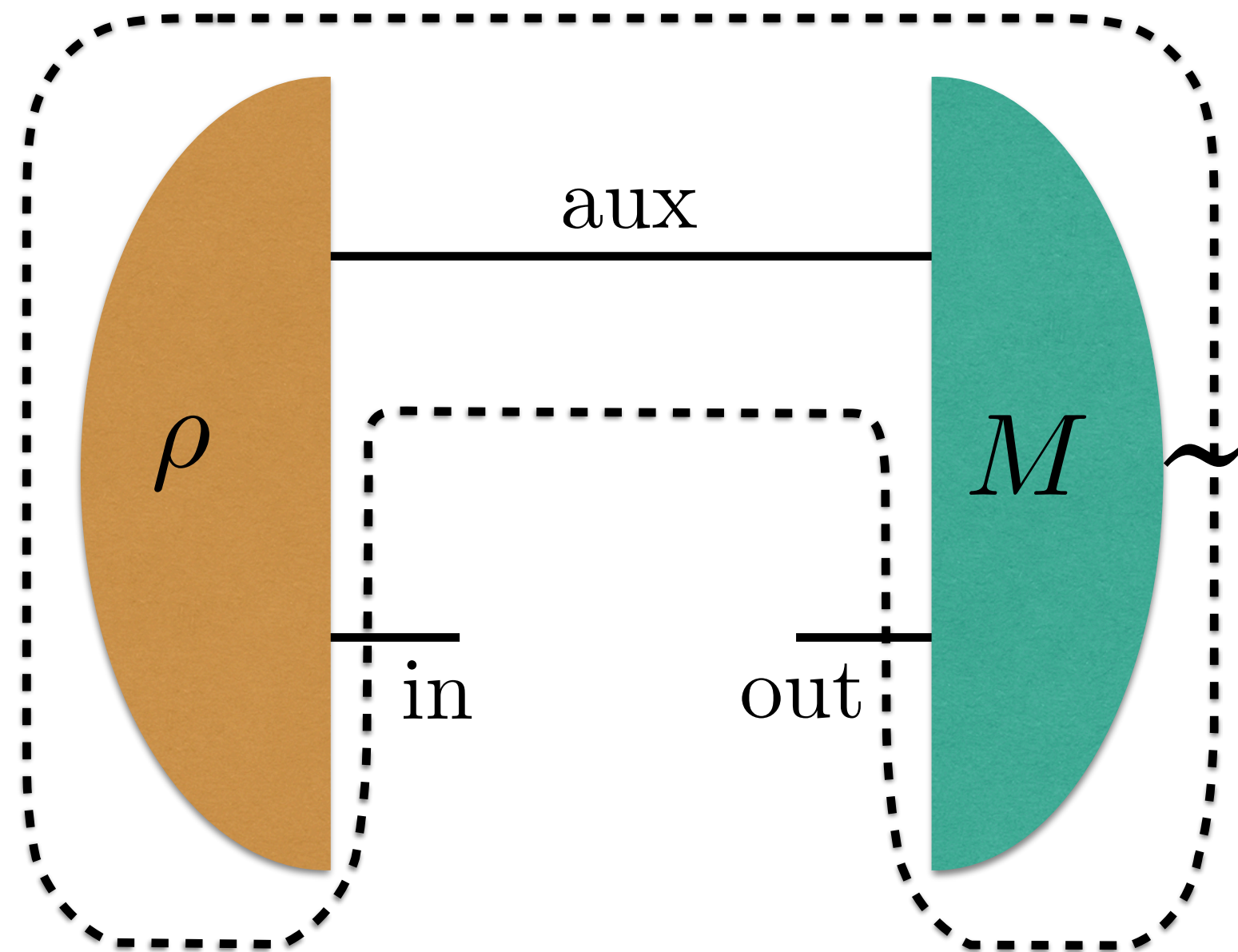
$$M_i \in L(H^{\text{out}}, \text{aux})$$

=



$$T_i \in L(H^{\text{in}}, \text{out})$$

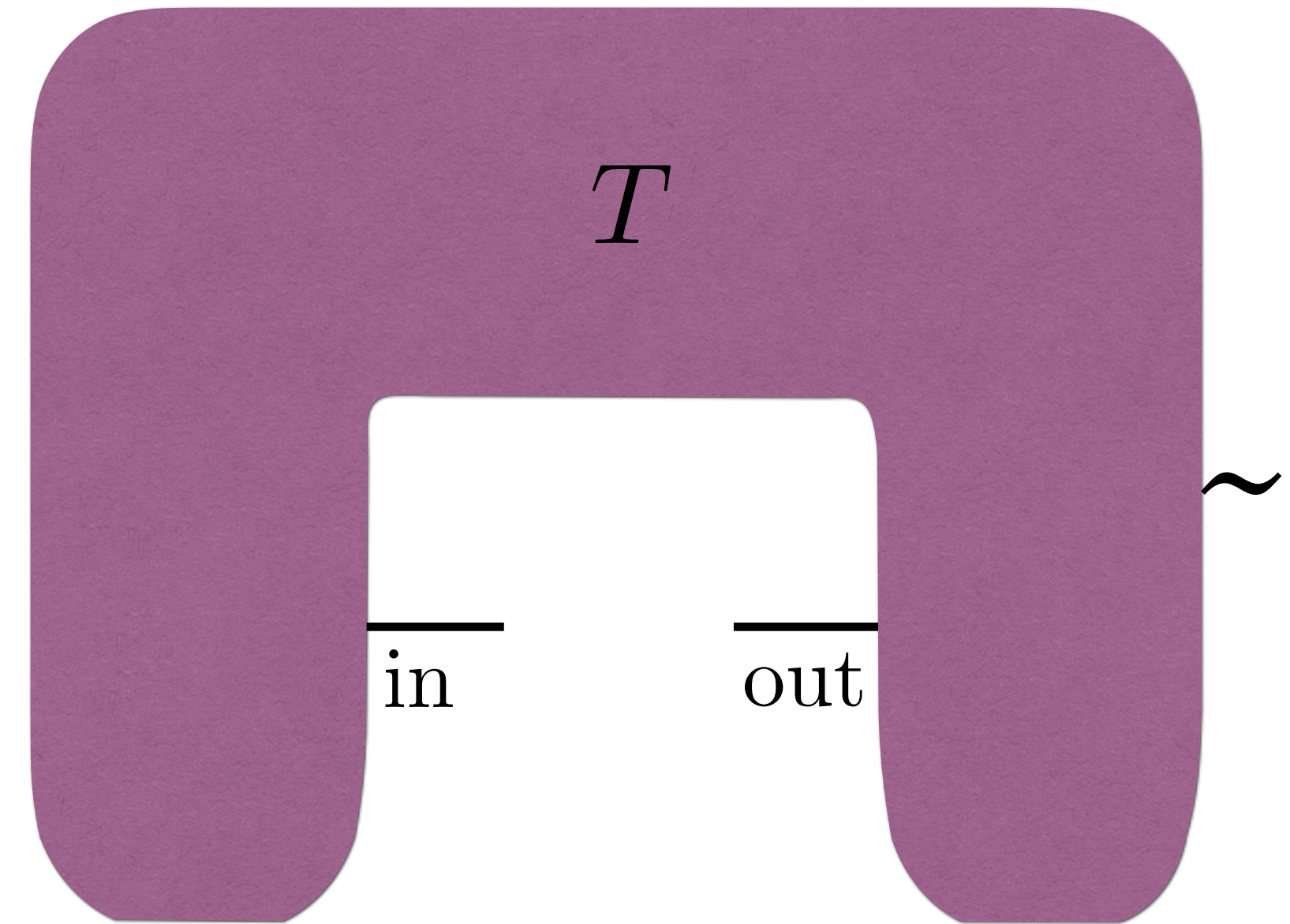
TESTERS



$$\rho \in L(H^{\text{in}}, \text{aux})$$

$$M_i \in L(H^{\text{out}}, \text{aux})$$

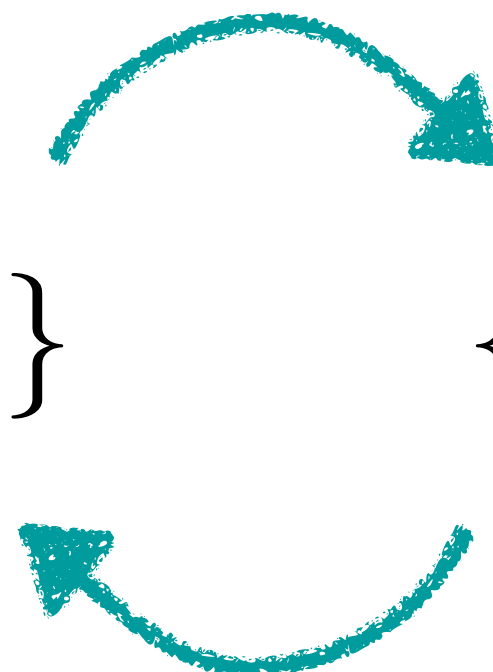
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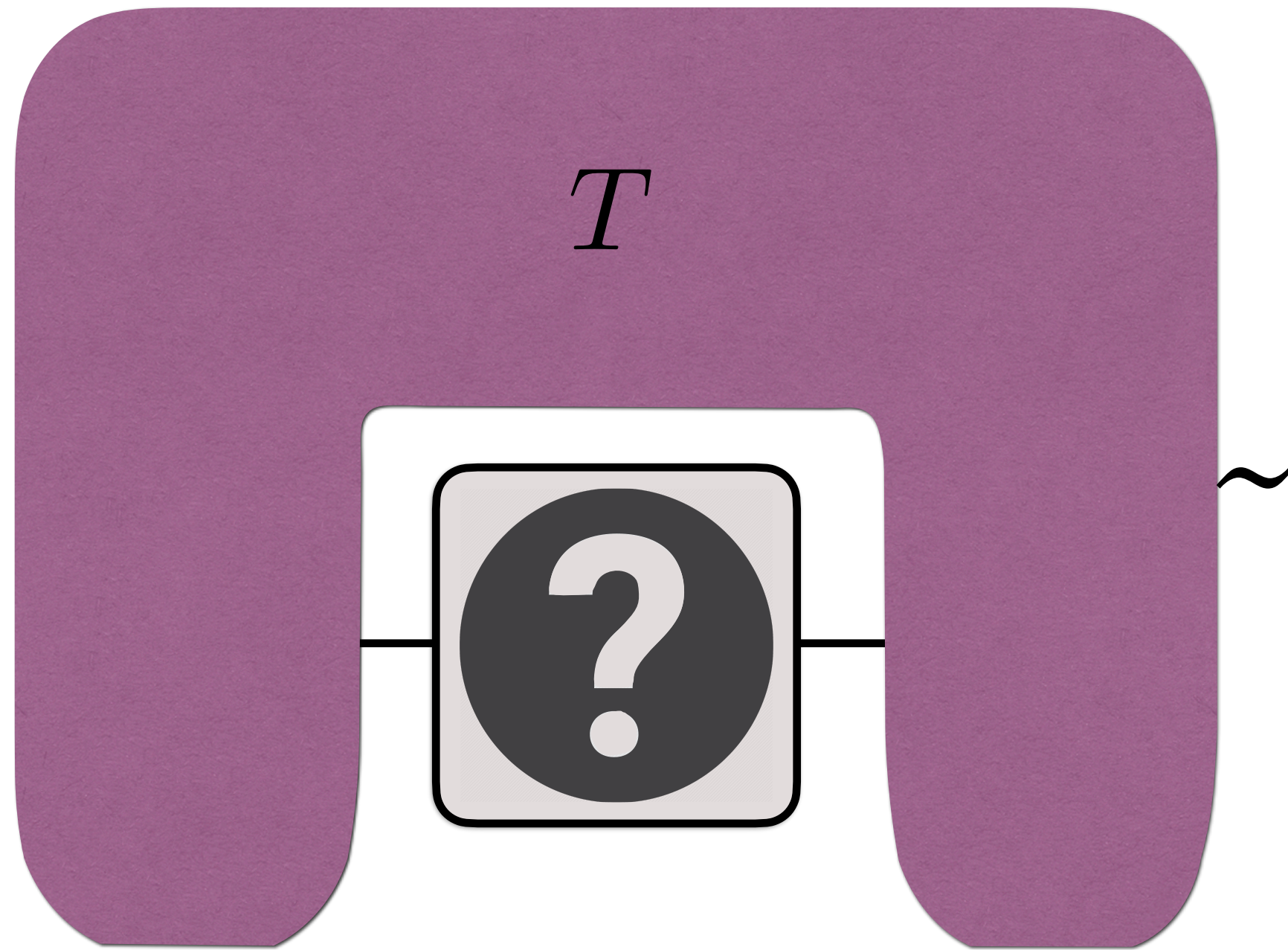
$$T_i \in L(H^{\text{in}}, \text{out})$$

$\rho, \{M_i\}$

$\{T_i\}$



TESTERS

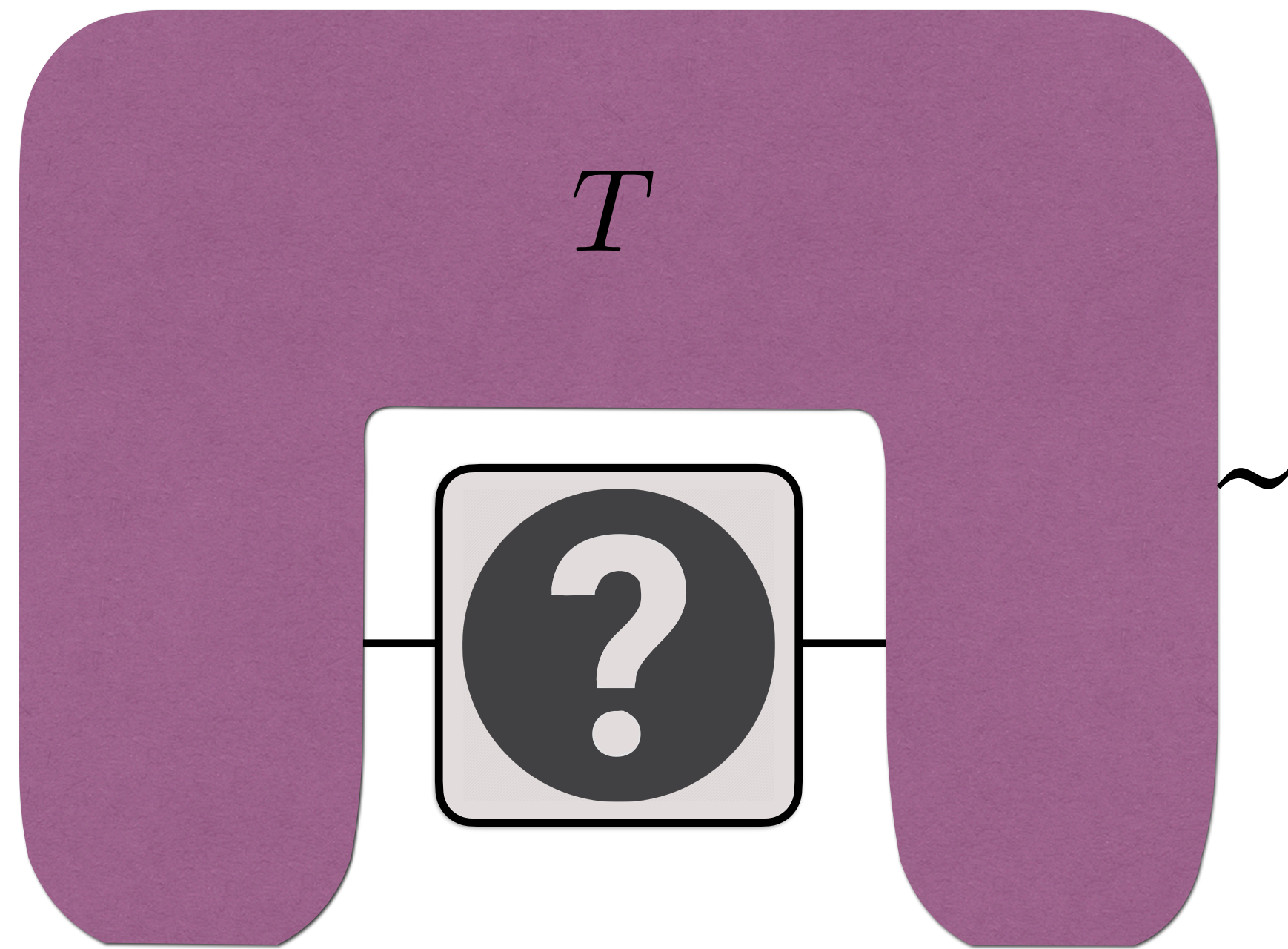


$$T = \{T_i\} \quad :$$

$$T_i \geq 0$$

$$\sum_i T_i = \rho_o^{\text{in}} \otimes I^{\text{out}}$$

TESTERS



$$T = \{T_i\} :$$

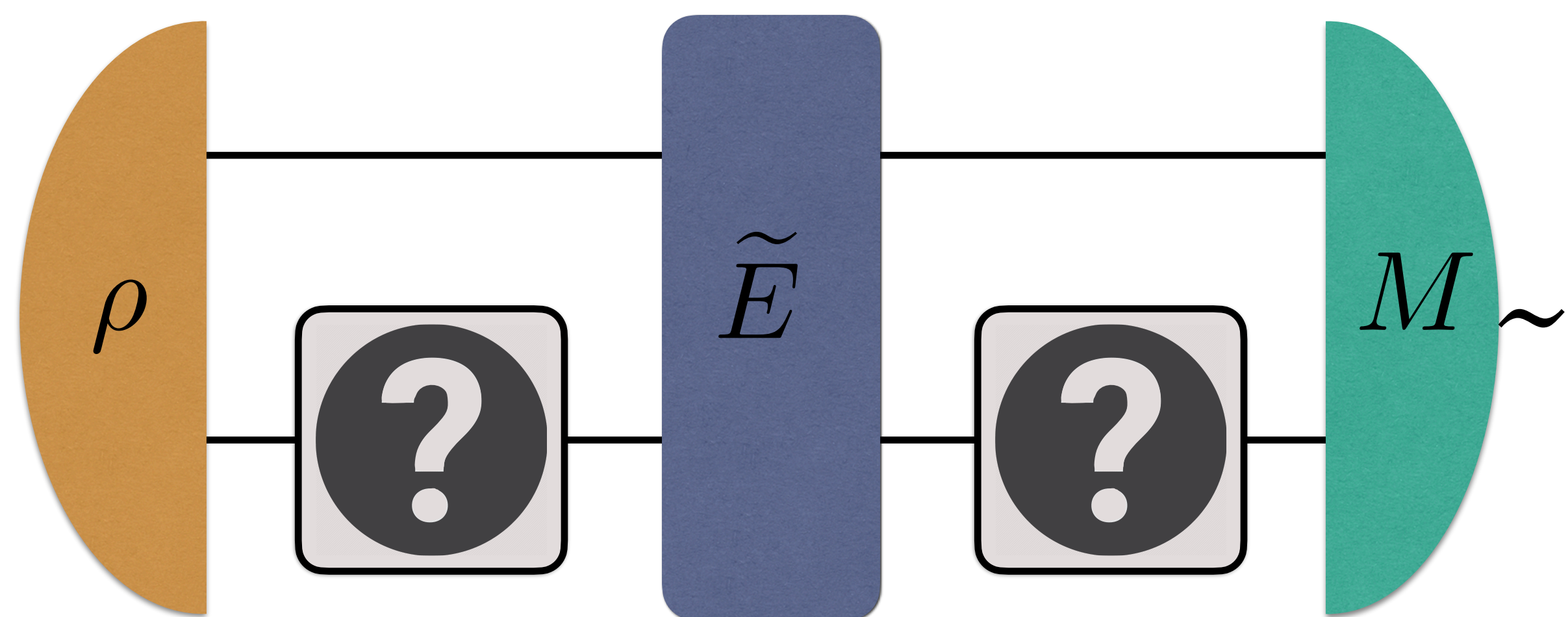
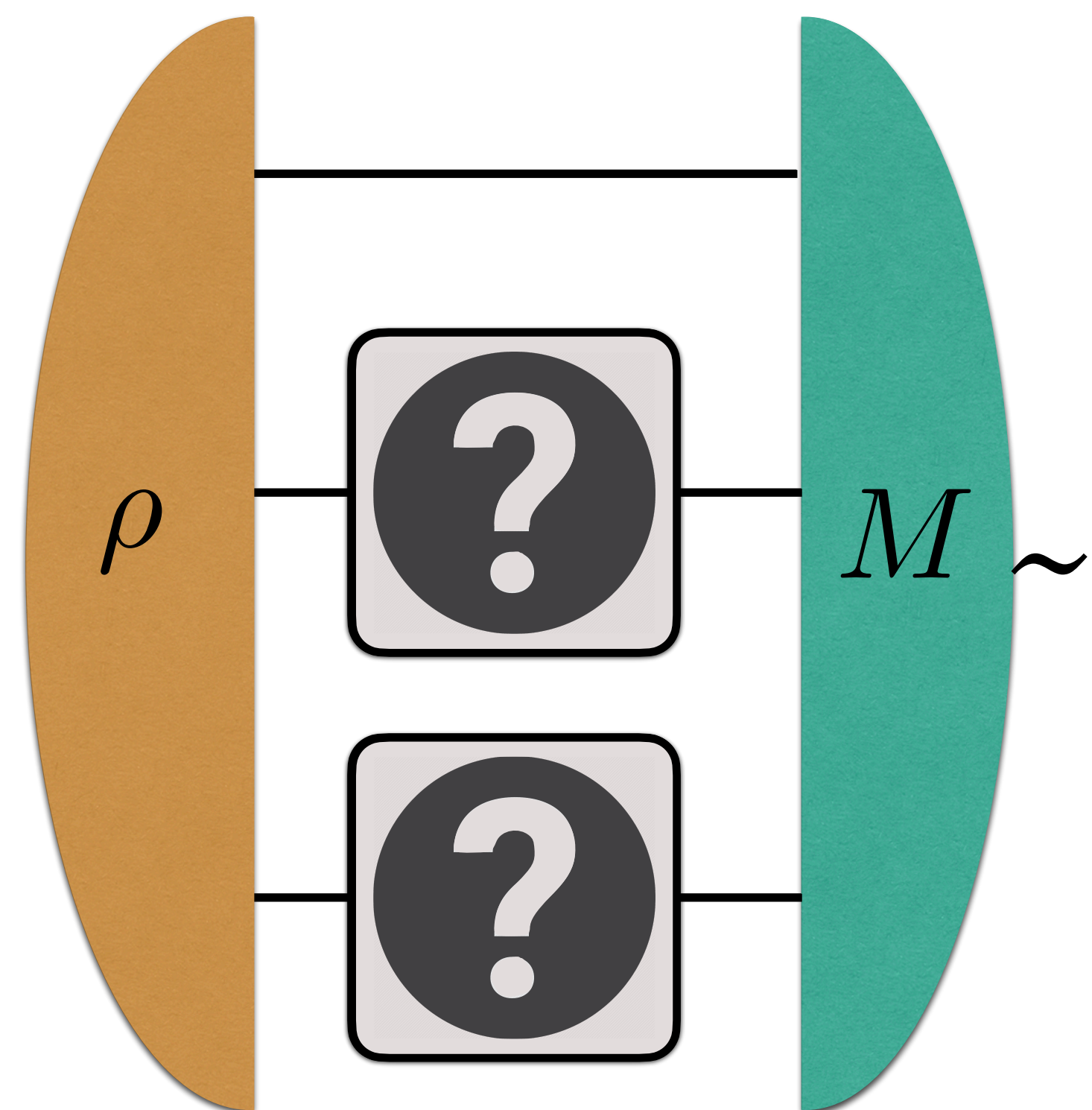
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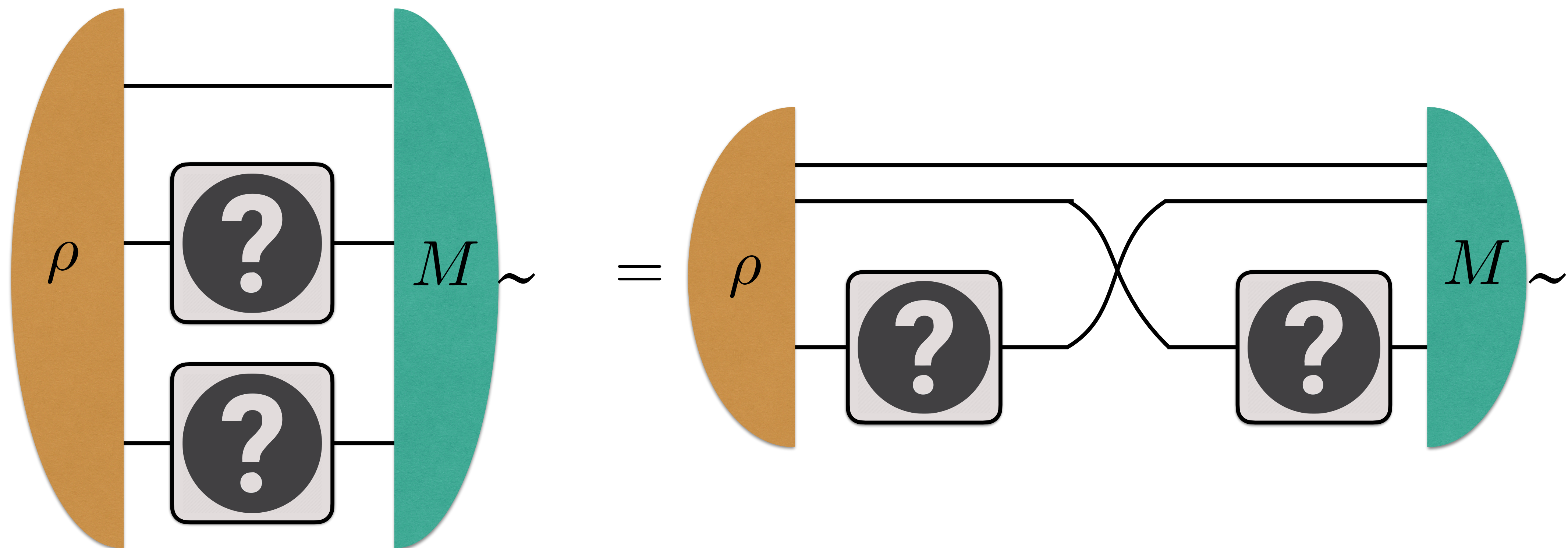
$$\sum_i T_i = \rho_o^{\text{in}} \otimes I^{\text{out}}$$

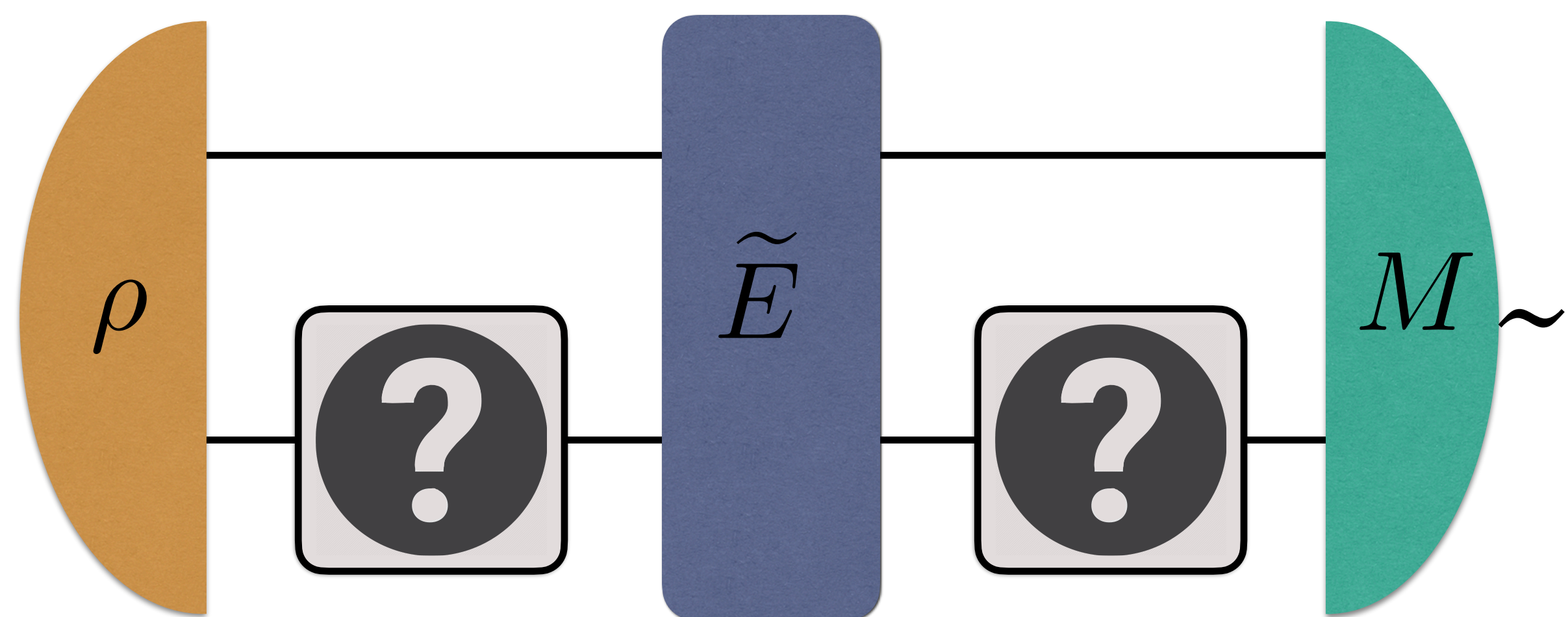
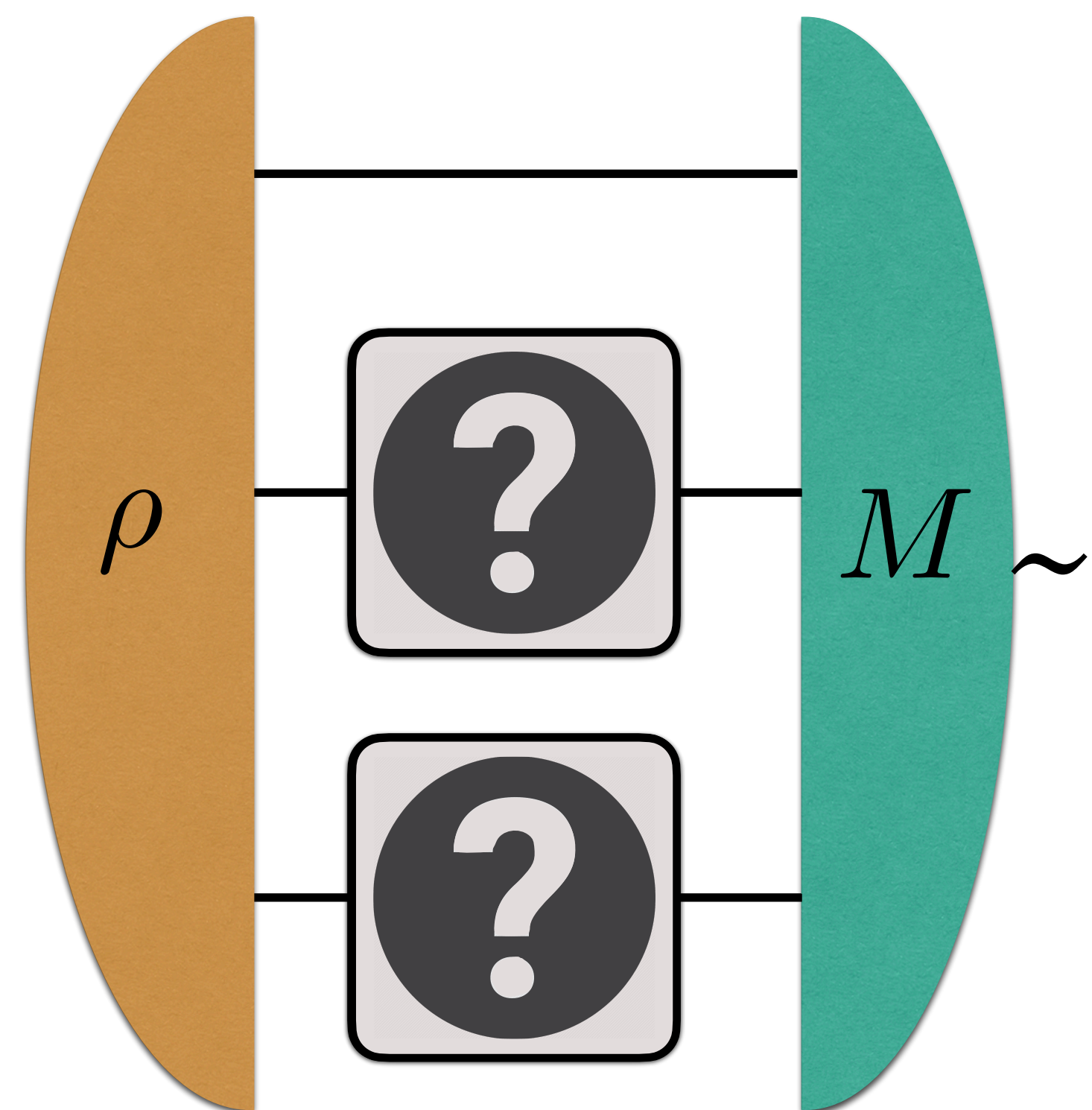
$$P = \max_{\{T_i\}} \sum_i p_i \text{Tr}(C_i T_i)$$

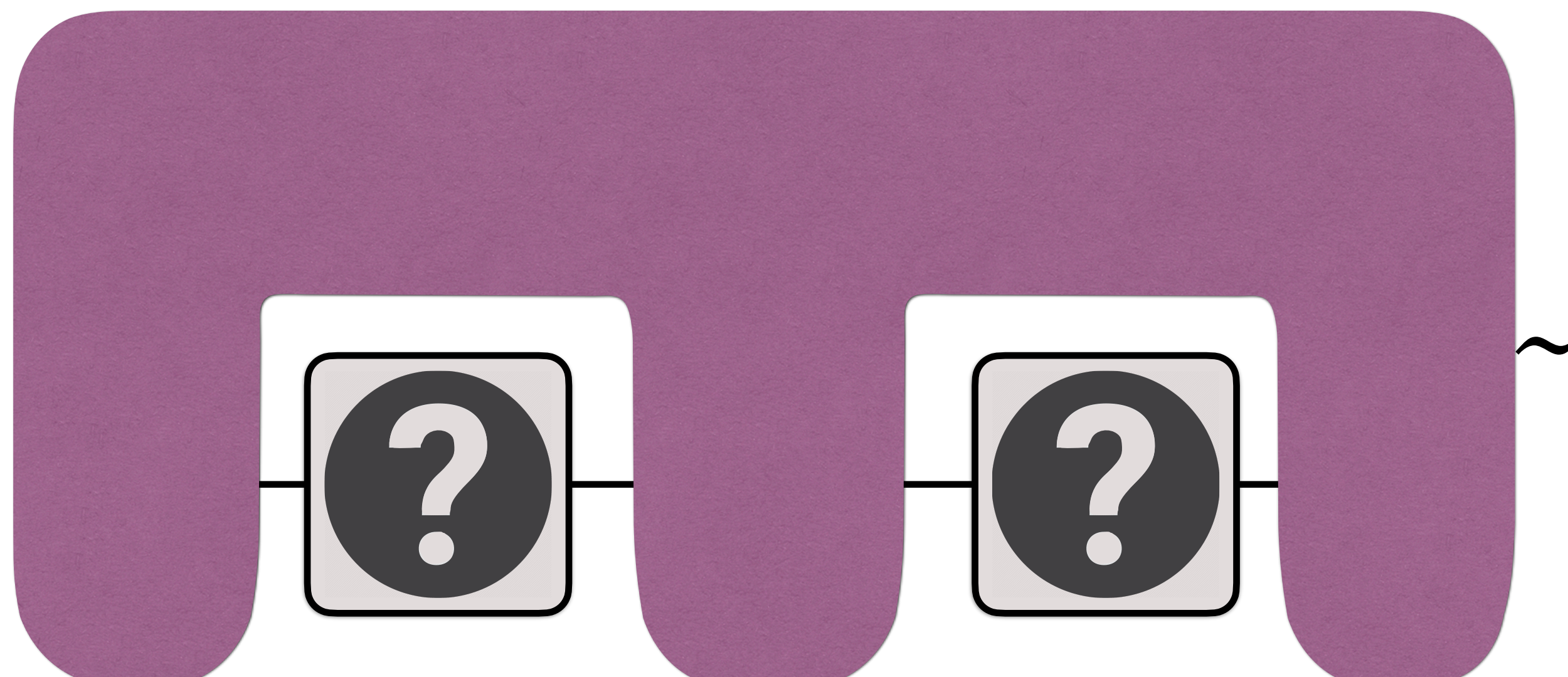
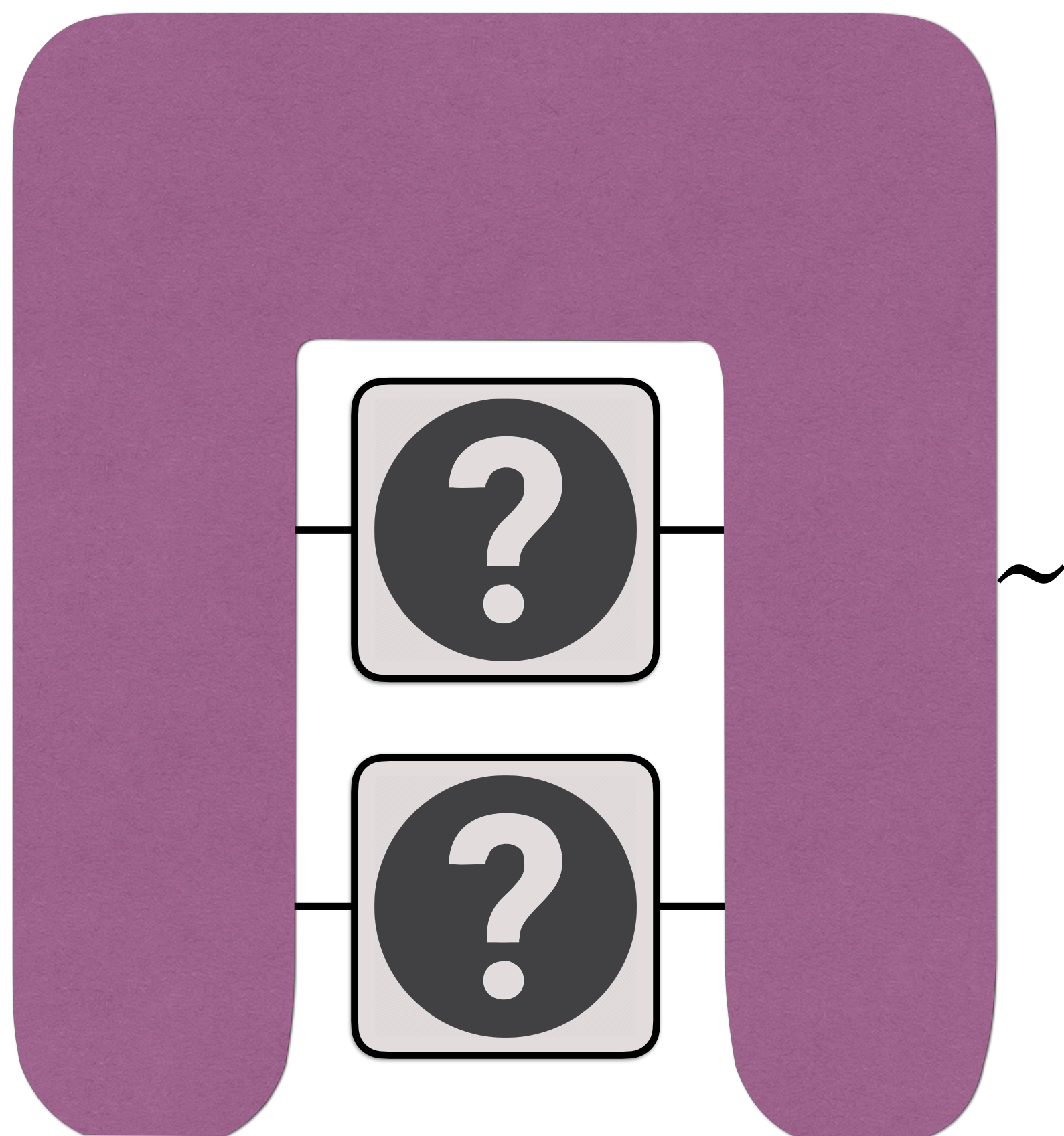
TWO COPIES

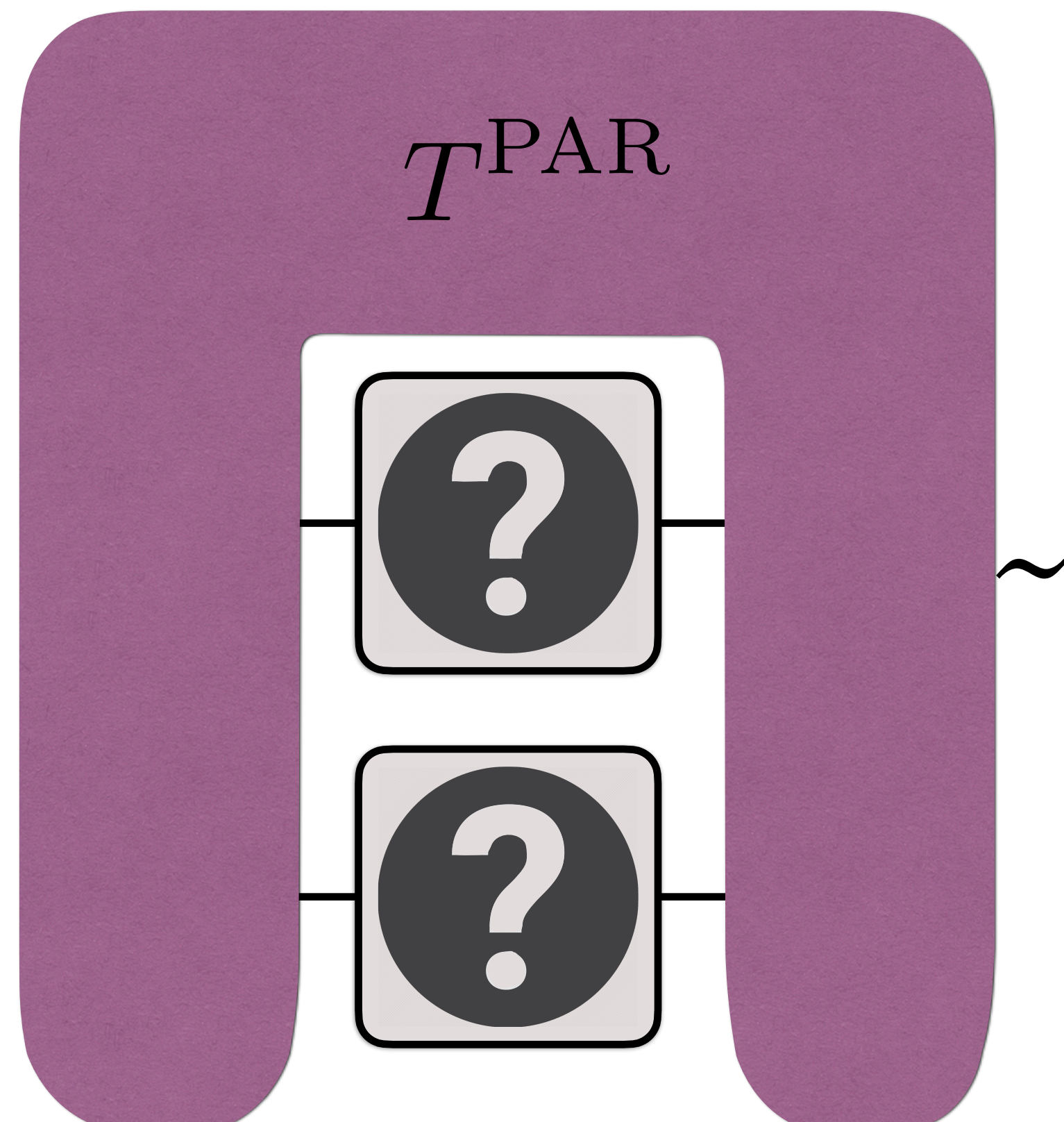




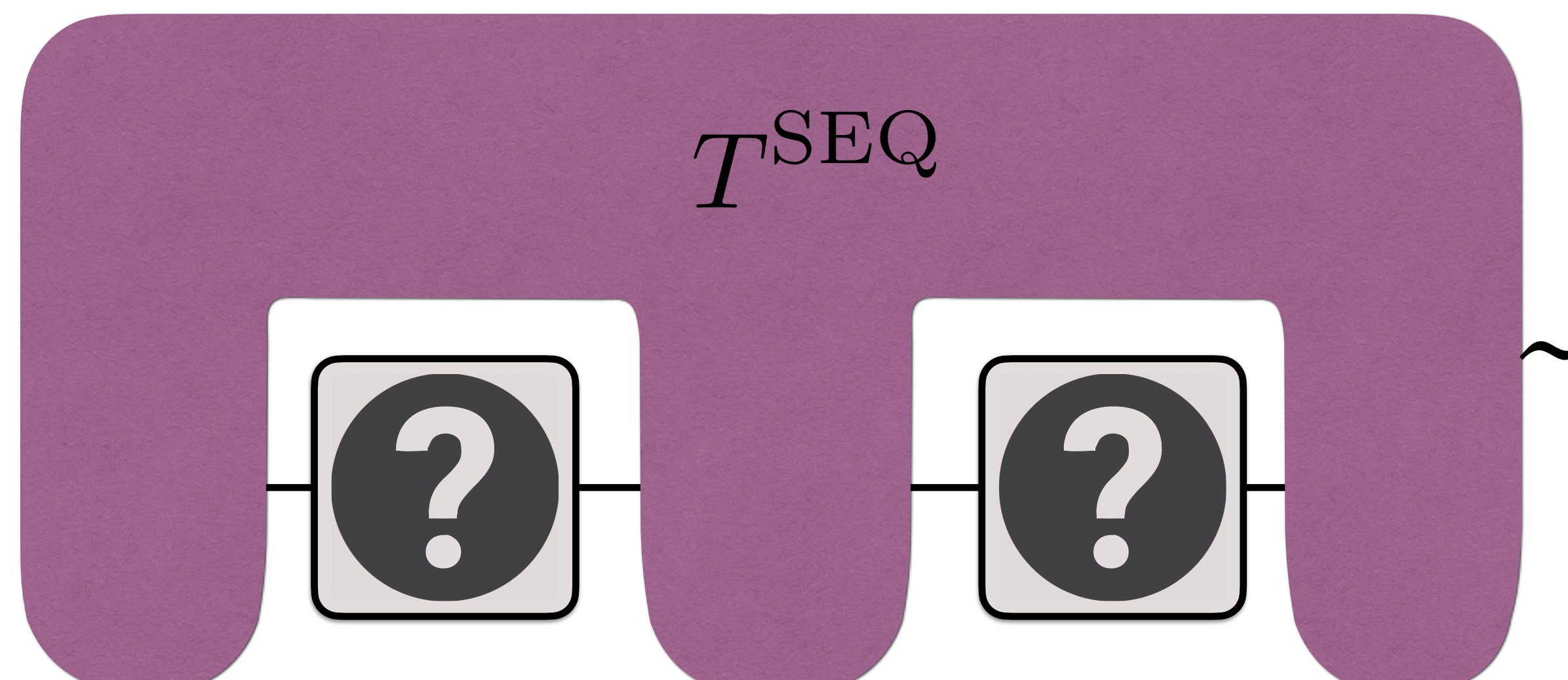
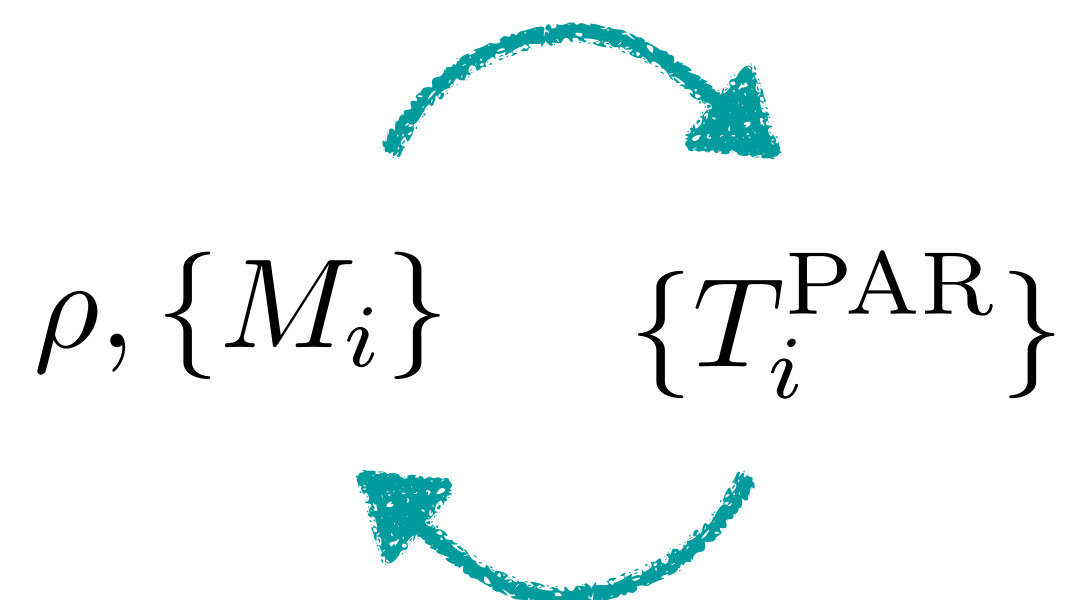




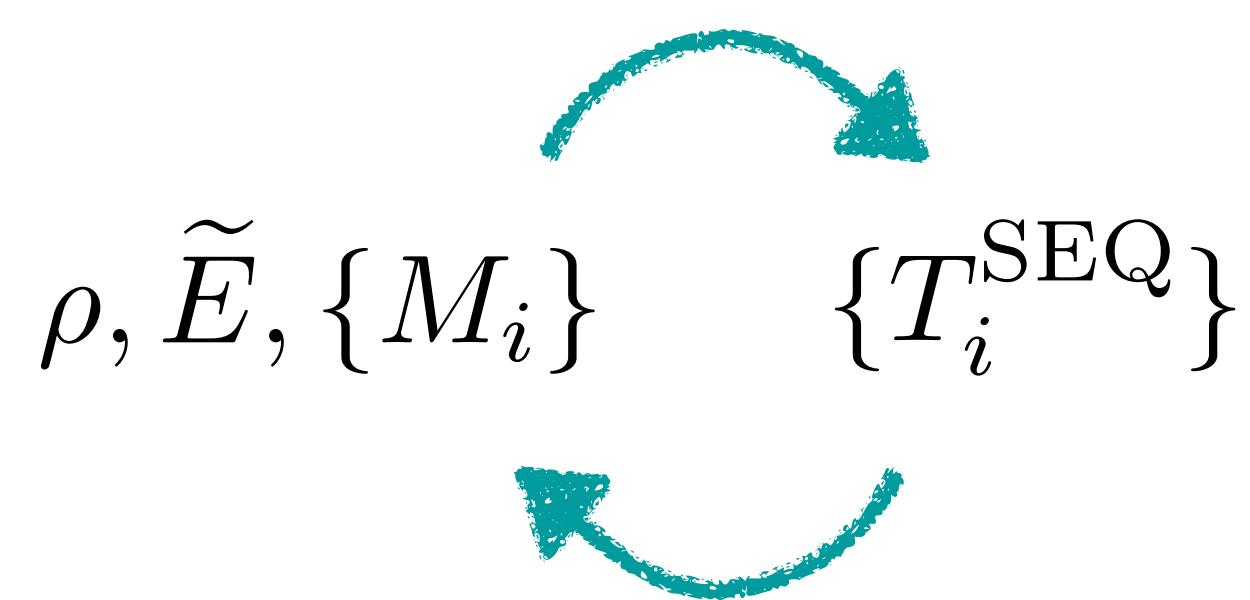




PARALLEL



SEQUENTIAL



PARALLEL

$$T^{\text{PAR}} = \{T_i^{\text{PAR}}\} :$$

$$T_i^{\text{PAR}} \geq 0$$

$$\sum_i T_i^{\text{PAR}} = W^{\text{PAR}}$$

SEQUENTIAL

$$T^{\text{SEQ}} = \{T_i^{\text{SEQ}}\} :$$

$$T_i^{\text{SEQ}} \geq 0$$

$$\sum_i T_i^{\text{SEQ}} = W^{\text{SEQ}}$$

$$T = \{T_i\} :$$

$$T_i \geq 0$$

$$\sum_i T_i = \rho_o^{\text{in}} \otimes I^{\text{out}}$$

PARALLEL

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PROCESS

PARALLEL

$$T^{\text{PAR}} = \{T_i^{\text{PAR}}\} :$$

$$T_i^{\text{PAR}} \geq 0$$

$$\sum_i T_i^{\text{PAR}} = W^{\text{PAR}}$$

$$P^{\text{PAR}} = \max_{\{T_i^{\text{PAR}}\}} \sum_i p_i \text{Tr} \left(C_i^{\otimes 2} T_i^{\text{PAR}} \right)$$

SEQUENTIAL

$$T^{\text{SEQ}} = \{T_i^{\text{SEQ}}\} :$$

$$T_i^{\text{SEQ}} \geq 0$$

$$\sum_i T_i^{\text{SEQ}} = W^{\text{SEQ}}$$

$$P^{\text{SEQ}} = \max_{\{T_i^{\text{SEQ}}\}} \sum_i p_i \text{Tr} \left(C_i^{\otimes 2} T_i^{\text{SEQ}} \right)$$

GENERAL TESTERS

GENERAL TESTERS

- ▶ Extracting probability distributions from channels:

GENERAL TESTERS

- ▶ Extracting probability distributions from channels:

The most general bilinear function $f : (C_1, C_2) \rightarrow \mathbb{R}$ that extracts valid probability distributions from a pair of Choi states of quantum channels $C_1 \in L(H^{I_1} \otimes H^{O_1})$ and $C_2 \in L(H^{I_2} \otimes H^{O_2})$ is

$$p(i|C_1, C_2) = \text{Tr}[(C_1 \otimes C_2) T_i^{\text{GEN}}],$$

*where $T^{\text{GEN}} = \{T_i^{\text{GEN}}\}$, $T_i^{\text{GEN}} \in L(H^{I_1} \otimes H^{O_1} \otimes H^{I_2} \otimes H^{O_2})$ is a **general tester**.*

GENERAL TESTERS

$$T^{\text{GEN}} = \{T_i^{\text{GEN}}\} \quad :$$

$$T_i^{\text{GEN}} \geq 0 \quad \forall i$$

$$\sum_i T_i^{\text{GEN}} = W^{\text{GEN}}$$

GENERAL TESTERS

$$T^{\text{GEN}} = \{T_i^{\text{GEN}}\} \quad :$$

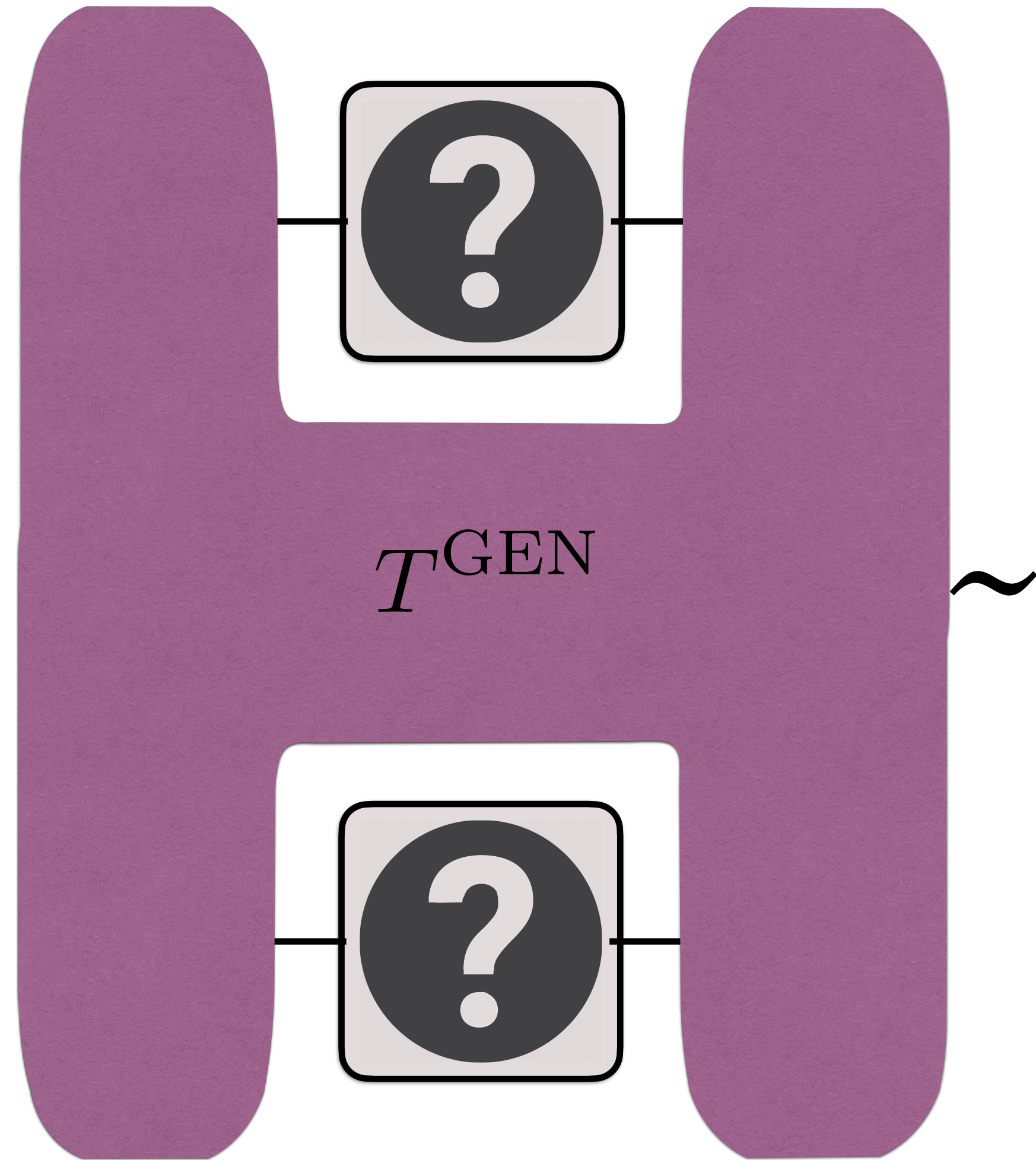
$$T_i^{\text{GEN}} \geq 0 \quad \forall i$$

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PROCESS MATRIX

GENERAL TESTERS



$$T^{\text{GEN}} = \{T_i^{\text{GEN}}\} :$$

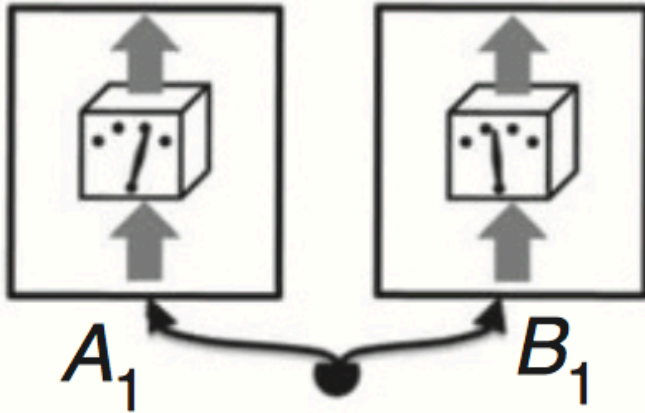
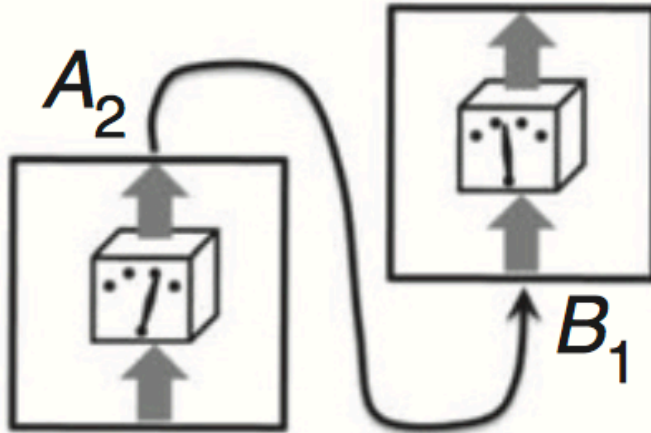
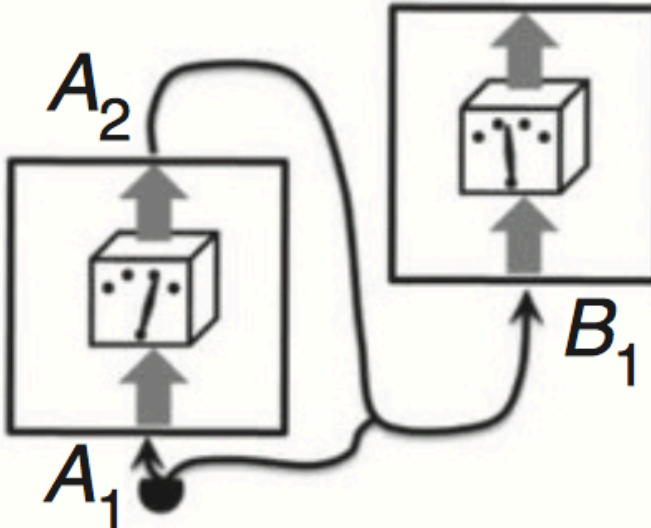
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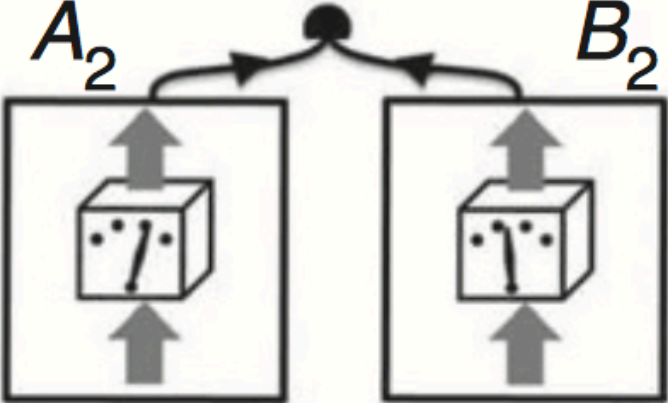
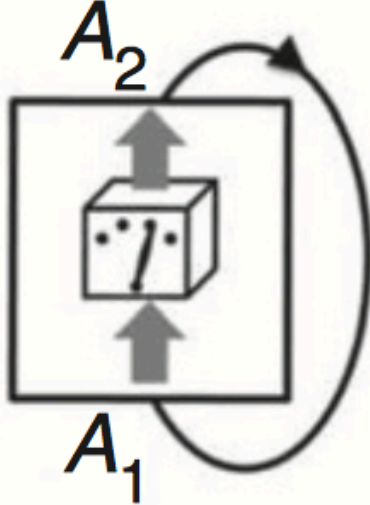
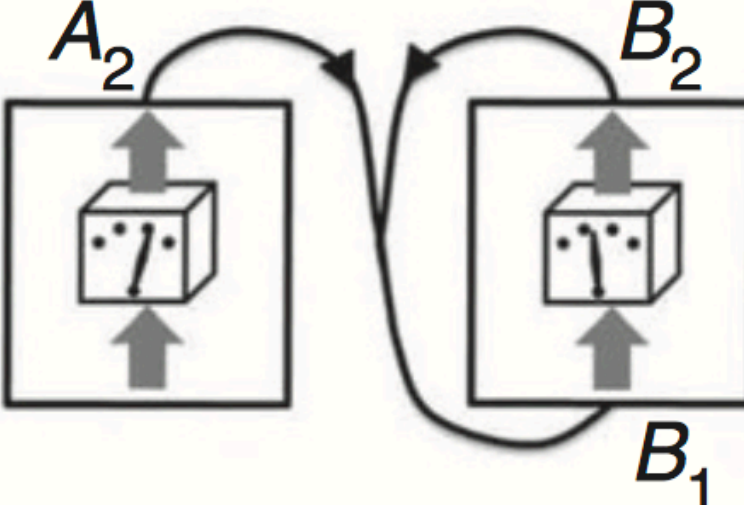
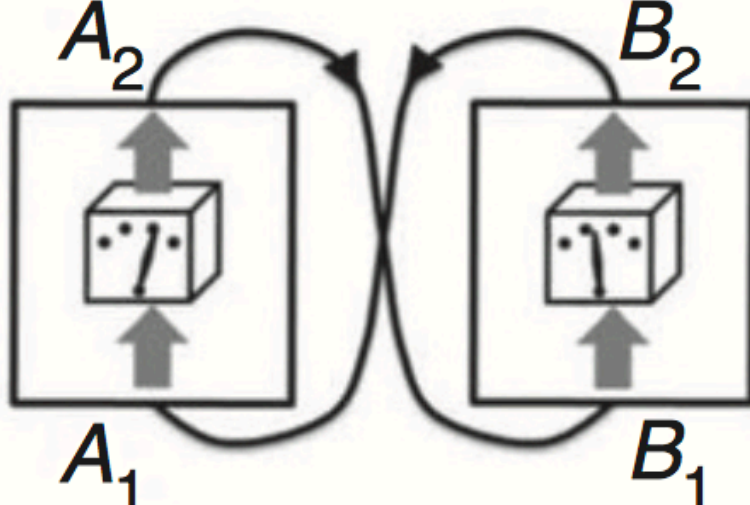


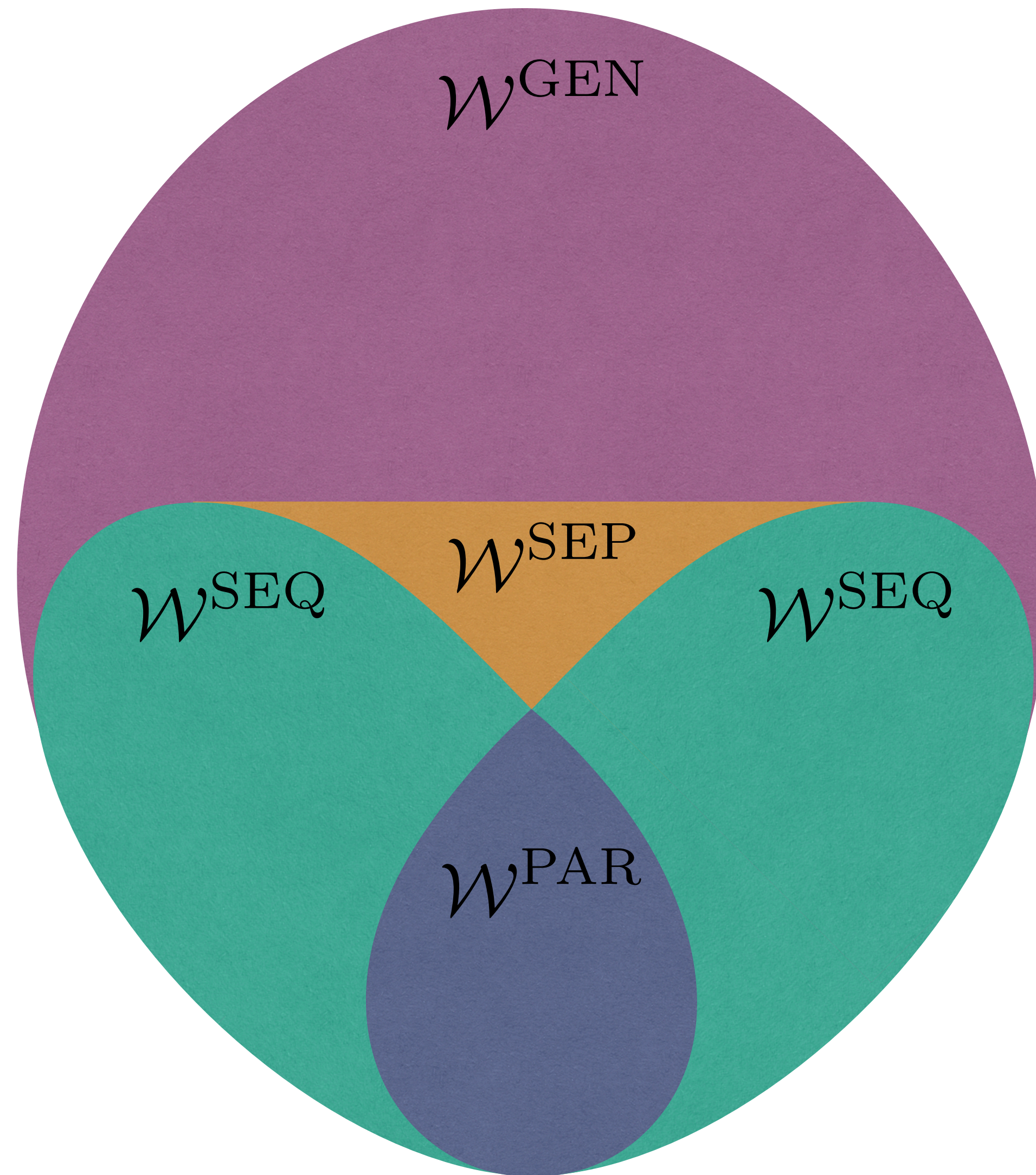
PROCESS MATRIX

ALLOWED

$B \not\leq A$	$A_1, B_1, A_1 B_1$	$A_2 B_1$	$A_1 A_2 B_1$
$A \not\leq B$		$A_1 B_2$	$A_1 B_1 B_2$
Causal order	States	Channels	Channels with memory
			

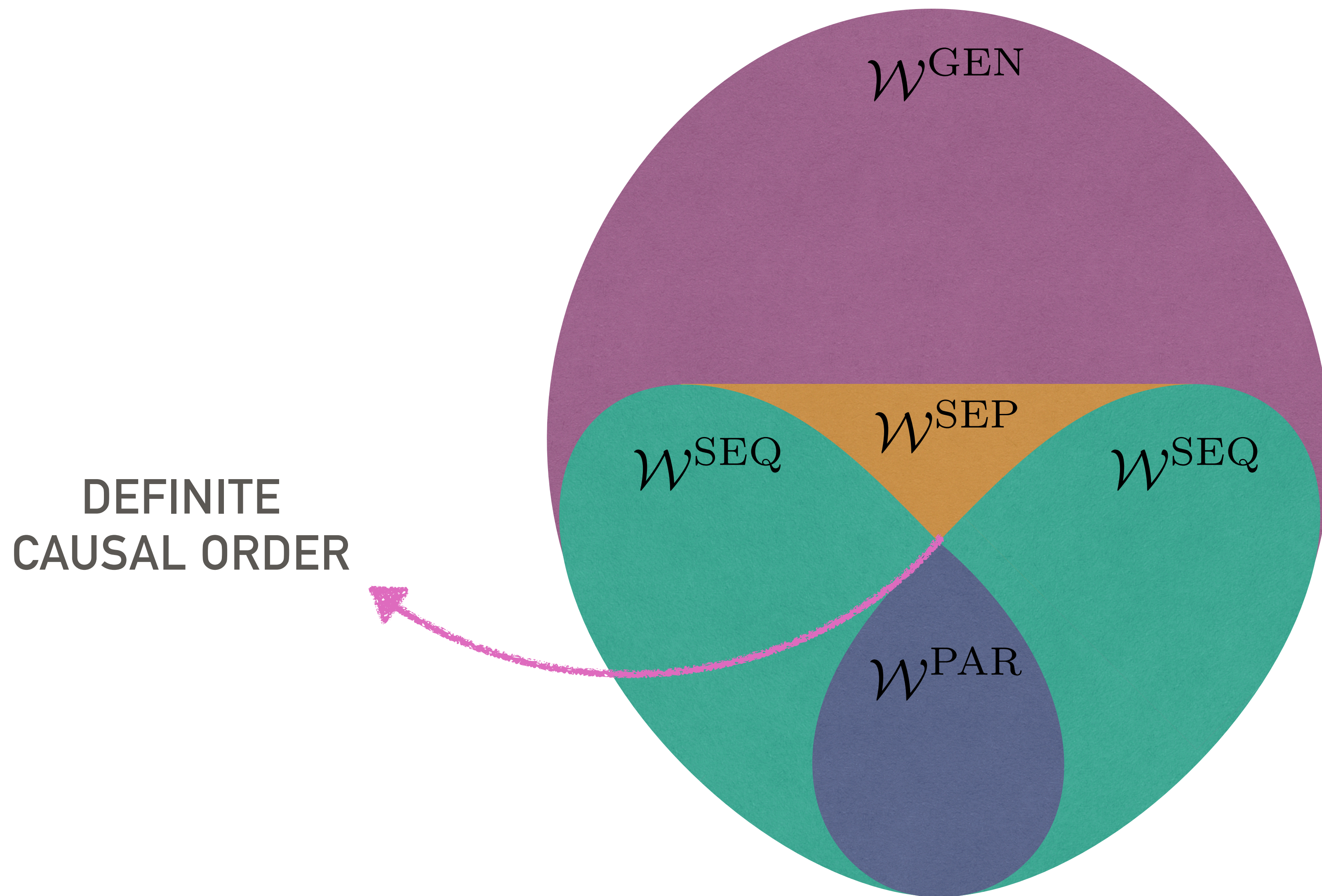
NOT ALLOWED

$A_2, B_2, A_2 B_2$	$A_1 A_2, B_1 B_2$	$A_1 A_2 B_2, A_2 B_1 B_2$	$A_1 A_2 B_1 B_2$
Postselection	Local loops	Channels with local loops	Global loops
			



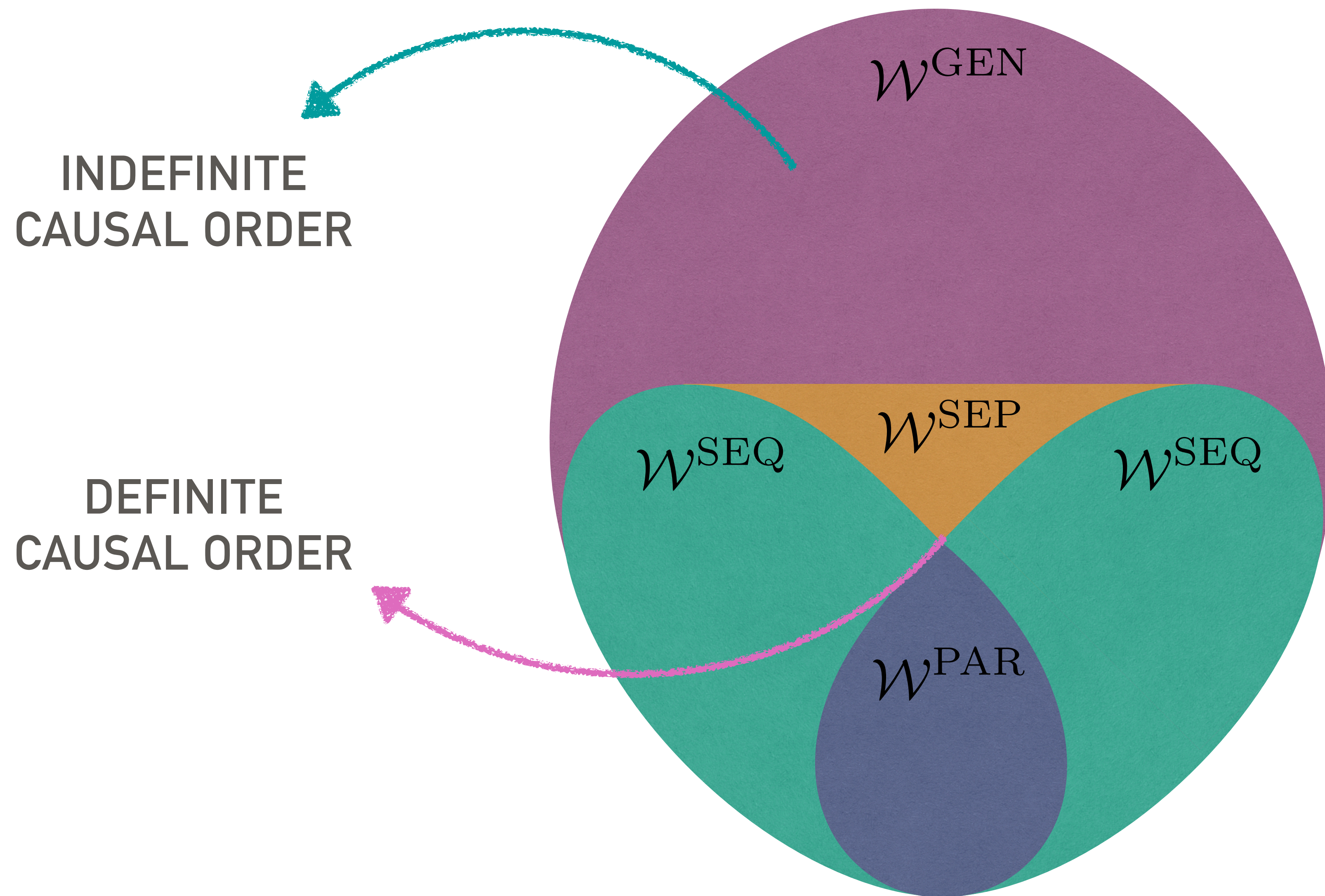
$$\begin{aligned}
 &W \geq 0 \\
 &\text{Tr}[W(C_1 \otimes C_2)] = 1 \\
 &\forall C_1, C_2
 \end{aligned}$$

Set of all processes $\mathcal{W}^{\mathcal{S}}$



$$\begin{aligned}
 W &\geq 0 \\
 \text{Tr}[W(C_1 \otimes C_2)] &= 1 \\
 \forall \quad C_1, C_2
 \end{aligned}$$

Set of all processes $\mathcal{W}^{\mathcal{S}}$

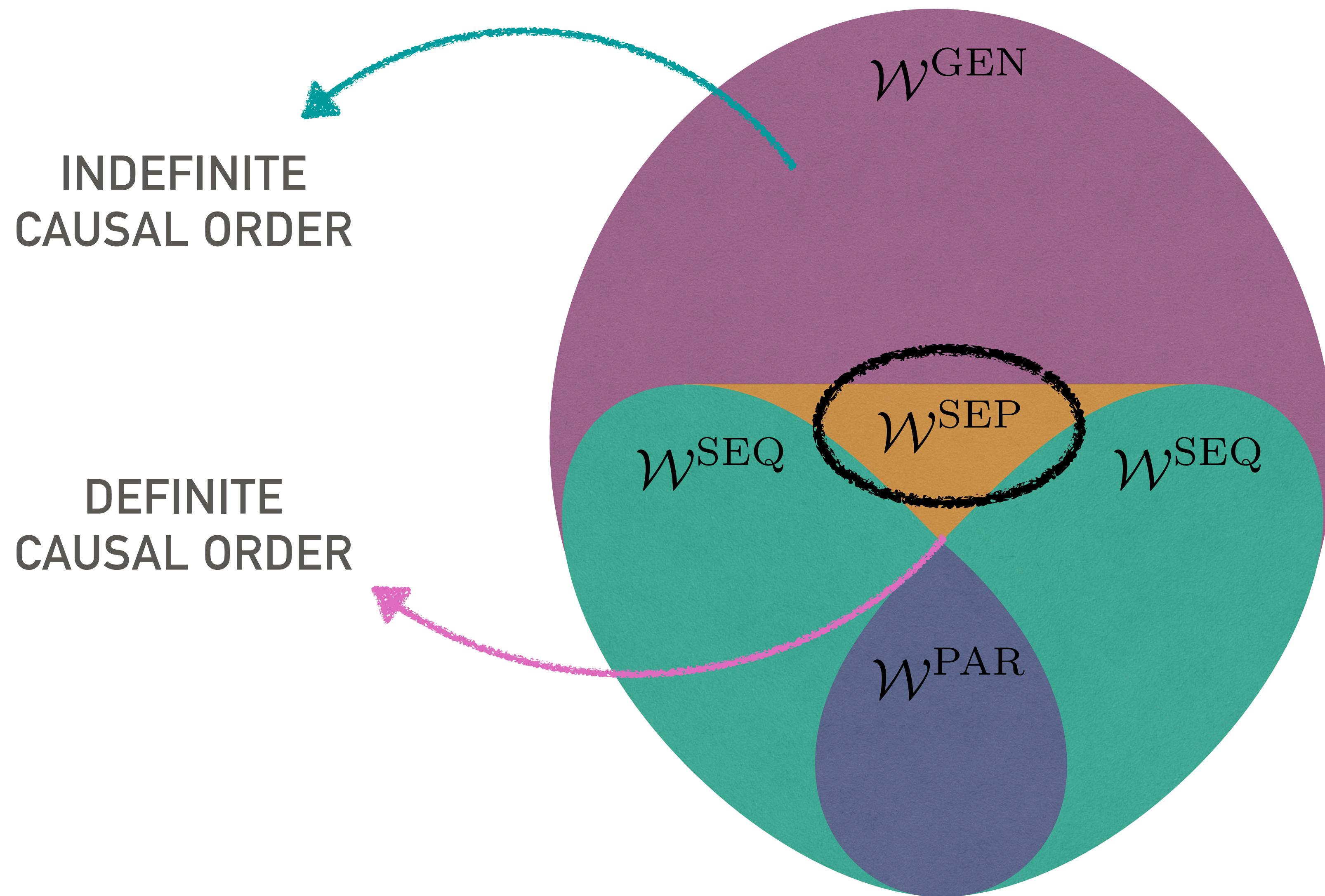


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Set of all processes $\mathcal{W}^S$



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Set of all processes $\mathcal{W}^S$

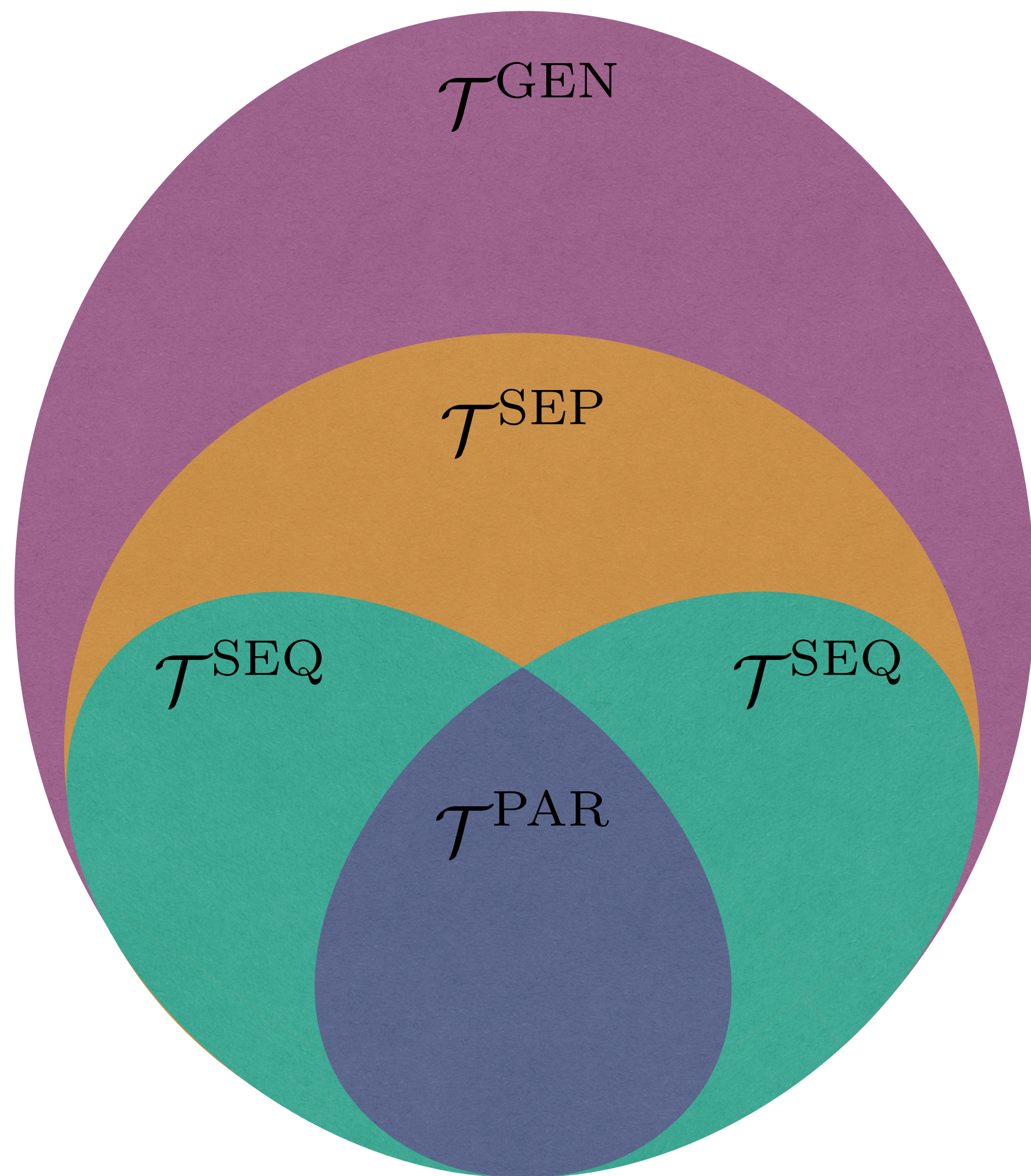
SEPARABLE TESTERS

SEPARABLE TESTERS

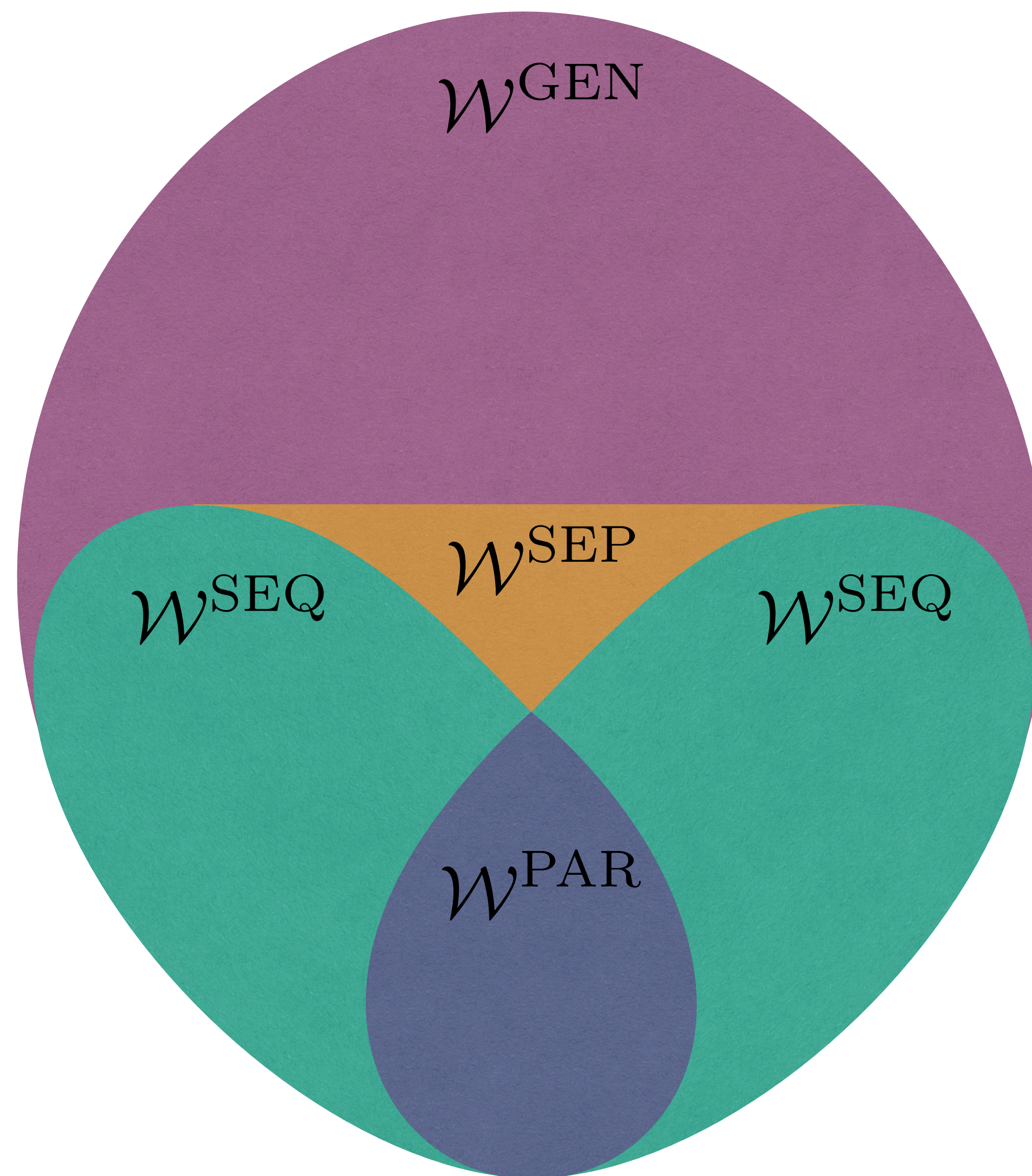
$$T^{\text{SEP}} = \{T_i^{\text{SEP}}\} \quad :$$

$$T_i^{\text{SEP}} \geq 0 \quad \forall i$$

$$\sum_i T_i^{\text{SEP}} = q W_{1 \prec 2}^{\text{SEQ}} + (1 - q) W_{2 \prec 1}^{\text{SEQ}} =: W^{\text{SEP}}$$



Set of all testers $\mathcal{T}^{\mathcal{S}}$



Set of all processes $\mathcal{W}^{\mathcal{S}}$

SEMIDEFINITE PROGRAMMING (SDP)

$$P^{\mathcal{S}} = \max_{\{T_i^{\mathcal{S}}\}} \sum_i p_i \operatorname{Tr} (C_i^{\otimes 2} T_i^{\mathcal{S}})$$

SEMIDEFINITE PROGRAMMING (SDP)

PRIMAL

$$P^{\mathcal{S}} = \max_{\{T_i^{\mathcal{S}}\}} \sum_i p_i \operatorname{Tr} (C_i^{\otimes 2} T_i^{\mathcal{S}})$$

given $\{p_i, C_i\}$

maximize $\sum_i p_i \operatorname{Tr}(T_i^{\mathcal{S}} C_i^{\otimes 2})$

subject to $\{T_i^{\mathcal{S}}\}$ is a tester with strategy $\mathcal{S}$.

SEMIDEFINITE PROGRAMMING (SDP)

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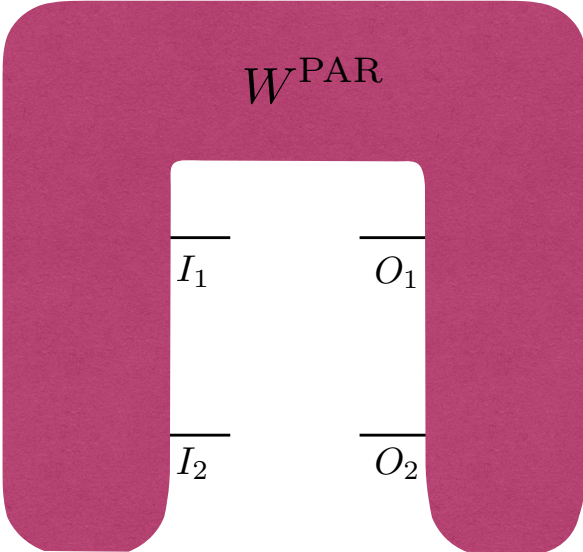
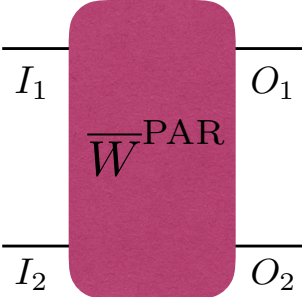
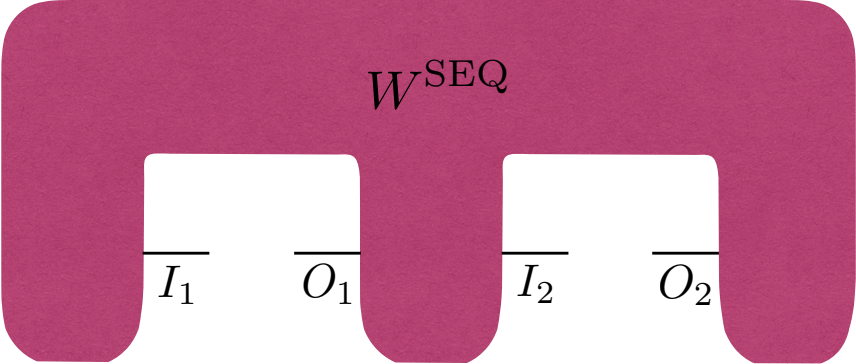
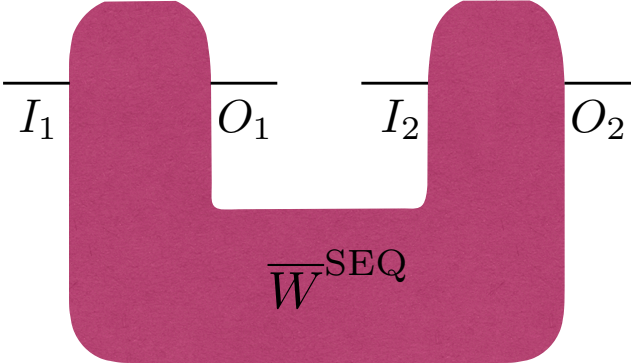
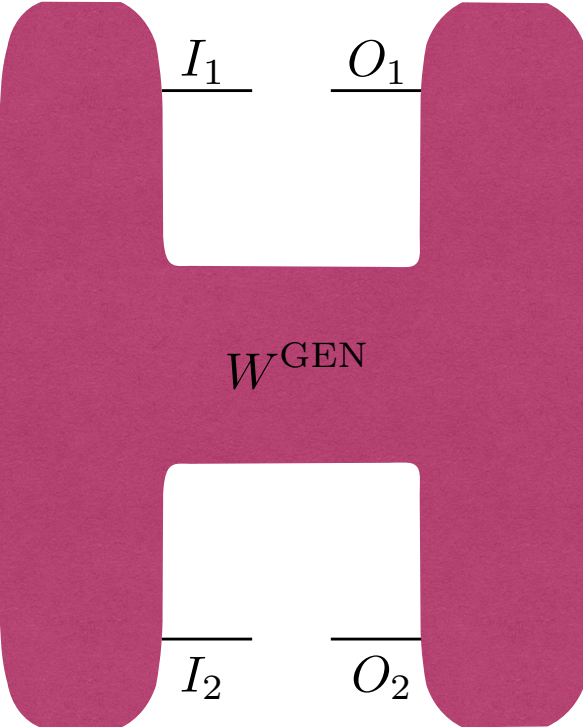
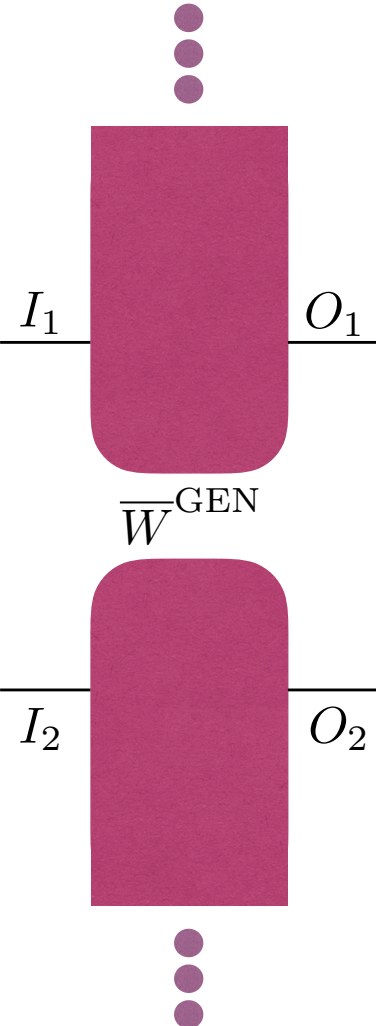
minimize λ

subject to $p_i C_i^{\otimes 2} \leq \lambda \overline{W}^{\mathcal{S}} \quad \forall i$

PRIMAL

DUAL

$$\text{Tr}(W \overline{W}) = 1 \quad \forall \quad W \in \mathcal{W}, \overline{W} \in \overline{\mathcal{W}}$$

	PROCESS		DUAL AFFINE (CHANNEL)	
PARALLEL		$\text{Tr}(W^{\text{PAR}}) = d_{O_1} d_{O_2}$ $W^{\text{PAR}} =_{O_1 O_2} W^{\text{PAR}}$		$\text{Tr}(\overline{W}^{\text{PAR}}) = d_{I_1} d_{I_2}$ $_{O_1 O_2} \overline{W}^{\text{PAR}} =_{I_1 O_1 I_2 O_2} \overline{W}^{\text{PAR}}$
SEQUENTIAL		$\text{Tr}(W^{\text{SEQ}}) = d_{O_1} d_{O_2}$ $W^{\text{SEQ}} =_{O_2} W^{\text{SEQ}}$ $_{I_2 O_2} W^{\text{SEQ}} =_{O_1 I_2 O_2} W^{\text{SEQ}}$		$\text{Tr}(\overline{W}^{\text{SEQ}}) = d_{I_1} d_{I_2}$ $_{O_2} \overline{W}^{\text{SEQ}} =_{I_2 O_2} \overline{W}^{\text{SEQ}}$ $_{O_1 I_2 O_2} \overline{W}^{\text{SEQ}} =_{I_1 O_1 I_2 O_2} \overline{W}^{\text{SEQ}}$
GENERAL		$\text{Tr}(W^{\text{GEN}}) = d_{O_1} d_{O_2}$ $_{I_1 O_1} W^{\text{GEN}} =_{I_1 O_1 O_2} W^{\text{GEN}}$ $_{I_2 O_2} W^{\text{GEN}} =_{O_1 I_2 O_2} W^{\text{GEN}}$ $W^{\text{GEN}} =_{O_1} W^{\text{GEN}} +_{O_2} W^{\text{GEN}} -_{O_1 O_2} W^{\text{GEN}}$		$\text{Tr}(\overline{W}^{\text{GEN}}) = d_{I_1} d_{I_2}$ $_{O_1} \overline{W}^{\text{GEN}} =_{I_1 O_1} \overline{W}^{\text{GEN}}$ $_{O_2} \overline{W}^{\text{GEN}} =_{I_2 O_2} \overline{W}^{\text{GEN}}$

$$P^{\mathcal{S}} = \max_{\{T_i^{\mathcal{S}}\}} \sum_i p_i \operatorname{Tr} (C_i^{\otimes 2} T_i^{\mathcal{S}})$$

$$P^{\text{PAR}} \leq P^{\text{SEQ}} \leq P^{\text{SEP}} \leq P^{\text{GEN}}$$

$$P^{\text{PAR}} < P^{\text{SEQ}} < P^{\text{SEP}} < P^{\text{GEN}}$$

STRICT HIERARCHY BETWEEN DISCRIMINATION STRATEGIES

MAIN EXAMPLE

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$k = 2$ copies, $N = 2$ candidates,

$$p_1 = p_2 = \frac{1}{2}$$

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$k = 2$ copies, $N = 2$ candidates,

$$p_1 = p_2 = \frac{1}{2}$$

$$\tilde{C}_1 = \tilde{C}_{\text{AD}}, \quad \tilde{C}_2 = \tilde{C}_{\text{BF}}$$

AMPLITUDE DAMPING

$$\tilde{C}_{\text{AD}}(\rho) = K_0 \rho K_0^\dagger + K_1 \rho K_1^\dagger$$

$$K_0 = |0\rangle\langle 0| + \sqrt{1 - \gamma}|1\rangle\langle 1|$$

$$K_1 = \sqrt{\gamma}|0\rangle\langle 1|$$

BIT FLIP

$$\tilde{C}_{\text{BF}}(\rho) = \eta \rho + (1 - \eta) X \rho X$$

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$P^{\text{PAR}} = 0.8346 < P^{\text{SEQ}} = 0.8447 < P^{\text{SEP}} = 0.8487 < P^{\text{GEN}} = 0.8514$$

COMPUTER-ASSISTED PROOF

Example: how to create a “valid” channel

- Take numerically imprecise matrix C from the solution of an SDP
- Truncate C and define $C \mapsto \frac{C + C^\dagger}{2}$
- Project C onto the subspace of valid channels, $C \mapsto L(C)$
- Find coefficient η such that $C \mapsto \eta C + (1 - \eta)\mathbb{I} \geq 0$
- Output $C \mapsto d_I \frac{C}{\text{Tr}(C)}$

$$\frac{8346}{10000} < P^{\text{PAR}} < \frac{8347}{10000}$$

$$\frac{8446}{10000} < P^{\text{SEQ}} < \frac{8447}{10000}$$

$$\frac{8486}{10000} < P^{\text{SEP}} < \frac{8487}{10000}$$

$$\frac{8514}{10000} < P^{\text{GEN}} < \frac{8515}{10000}$$

QUANTUM REALISATION OF TESTERS

PARALLEL

SEQUENTIAL

SEPARABLE

GENERAL

QUANTUM REALISATION OF TESTERS

PARALLEL

states, measurements

SEQUENTIAL

SEPARABLE

GENERAL

QUANTUM REALISATION OF TESTERS

PARALLEL

states, measurements

SEQUENTIAL

states, channels, measurements

SEPARABLE

GENERAL

QUANTUM REALISATION OF TESTERS

PARALLEL

states, measurements

SEQUENTIAL

states, channels, measurements

SEPARABLE

coherent quantum control
of causal orders¹

GENERAL

¹ J. Wechs, H. Dourdent, A. A. Abbott, C. Branciard, arXiv: 2101.08796 [quant-ph] (2021)

QUANTUM REALISATION OF TESTERS

PARALLEL

states, measurements

SEQUENTIAL

states, channels, measurements

SEPARABLE

coherent quantum control
of causal orders¹

GENERAL

process matrices, measurements

¹ J. Wechs, H. Dourdent, A. A. Abbott, C. Branciard, arXiv: 2101.08796 [quant-ph] (2021)

CONCLUSIONS

CONCLUSIONS

- Unified tester formalism that includes indefinite-causal-order strategies.
- Strict hierarchy between discrimination strategies in the simplest scenario (2 copies, 2 candidates, qubit channels).
- Method of computer-assisted proofs readily applicable to quantum information problems.

THANK YOU!

EXTRA

PARALLEL PROCESSES

$$W^{\text{PAR}} \in L(H^{I_1} \otimes H^{O_1} \otimes H^{I_2} \otimes H^{O_2})$$

$$W^{\text{PAR}} \geq 0$$

$$\text{Tr}(W^{\text{PAR}}) = d_{O_1} d_{O_2}$$

$$W^{\text{PAR}} =_{O_1 O_2} W^{\text{PAR}}$$

SEQUENTIAL PROCESSES

$$W_{1\prec 2}^{\text{SEQ}} \in L(H^{I_1} \otimes H^{O_1} \otimes H^{I_2} \otimes H^{O_2})$$

$$W_{1\prec 2}^{\text{SEQ}} \geq 0$$

$$\text{Tr}(W_{1\prec 2}^{\text{SEQ}}) = d_{O_1} d_{O_2}$$

$$W_{1\prec 2}^{\text{SEQ}} =_{O_2} W_{1\prec 2}^{\text{SEQ}}$$

$$I_2 O_2 W_{1\prec 2}^{\text{SEQ}} =_{O_1} I_2 O_2 W_{1\prec 2}^{\text{SEQ}}$$

GENERAL PROCESSES

$$W^{\text{GEN}} \in L(H^{I_1} \otimes H^{O_1} \otimes H^{I_2} \otimes H^{O_2})$$

$$W^{\text{GEN}} \geq 0$$

$$\text{Tr}(W^{\text{GEN}}) = d_{O_1} d_{O_2}$$

$$I_1 O_1 W^{\text{GEN}} = I_1 O_1 O_2 W^{\text{GEN}}$$

$$I_2 O_2 W^{\text{GEN}} = O_1 I_2 O_2 W^{\text{GEN}}$$

$$W^{\text{GEN}} + O_1 O_2 W^{\text{GEN}} = O_1 W^{\text{GEN}} + O_2 W^{\text{GEN}}$$